

GeoForge Desktop

User manual

Version 0.6.55 · Windows

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0 Quick start

Edition 0.6.55 (Windows). The full manual preserves the detailed real FSM2 walkthrough and screenshots captured for 0.6.54. Current fixes and SHAW/CRHM/VIC verification are described in Chapter 11. For the shortest route, open **Guide → Quickstart · 3 pages**; calibration has its own guide and Chapter 10. Screenshots are from the real application, not invented screens.

In this chapter you will follow the whole path from installing GeoForge Desktop 0.6.55 to a model run that GeoForge marks **Completed**, in eight steps. Each step names the chapter that explains it in detail.

The screenshots show the Windows build in English. The example is the FSM2 snow model run with a DeepSeek API key, the one case tested end to end on Windows for 0.6.54.

Three ideas worth remembering

- 1. **Chat is the entry point.** You start every task by describing it in a chat. GeoForge plans from what you write.
- 2. **A KI describes one scientific model.** A KI (Knowledge Infrastructure) is GeoForge’s packaged knowledge about a model such as FSM2: how to install, prepare, run and check it. Having the KI does not mean the model software is on your computer yet.
- 3. **Every chat is one project folder.** Each chat has its own folder on your computer for its inputs, model runs, outputs, plots and run records. A run record is a signed file GeoForge writes for each step it runs, naming the command and the files read and written.

Step 1: Install and open GeoForge (Chapter 1)

	Windows	macOS
Release to use	windows-v0.6.55	v0.6.54
Where GeoForge appears	A page in your default web browser, plus an icon in the system tray	Its own window
How to quit	Right-click the tray icon, then choose Exit GeoForge	Close the window

On Windows:

- 1. Open the [windows-v0.6.55 release page](#).
 - 2. Download `GeoForge-Desktop-Setup-v0.6.55-Windows-x64.exe`.
 - 3. Double-click the downloaded file and follow the installer to the end. You do not need administrator rights. If Windows blocks the file, see Chapter 1.
 - 4. Open **GeoForge Desktop** from the Start menu.
- Check that GeoForge opens as a page in your web browser and that its icon appears in the system tray.

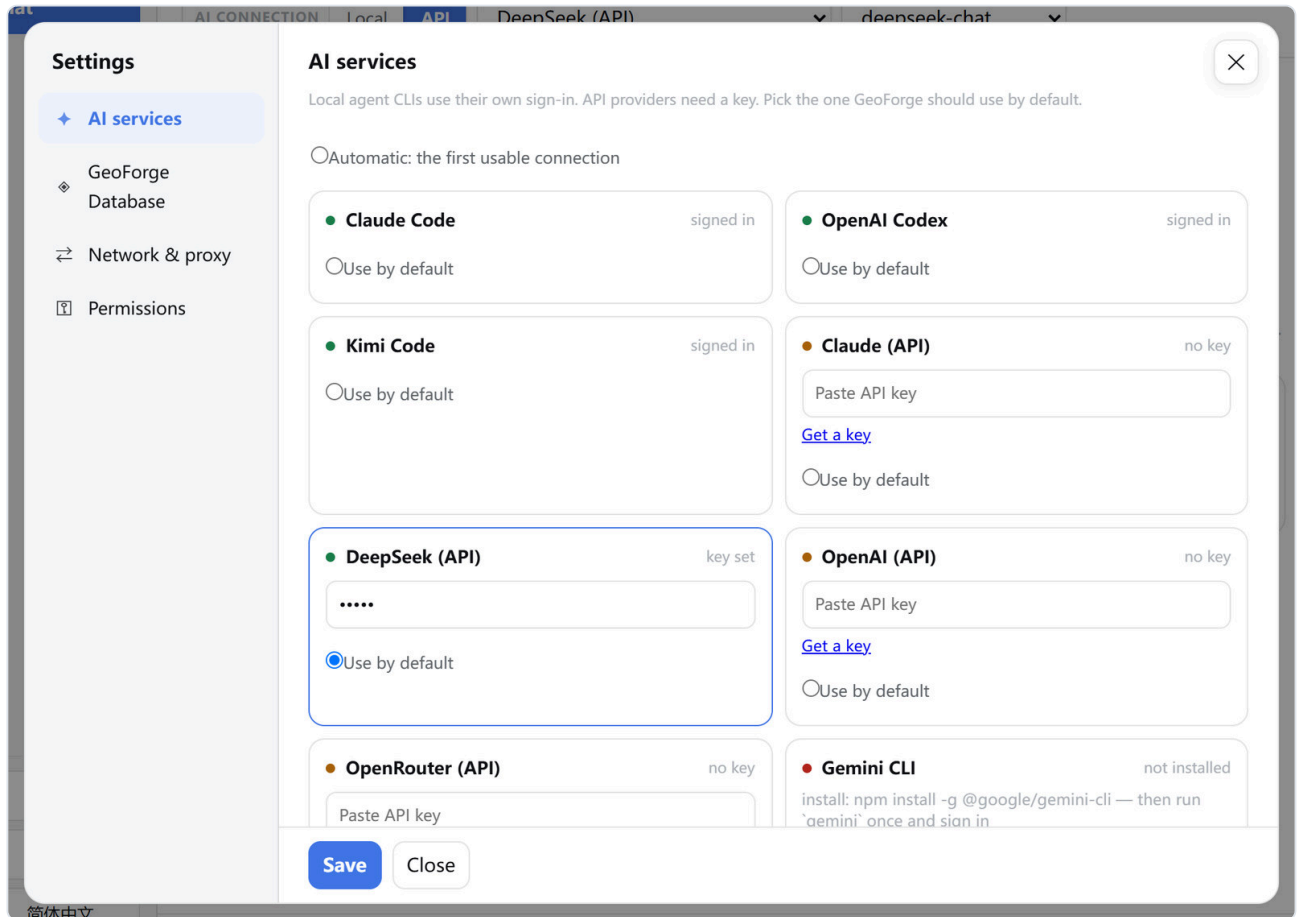
Closing the browser tab does not quit GeoForge; it keeps running in the tray. To quit, right-click the tray icon and choose **Exit GeoForge**.

On macOS: install from the `v0.6.54` release as Chapter 1 describes. The macOS 0.6.54 build dates from 19 September 2026 and lacks some later changes in the Windows build, so some screens in this manual look different on a Mac.

Step 2: Connect an AI (Chapter 2)

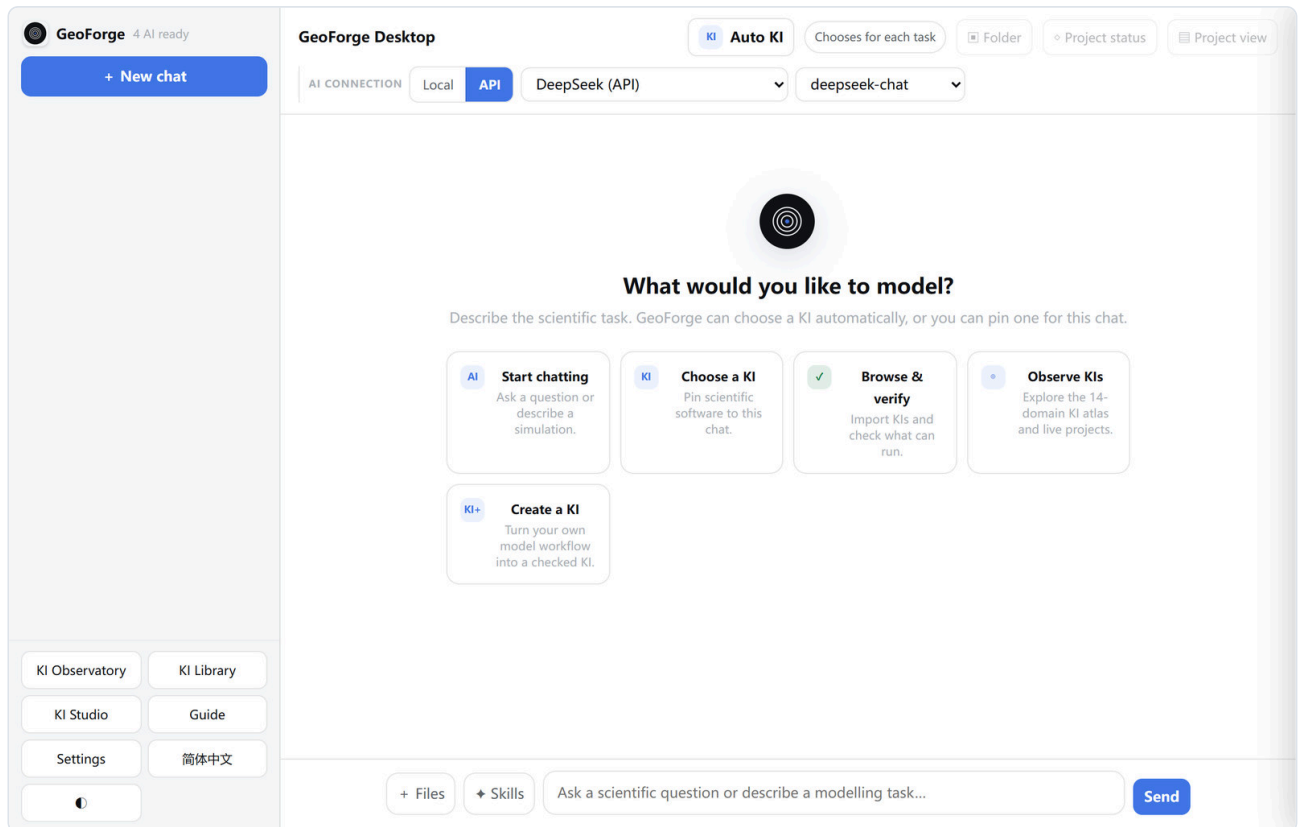
GeoForge needs an AI to plan and run work. This example uses a DeepSeek API key. The GeoForge Database (Chapter 3) is optional and not needed here.

1. Click **Settings** at the bottom of the sidebar.
2. Click **AI services** if that page is not already open.
3. On the **DeepSeek (API)** card, paste your key into the **Paste API key** field.
4. On the same card, select **Use by default**.
5. Click **Save**.



Check that the word "saved" appears briefly next to **Close**, and that the **DeepSeek (API)** card reads **key set** with **Use by default** selected.

6. Click **Close**.



Check that the top of the sidebar reads a number followed by **AI ready** (here **4 AI ready**), not **AI setup needed**.

To use an agent installed on your computer instead (such as Claude Code, OpenAI Codex or Kimi Code):

1. Sign in to the agent in PowerShell (Windows) or Terminal (macOS), as the agent's own instructions say.
2. In **Settings** → **AI services**, click **Recheck local CLIs**.

Check that the agent's card reads **signed in**.

Caution: In 0.6.54 only the DeepSeek API path was tested end to end on Windows. Local agents were not. On Windows, Kimi Code does not start until you choose **Enable Kimi with full computer access** under **Settings** → **Permissions** (Chapter 2 explains the risk).

Step 3: Start a new chat and choose the model (*Chapters 4 and 5*)

1. Click + **New chat**.

New project chat

×

Each chat has its own project folder for memory, inputs, model runs, outputs, and plots.

Project name

Alptal snow example

Create the project inside

D:\GeoForge-Manual\projects

Choose...

Project folder (created automatically)

D:\GeoForge-Manual\projects\2026-10-01-Alptal-snow-example--{id}

Edit the suggested name or keep it. GeoForge creates the folder automatically, including missing parent folders. {id} becomes a unique ID so projects with the same name stay separate. Existing files are not changed.

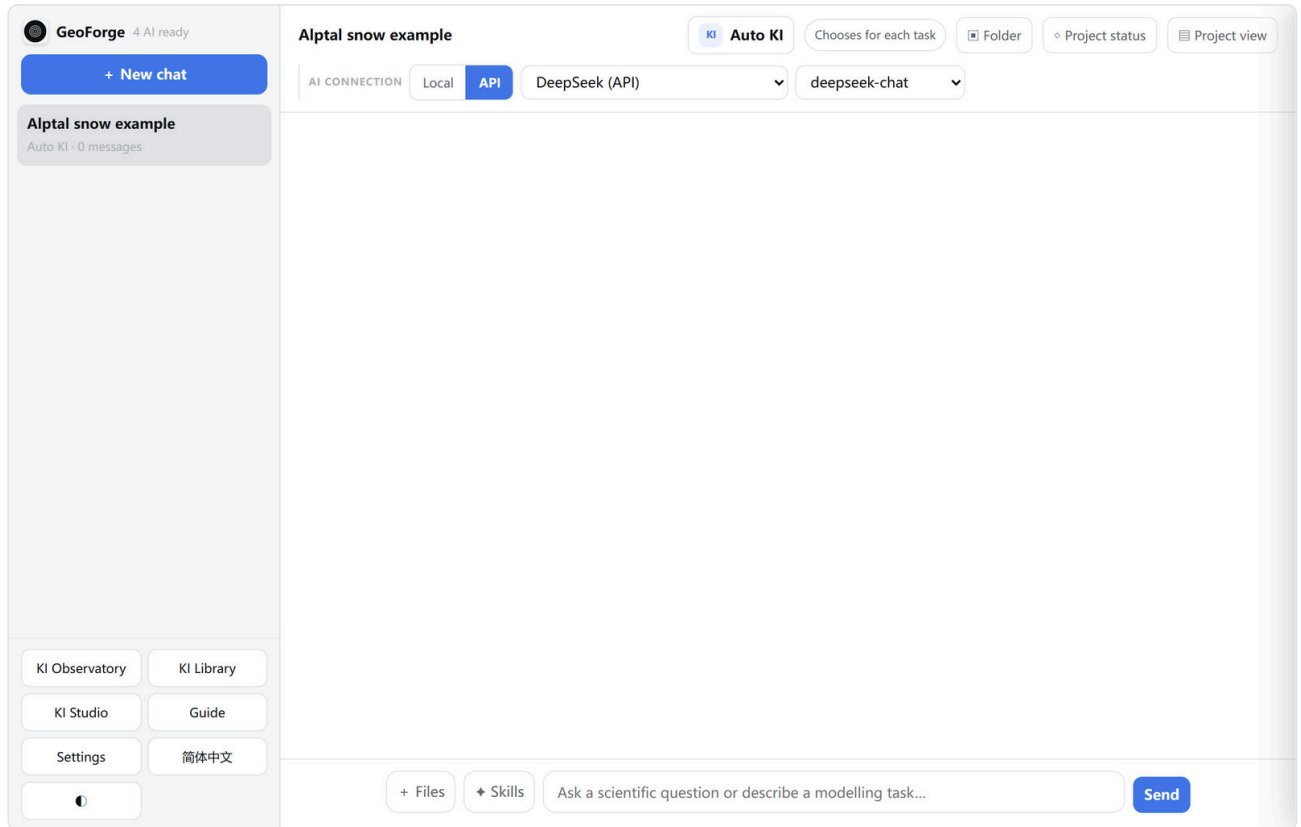
Create chat

Use default location

Cancel

Check that **Project name** is filled in and that **Project folder (created automatically)** shows where the project will go. {id} is replaced by a unique code.

2. Keep the suggested **Project name**, or type a new one.
3. Keep the folder in **Create the project inside**, or type another path (for example `D:\GeoForge-Manual\projects`). On Windows, **Choose...** cannot open a folder picker, so type or paste the path.
4. Click **Create chat**.



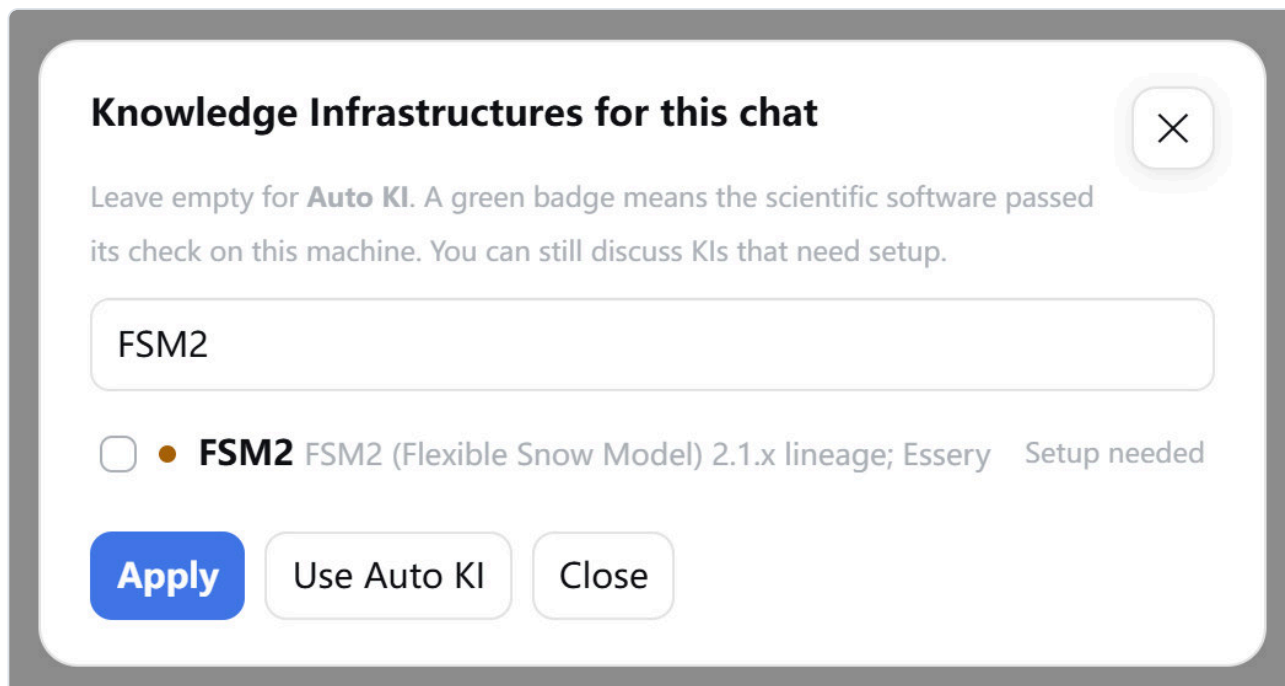
Check that the chat name appears in the header and in the sidebar, with **Auto KI • 0 messages** under it.

5. Under **AI CONNECTION**, click **API** to use an API key, or **Local** to use an agent on your computer.
6. In the first list next to it, choose the provider (here **DeepSeek (API)**).
7. In the second list, choose the model (here **deepseek-chat**).

Tip: On Windows, each start of GeoForge uses a new local address, so the page can forget your last **Local/API** choice. Check **AI CONNECTION** before you send the first message.

To let GeoForge choose the KI from your request, leave **Auto KI** as it is and go to Step 4. To choose the KI yourself:

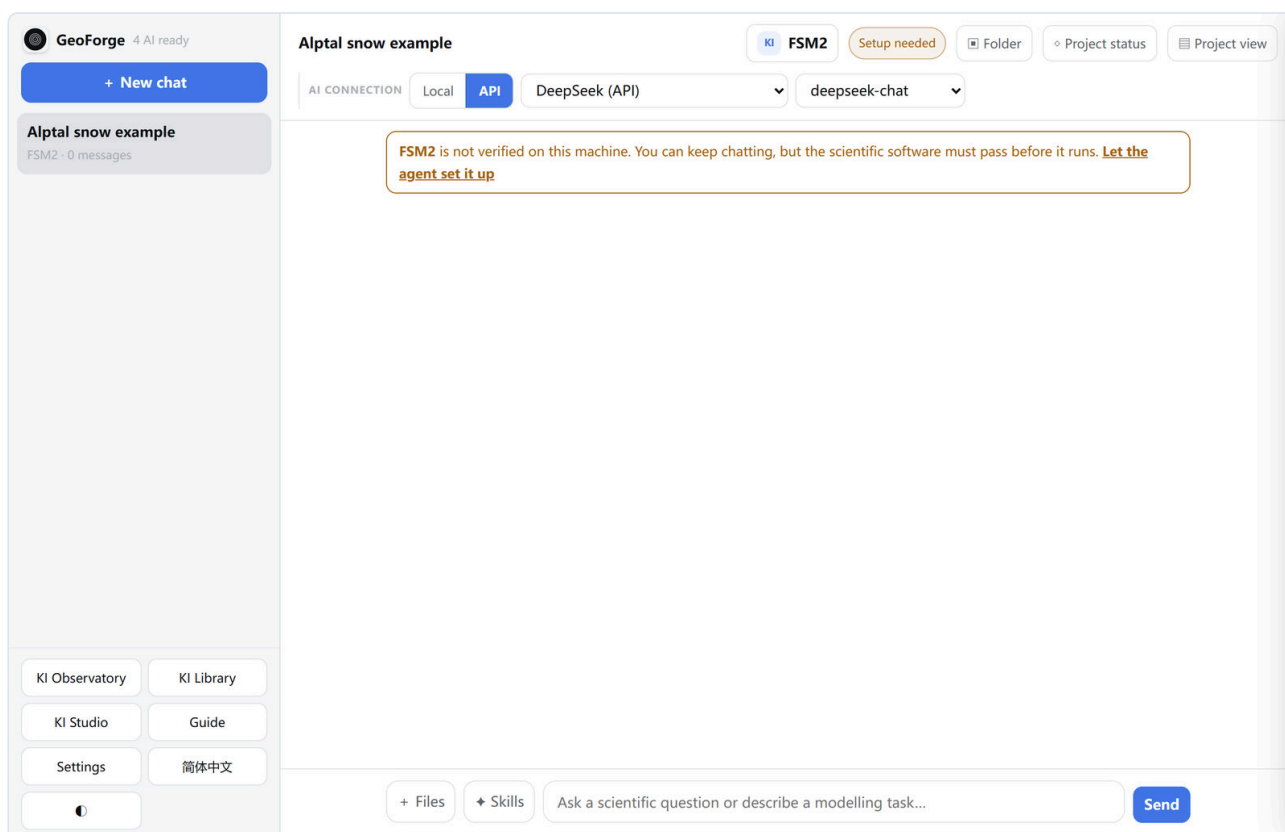
1. Click **Auto KI** in the header.
2. Type the model name, for example `FSM2`, in **Search KIs....**



Check that the model's row appears. **Setup needed** means the model software is not yet verified on this computer.

3. Tick the box next to the model.

4. Click **Apply**.



Check that the header shows **FSM2** with **Setup needed**, and that a banner says FSM2 is not verified on this machine. You do not need to set it up now: after you approve the plan, GeoForge sets up the software before it runs the model.

Caution: After you send the first message, the KI, the **Local/API** choice, the provider and the model are fixed for this chat. To change any of them, start a new chat.

Step 4: Describe your task (*Chapter 5*)

1. Click in the message box (**Ask a scientific question or describe a modelling task...**).
2. Type your request. Name the place, the period, the process, the outputs you want, and any data you have or do not want.
3. Click **Send**.

Check that **Send** changes to **Working...** and that a bar with a **Stop** button appears under the message box while the AI works.

This is the request used for the worked example in Chapter 5 (SWE is snow water equivalent):

Run an end-to-end FSM2 snow simulation using the official Alptal example case that ships with the FSM2 repository (its sample meteorological forcing and namelist, winter 2004-2005, two points: open and forest) with the default physics options. Produce the snow depth and snow water equivalent (SWE) time series with a plot, and summarise peak SWE and the melt-out date for each point. No GeoForge Database data is needed. Use only the FSM2 KI's own tools for every step.

A plain question such as "What is SWE?" gets an answer but does not start a plan. GeoForge decides by simple keywords: a message that mentions a model, a simulation or a run starts planning.

Step 5: Answer the planning questions (*Chapter 5*)

While the AI plans, GeoForge allows it only to read and to draft the plan. Downloads, installations and model runs wait for your approval. When the AI needs a decision, a card with a **Needs you** tag opens. Cards come one at a time.

1. Read the question on the card.
2. Select one option. To answer in your own words, select **Use my own answer / file** and type your answer in **Optional note for the agent**.

The option set above rests on `kiss.windows.yaml` + `install.windows.md` + the tool default, not on reading `OPTS.h` from disk. I will confirm this at execution. I have also not yet confirmed that `met\_Alptal\_0405.txt` and `nlst\_Alptal.txt` are physically present at the installed repo path — that is a file existence check for the acquisition step, and if they are absent the fallback is re-running the pinned build step, not fabricating forcing.

This run is the KI's official example case; per `docs/validation\_convention.yaml` and SKILL.md §11 it is the source of the KI's `example\_validated` status, not a validation against observations for your site. A separate decision about observational validation can be asked later if you want it — it is deliberately not bundled here.

☒ **A — Use the verified pre-built FSM2.exe with its default physics options (recommended)**
 Run the already-built, machine-verified binary. Physics = the official default set recorded in kiss.windows.yaml / run_fsm2.py DEFAULT_OPTIONS (ALBEDO=2, CANINT=1, CANMOD=1, CANRAD=1, CANUNL=1, CONDUCT=1, DENSTY=1, EXCHNG=1, HYDROL=1, SGRAIN=1, SNFRAC=1, DRIV1D=1, SWPART=0, ZOFFST=0, PROFNC=0). No compilation, no toolchain needed; reproduces the KI's recorded 5,832-row Alptal example.

☐ **B — Recompile from the pinned source with the same default options, in a copied work directory**
 Build a fresh binary from commit f0c86be1 with the same official option set, using the staged gfortran 16.2.0. Adds a compile step and copies source/repo/src first so the shared installation's OPTS.h and binary are not modified. Physically identical results to A.

☐ **C — Recompile with a different physics option set (I will specify which switches)**
 Choose non-default switches, e.g. SWPART=1 (solar-position diffuse/direct split, needs a correct lon-derived `noon` in &drive), CANMOD=2 (two canopy layers), or SNFRAC=2/3. This is no longer the KI's default configuration and its output cannot be compared directly to the KI's recorded example validation.

Check that your option is highlighted. The question in your project will differ from this example.

3. Click **Continue with this choice** at the bottom of the card.

4. Repeat for each new card.

Answer on the card itself: a reply typed in the chat is not saved as your answer. If you close a card with **Not now** or ✕, reopen it with ◇ **Project status** → **Respond**.

Step 6: Approve the plan (*Chapter 5*)

When planning is finished, the card **Approve the plan?** opens. Its sections are **What I understood**, **Data**, **Steps**, and, when they apply, **Data sources**, **your pick**, **Scientific decisions**, **Waiting on you** and **Not ready to execute yet**.

?
Approve the plan?
Needs you
X

WHAT I UNDERSTOOD

Run an end-to-end FSM2 snow simulation using the official Alptal example case that ships with the FSM2 repository (its sample meteorological forcing and namelist, winter 2004-2005, two points: open and forest) with the default physics options. Produce the snow depth and snow water equivalent (SWE) time series with a plot, and summarise peak SWE and the melt-out date for each point. No GeoForge Database data is needed. Use only the FSM2 KI's own tools for every step.

KI · FSM2

area · Alptal, Switzerland (47.05 N, ~8.7 E) — the FSM2 upstream example site, one open point and one forested point

period · 2004-12-01 00:00 to 2005-09-01 00:00 (winter 2004-2005; 5832 hourly records)

DATA · 10

► The run prepares itself (10) already on disk · made by a step

STEPS · 3

1. **FSM2:s6_run_official_example** run · run_fsm2.py env 4
2. **FSM2:s7_parse_official_example** postprocess · parse_fsm2_output.py env 5
3. **FSM2:postflight_check** check · run by GeoForge (preflight)

SCIENTIFIC DECISIONS

- physics_options: A: verified pre-built FSM2.exe with the upstream default physics options (ALBEDO=2, CANINT=1, CANMOD=1, CANRAD=1, CANUNL=1, CONDCT=1, DENSTY=1, EXCHNG=1, HYDROL=1, SGRAIN=1, SNFRAC=1, DRIV1D=1, SWPART=0, ZOFFST=0, PROFNC=0) / B: recompile from source with a different switch combination (factorial experiment) → **Use the verified pre-built FSM2.exe with default physics options; no recompilation.** decided

From your answer: FSM2 physics options for the Alptal example run → A — Use the verified pre-built

Check **What I understood**, **Data** and the three **Steps**, and that the card shows no **Not ready to execute yet** section.

1. Read **What I understood** and **Steps**, and check that they match your request.
2. Read **Data**. Each row names an input; when a specific dataset is chosen for it, that dataset follows a ← sign.
3. Select **Approve and start**.
4. Click **Continue with this choice**.

To change the plan instead, select **Modify the plan**, write what to change in **Optional note for the agent**, and click **Continue with this choice**. The AI revises the plan and shows a new card.

If a row under **You** in **Data** asks for your own file, approval waits for that file. Upload it with the row's **Upload files** button in ◇ **Project status** → **Data in this plan** (Chapter 6).

Check the actual plan: data sources, required inputs and executable steps must match your request. If they disagree, or if **Not ready to execute yet** appears, choose **Modify the plan**. Chapter 9 explains how to diagnose the current approval state.

Step 7: Let GeoForge fetch the data and run (Chapters 3, 4 and 5)

After you approve, GeoForge downloads the approved data itself and shows progress in a banner above the message box. The FSM2 example uses the forcing data that ships with FSM2, so nothing is downloaded. If the model software is not yet verified on this computer, GeoForge sets it up first, then runs the approved steps.

1. Watch the banner above the message box and the AI's messages in the chat.

2. If the banner shows **Start the approved run**, click it.

Current Windows behaviour: API software setup installs and verifies the environment. Execute the scientific example later through the separately approved project plan.

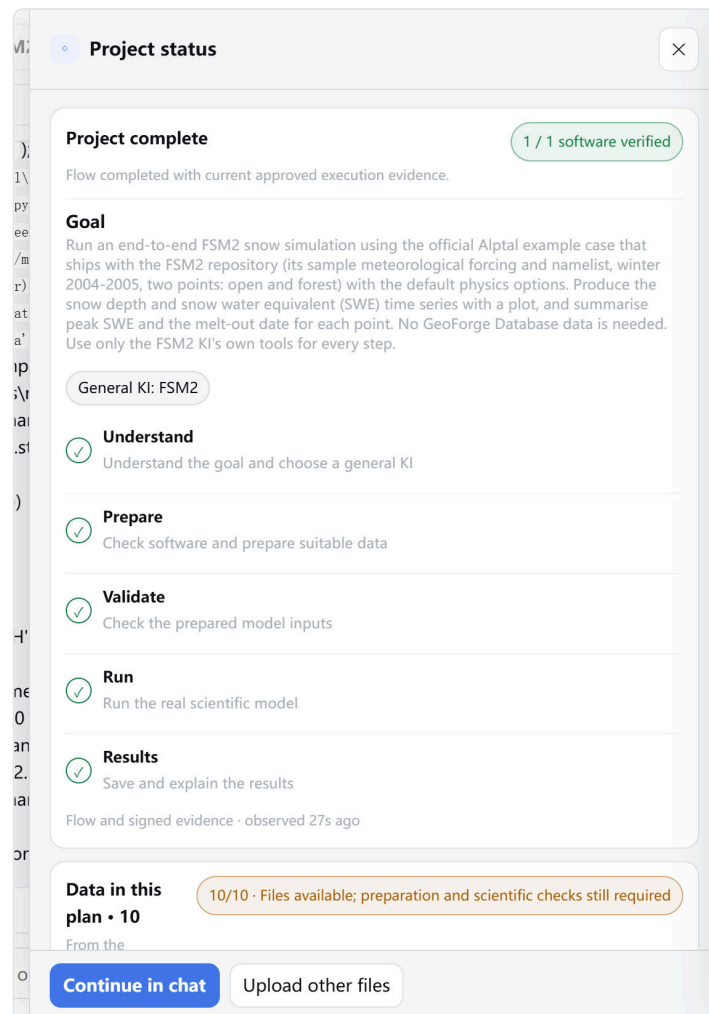
Some large GeoForge Database files must be downloaded by hand from Baidu Pan, a cloud-storage service. ◇ **Project status** then shows a manual download card:

1. Click **Open download link** and download the file, using the **Extraction code** shown on the card.
2. Wait until the download has finished completely.
3. Copy the files into the folder shown after **Place at** (**Copy path** copies it).
4. Click **Files are in place, continue**.

To stop the current work, click **Stop** in the bar under the message box. On Windows, a job the agent started in the background can keep running after **Stop**; check Task Manager (Chapter 9).

Step 8: Check Completed and read the results (*Chapters 5 and 6*)

1. Click ◇ **Project status** in the header.



Check that the panel title reads **Project complete** and that all five steps show a tick. The ◇ **Project status** button turns green.

2. Close the panel with ✕.

3. Read the AI's summary of the results in the chat.
4. Click  **Folder** to open the project folder in File Explorer (Finder on macOS).
5. Click  **Project view** to see the result panels.

After **Completed**, you can still ask the AI about the results in this chat, but it can no longer run, download or change inputs. For another period or scenario, start a new chat.

What "Completed" means, and what it does not

Completed means GeoForge found a passing, signed run record for every step of the approved plan, and the outputs passed GeoForge's automatic checks. For example: the run reported no error, the output files exist and are not empty, they contain no NaN or infinite values, the time axis is complete, and the main output is not all zero or one constant value. Work the AI only describes in the chat does not count.

Completed does not mean the science is validated. Nothing was compared with observations unless your plan included it. Downloaded data were not checked scientifically. Benchmark scores quoted from a KI are not scores of your run.

That is why a completed project can still show "Produced; scientific checks pending" in **Data in this plan**, and "—experimental / unvalidated" on **Project view** panels that no passing run record covers. Ask the AI: "What did you assume, what is measured, and what is generated?"

If something goes wrong

What you see	What to do
AI setup needed, Connect an AI to begin. , or "No API connection yet. Open AI Settings and add a key."	Open Settings → AI services and add a key or sign in to an agent. "AI Settings" is the old name of that page. See Chapter 2.
Send stays grey although a key is saved.	Under AI CONNECTION , click API (or Local) and choose a provider that is ready.
You closed the browser tab on Windows.	GeoForge is still running. Double-click the tray icon, or right-click it and choose Open GeoForge . Do not start it again from the Start menu: that starts a second copy.
"Browser mode: type the folder path here. The desktop build opens the system folder picker."	This appears on Windows when you click Choose.... Type or paste the folder path into Create the project inside .
Wrong model or AI, and the first message is already sent.	Start a new chat and choose before you send.
A question or approval card disappeared.	Click  Project status , then Respond .
"GeoForge verification: not complete yet" and no Completed .	Ask the AI to regenerate or remove the files the message lists. Never delete your own input files. See Chapter 9.

Where to go next

Chapter	Read it to...
1 Install GeoForge	install, start and quit on Windows or macOS
2 Connect an AI	set up keys, local agents, proxy and permissions
3 The GeoForge Database	activate the data catalogue and understand its downloads
4 Install a scientific model	set up or verify model software, or use a copy you have
5 Your first project	follow one full run, from request to Completed
6 Project files	find outputs, plots and run records
7 More tools	use the KI Library, KI Observatory, KI Studio and other tools
8 Language, appearance and keyboard	switch language (简体中文), theme (🌑) and learn the keys
9 Troubleshooting and known issues	fix problems and check current Windows limits

1 Install and first launch

In this chapter you will download GeoForge Desktop 0.6.55 for Windows (macOS notes describe 0.6.54), check the file, install it on Windows or macOS, start it for the first time, and find your way around the main screen. You will also learn where GeoForge keeps your files, and how to upgrade or uninstall it.

What you need

	Windows	macOS
Computer	64-bit (x64) Windows PC. This build was tested on Windows 11.	Mac with Apple Silicon (M-series). Intel Macs are not supported by this release.
Release to use	windows-v0.6.55	v0.6.54
Download size	See the current release Assets for exact bytes	See the macOS release Assets
Disk space for the program	About 400 MB	A few hundred MB
Administrator rights	Not needed	Not needed; macOS may ask for your password when you allow the app to open
Included	The app, its own Python, and all 127 KIs	The app and all 127 KIs

A KI (Knowledge Infrastructure) is GeoForge's packaged knowledge about one scientific model: how to install, prepare, run and check it. Having the KI does not mean the model software is on your computer.

Installing GeoForge does **not** install any scientific model, and it does not give you an AI. You connect an AI in Chapter 2 and install model software in Chapter 4. Each model you install later needs its own disk space, often several GB.

Choose the right download

GeoForge is published on GitHub. Windows and macOS use **different release pages**, and this Windows manual documents 0.6.55; the macOS notes retain their earlier version.

1. Open the release page for your system:
 - **Windows:** <https://github.com/lzwei196/KISS-Knowledge-Infrastructure-for-Scientific-Simulation/releases/tag/windows-v0.6.55> (title "GeoForge Desktop 0.6.55 for Windows").
 - **macOS:** <https://github.com/lzwei196/KISS-Knowledge-Infrastructure-for-Scientific-Simulation/releases/tag/v0.6.54> (title "GeoForge Desktop v0.6.54 — macOS Apple Silicon").
2. Scroll down to **Assets**. If the list is collapsed, click **Assets** to open it.
3. Click the program file for your system (see the tables below). Your browser downloads it.
4. Click the checksum file to download it too: `SHA256SUMS-Windows.txt` on Windows, `SHA256SUMS.txt` on macOS.

Check that the page title reads "GeoForge Desktop 0.6.55 for Windows" and that the file names contain `v0.6.55-Windows-x64` .

Windows files (windows-v0.6.55)

File	What it is	Do you need it?
GeoForge-Desktop-Setup-v0.6.55-Windows-x64.exe	Installer	Yes, recommended for most people
GeoForge-Desktop-v0.6.55-Windows-x64.zip	Portable copy, no installation	Only if you cannot or do not want to install software
SHA256SUMS-Windows.txt	Checksums for the two files above	Yes, to check your download
Windows-release-validation.json , release-manifest.json , DESKTOP_CHANGELOG.md	Test record and release notes	No

macOS files (v0.6.54)

File	What it is	Do you need it?
GeoForge-Desktop-macos-arm64.app.zip	The app	Yes
SHA256SUMS.txt	Checksums	Yes, to check your download
kiss-macos-arm64.tar.gz	Optional command-line tool	No
kiss-ki-packages.tar.gz	The KIs on their own	No, the app already contains them
release-manifest.json , DESKTOP_CHANGELOG.md	Release notes	No

Caution: On Windows, use only the files from windows-v0.6.55 . The macOS page v0.6.54 has no Windows files. Some older release pages also list Windows files (for example v0.6.52 , whose title says macOS); those are older Windows builds, not 0.6.55.

The macOS app on v0.6.54 was built on 19 September 2026. Windows 0.6.55 contains later planning, native-model and calibration fixes. This manual follows the Windows build, so a few planning screens in later chapters can look different on a Mac.

Check the download (SHA-256)

GeoForge is not signed by a publisher that Windows or Apple recognises, so your computer cannot vouch for the file. A checksum (SHA-256) is a 64-character fingerprint of a file: if even one byte changes, the fingerprint changes. Comparing it with the published value confirms that the file arrived complete and unchanged. Do this before you run anything.

On Windows

- 1. Open File Explorer and go to the folder with the downloaded file (usually **Downloads**).
- 2. In the File Explorer address bar, type powershell and press Enter. PowerShell opens in that folder.
- 3. Type the command for your file and press Enter:

```
Get-FileHash .\GeoForge-Desktop-Setup-v0.6.55-Windows-x64.exe -Algorithm SHA256
```

For the portable copy, use .\GeoForge-Desktop-v0.6.55-Windows-x64.zip instead.

- 4. Open SHA256SUMS-Windows.txt in Notepad.

5. Compare the **Hash** that PowerShell printed with the line for the same file name. Capital and small letters do not matter; every character must match.

Check that the two 64-character values are identical.

On macOS

1. Open **Terminal** (Applications → Utilities → Terminal).
2. Type this command and press Return:

```
shasum -a 256 ~/Downloads/GeoForge-Desktop-macos-arm64.app.zip
```

3. Open `SHA256SUMS.txt` and compare the printed value with the line for `GeoForge-Desktop-macos-arm64.app.zip`. Every character must match.

Get the checksum for this release

Use `SHA256SUMS-Windows.txt` from the **same windows-v0.6.55 release** as the installer or portable archive. Release artifacts have different hashes; do not compare a current download with an older edition's checksum. The Windows checksum file is available at:

<https://github.com/lzwei196/KISS-Knowledge-Infrastructure-for-Scientific-Simulation/releases/download/windows-v0.6.55/SHA256SUMS-Windows.txt>

For the separate macOS 0.6.54 release, use that release's `SHA256SUMS.txt`.

If the values differ, delete the file and download it again. Do not run a file whose checksum does not match.

Install on Windows with the installer

The installer puts GeoForge in your own user folder, so it needs no administrator rights. The installer wizard is in English only.

1. Double-click `GeoForge-Desktop-Setup-v0.6.55-Windows-x64.exe`. Windows usually shows a blue box, **Windows protected your PC**. This is Microsoft Defender SmartScreen, Windows' check for downloaded programs. It appears because the installer carries no publisher signature; continue only if the checksum matched.
2. Click **More info**.
3. Check that the app is `GeoForge-Desktop-Setup-v0.6.55-Windows-x64.exe` and the publisher is **Unknown publisher**.
4. Click **Run anyway**.
5. On **Select Destination Location**, click **Next** to keep the suggested folder. The default is `C:\Users\you\AppData\Local\Programs\GeoForge Desktop` (written `%LOCALAPPDATA%\Programs\GeoForge Desktop`). To use another folder you can write to, type it in the box before you click **Next**.

Check that the path is inside your own user folder (`C:\Users\you\...`).
6. On **Select Additional Tasks**, tick **Create a desktop shortcut** if you want a desktop icon. It is not ticked by default.
7. Click **Next**.
8. On **Ready to Install**, click **Install**.

9. On **Completing the GeoForge Desktop Setup Wizard**, click **Finish**. Leave **Launch GeoForge Desktop** ticked so that GeoForge starts, then continue with [First launch on Windows](#).

To start GeoForge later, open the Start menu, type `GeoForge`, and click **GeoForge Desktop**. You can also find it under **All apps**, in the **GeoForge Desktop** folder. If you ticked the option in step 6, you can use the desktop shortcut.

Use the portable zip on Windows

The portable copy runs without installation. Use it when you cannot install software, or want to try GeoForge without changing anything on the PC.

1. In File Explorer, right-click `GeoForge-Desktop-v0.6.55-Windows-x64.zip` and choose **Extract All...**
2. In the destination box, type a short folder, for example `D:\GeoForge-Manual\App`.
3. Click **Extract**.
4. Open the extracted folder `GeoForge Desktop 0.6.55 Windows` (inside `D:\GeoForge-Manual\App`). It contains `GeoForge Desktop.exe`, `geoforge-agent-bridge.exe` and the folder `_internal`.
5. Double-click `GeoForge Desktop.exe`. If SmartScreen warns you, click **More info**, then **Run anyway**, as for the installer.

Check that `GeoForge Desktop.exe` and `_internal` are in the same folder.

Caution: Extract the whole zip first. Do not start the app from inside the zip window, and never move `GeoForge Desktop.exe` away from `_internal`; the program needs every file in that folder. For a desktop icon, right-click `GeoForge Desktop.exe` → **Send to** → **Desktop (create shortcut)** (Windows 11: **Show more options** first).

The portable copy uses the same settings and project folders as an installed copy (see [Where GeoForge keeps things](#)). Do not run both at the same time.

Install on macOS

1. In **Downloads**, double-click `GeoForge-Desktop-macos-arm64.app.zip`. Finder unpacks `GeoForge Desktop.app`.
2. Drag `GeoForge Desktop.app` into **Applications**.

The app is not notarised by Apple (Apple has not checked and approved it), so macOS blocks its first launch. Allow it once, in one of the two ways below.

Allow the app to open with Terminal

This is the method in the release notes.

1. Open **Terminal** (Applications → Utilities → Terminal).
2. Run this command:

```
xattr -dr com.apple.quarantine "/Applications/GeoForge Desktop.app"
```

3. Open **GeoForge Desktop** from **Applications**. Continue with [First launch on macOS](#).

Allow the app to open without Terminal

1. Double-click **GeoForge Desktop** in **Applications**. macOS says it could not verify the app.
2. Click **Done** (on older macOS, **OK**). Do not click **Move to Trash**.

3. Open **System Settings** → **Privacy & Security**.
4. Scroll down to the Security message saying that "GeoForge Desktop" was blocked.
5. Click **Open Anyway**.
6. Confirm with **Open Anyway** and your password when macOS asks. GeoForge opens. Continue with [First launch on macOS](#).

You must allow the app again after every update, because each new build is a new download. macOS may also ask again for access to **Documents**, **Desktop** or **Downloads**. Keeping projects outside those folders and outside iCloud reduces these prompts.

First launch on Windows

On Windows, GeoForge has no window of its own. When it starts, it runs a small local server (a program that serves pages only to your own computer) and shows its pages in your **default web browser**. A GeoForge icon also appears in the notification area (system tray) at the bottom right of the screen.

1. Start GeoForge (installer finish page, Start menu, desktop shortcut, or `GeoForge Desktop.exe` in the portable folder).
2. Wait a few seconds. A browser tab titled **GeoForge Desktop** opens at an address like `http://127.0.0.1:52814/` . This tab **is** the app.
3. Find the tray icon. If you do not see it, click the ^ arrow next to the clock. Hovering over the icon shows "GeoForge Desktop — double-click to open".
4. Optional: drag the icon from the ^ area onto the taskbar so it stays visible.

Check that the menu shows exactly two items, **Open GeoForge** and **Exit GeoForge**.

Check that the address starts with `http://127.0.0.1:` and that the sidebar reads **AI setup needed**. Both banners are normal at this point (see [What the first screen tells you](#)).

Open, close and quit

You do this	What happens
Close the browser tab or the browser	GeoForge keeps running in the tray. Work in progress (an agent, a download, a model run) continues.
Double-click the tray icon, or right-click it → Open GeoForge	The GeoForge page opens again in your browser.
Right-click the tray icon → Exit GeoForge	GeoForge first stops everything it started (agents, downloads, calibration, and the models or compilers they launched), then quits. A model run in progress is interrupted.
Start GeoForge again from the Start menu while it is running	A second copy starts, with a second tray icon and a different address. Avoid this: both copies share the same settings and projects. Use the tray icon to reopen the page instead.

Always quit with **Exit GeoForge**, not with Task Manager, so that running work is stopped cleanly. The tray menu is in English only.

Caution: Known issue in 0.6.54: a long job that an AI agent started in the background through Git Bash (the command window some agents, such as Claude Code, use on Windows) can keep running after **Exit GeoForge**. If your computer stays busy after you quit during a long run, open Task Manager (Ctrl+Shift+Esc) and end the leftover model or compiler program.

What changes at every launch

- **The address changes.** The number after `127.0.0.1:` is new each time. Do not bookmark it; use the tray icon. Old tabs from an earlier launch stop working, so close them.
- **Some preferences reset.** Your browser treats each new address as a new site. The interface language goes back to your browser's language, the theme (☾) to your system's light or dark setting, and the **Local | API** switch to **Local**. If you use only an API key (for example DeepSeek), click **API** under **AI CONNECTION** before you send the first message in a new chat. See Chapter 8, Language, appearance and keyboard.

Your chats, projects, API keys and **Settings** are stored by GeoForge itself, so they are kept.

First launch on macOS

- GeoForge opens in its own window, titled **GeoForge Desktop**, with an icon in the Dock. The main screen is laid out the same way as on Windows.
- **Closing the window quits GeoForge**, as **Quit** (Command-Q) does. Any running agent, software setup or model run stops. During long installs and runs, keep the window open or minimise it.
- At each launch the Mac app checks the official KI library and may show a small notice at the bottom right (for example **KI library checked**). Click **Dismiss** to close it. Chapter 7 explains KI updates.
- As on Windows, the language, theme and **Local | API** choices can reset at each start (Chapter 8).

What the first screen tells you

Until you connect an AI, the start screen shows two signs. Both are normal on a new installation:

- Next to **GeoForge** at the top of the sidebar: **AI setup needed**.
- Above the start screen, the banner **Connect an AI to begin**. "Open AI Settings to add a key or configure a local agent." The banner says "AI Settings"; the button you need is **Settings** at the bottom of the sidebar.

You may also see a second message above the start screen. It depends on the **Local | API** switch under **AI CONNECTION**. **Local** means a local agent: an AI command-line program, such as Claude Code or Codex, installed and signed in on this computer. **API** means a key from an AI provider, such as DeepSeek.

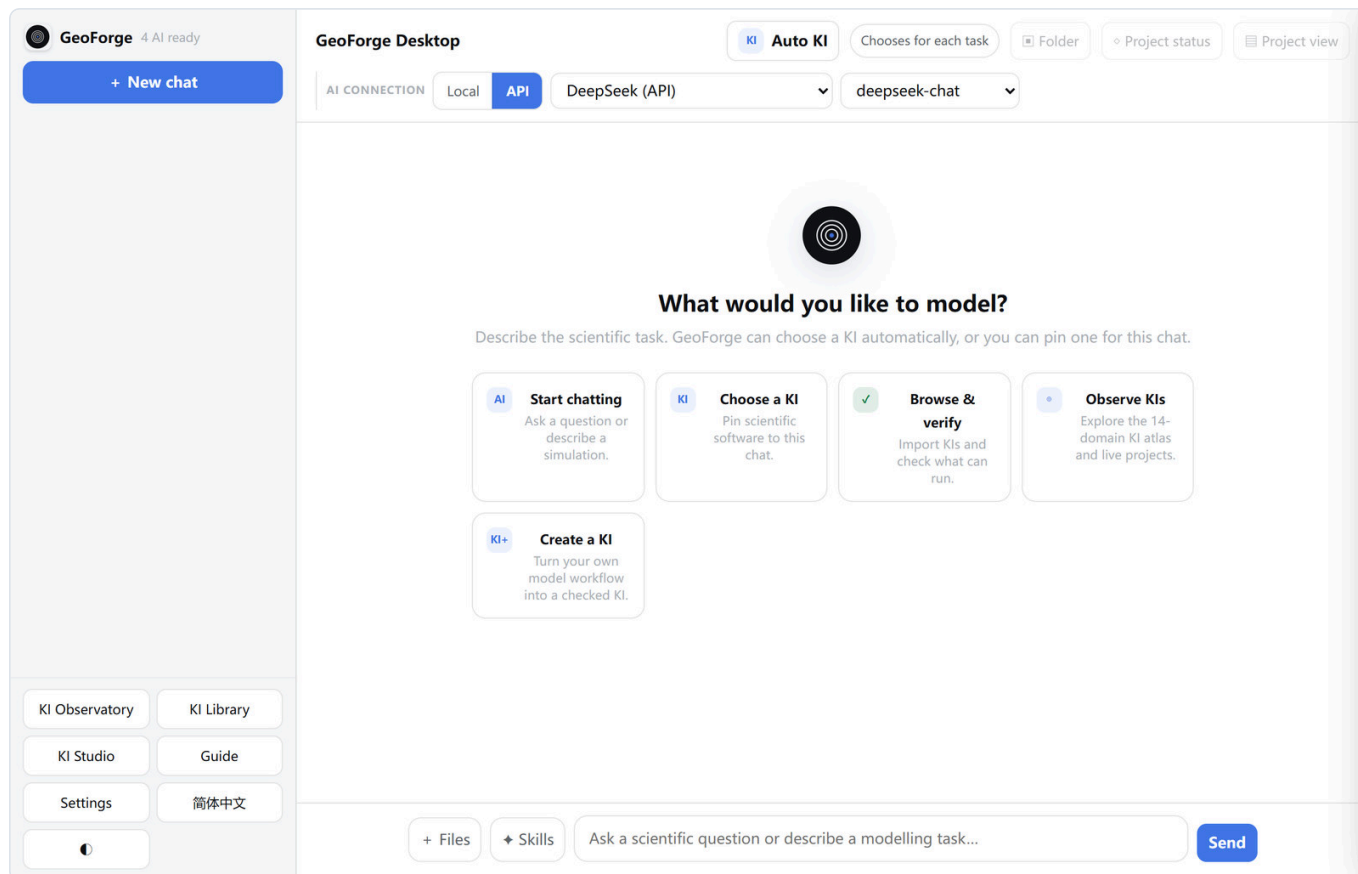
Message	Meaning
"No local agent CLI is installed. Open AI Settings for setup details." with Recheck	Local is selected and no local agent program is installed.
"Local CLIs are installed but not ready — ..." with Recheck	Local is selected. An agent program is installed but not signed in, or needs an update. The message names it.
"No API connection yet. Open AI Settings and add a key."	API is selected and no API key is saved.

These messages go away when the selected side has an AI that is ready. If you use only an API key, add it (Chapter 2), then click **API**.

Once an AI is connected, the sidebar shows, for example, **1 AI ready**, and the **Connect an AI to begin**. banner disappears. Chapter 2, Connect an AI, shows how.

A tour of the main screen

This is the screen you will use most. Here it is with four AIs already connected and no chat open.



Check that the sidebar shows **N AI ready** (here **4 AI ready**), and that **AI CONNECTION** shows the provider and model you expect (here **API**, **DeepSeek (API)** and **deepseek-chat**).

Sidebar (left)

Control	What it does
GeoForge and N AI ready / AI setup needed	How many AI connections are ready to use.
+ New chat	Starts a new project chat. A dialog, New project chat , asks for the project name and folder (Chapter 6).
Chat list	Your chats, below + New chat (empty at first). Each chat is one project with its own folder. Hovering over a chat shows X . It archives the chat after you confirm: the project folder is moved into an <code>_archived</code> folder beside it, and nothing is deleted.
KI Observatory , KI Library , KI Studio	Open other GeoForge pages (Chapter 7). To return, use the arrow at the top left of the page or the browser's Back button. In KI Studio the arrow leads to KI Library .
Guide	Opens How GeoForge works , a three-step summary with an Open KI Library button. This manual covers much more.
Settings	Opens the Settings window: AI services , GeoForge Database , Network & proxy , Permissions (Chapters 2 and 3).
简体中文	Switches the interface to Chinese and reloads the page. In Chinese mode the same button reads English .
	Switches between light and dark.

Header (top)

Control	What it does
Title	The chat's name; GeoForge Desktop when no chat is open.
Auto KI (with the KI badge)	Chooses the KI (the model) for this chat. Auto KI lets GeoForge choose for each task (Chapter 4). It is fixed after the chat's first message.
Status pill	Chooses for each task with Auto KI. With a chosen KI, it shows whether that model's software is verified on this computer.
▣ Folder	Opens this chat's project folder in File Explorer (Finder on macOS).
◇ Project status	Shows what GeoForge is doing in this project and what it needs from you (Chapter 6).
☰ Project view	Opens this chat's result panels (Chapter 6).
AI CONNECTION: Local API, provider list, model list	Chooses the AI for a new chat: a local agent program or an API key, then the provider and model. They are fixed after the chat's first message (Chapter 2).

▣ **Folder**, ◇ **Project status** and ☰ **Project view** stay grey until a chat is open.

Centre

With no chat open, the centre shows **What would you like to model?** and five starter cards:

Card	Opens
Start chatting	Puts the cursor in the message box.
Choose a KI	The KI picker, to pin a model to the chat.
Browse & verify	KI Library.
Observe KIs	KI Observatory.
Create a KI	KI Studio.

When a chat is open, the centre shows the conversation instead.

Message area (bottom)

Control	What it does
+ Files	Attaches files to your next message.
⚡ Skills	Chooses extra skills for this chat.
Message box ("Ask a scientific question or describe a modelling task...")	Where you type. Enter sends; Shift+Enter starts a new line.
Send	Sends the message. It stays grey until the selected AI is ready.

Tip: If you send a message that contains Chinese characters, the interface switches to Chinese automatically. Click **English** in the sidebar to switch back.

Where GeoForge keeps things

By default, GeoForge keeps your projects outside the program folder. Updating or uninstalling the program therefore does not touch your work.

What	Windows	macOS
Program	%LOCALAPPDATA%\Programs\GeoForge Desktop (or the folder where you extracted the portable zip)	/Applications/GeoForge Desktop.app
Settings: API keys, proxy, default AI	%APPDATA%\KISS\settings.json	~/Library/Application Support/KISS/settings.json
Other app data: KI updates (ki-updates), KIs you import (user_models), Database catalogue copy (database), KI Studio work (ki-studio)	%APPDATA%\KISS\	~/Library/Application Support/KISS/
GeoForge Database activation token	Windows Credential Manager → Windows Credentials → Generic Credentials → com.geoforge.desktop:observation-activation-token	A private file under ~/Library/Application Support/KISS/secrets/
Work folder: projects and model installs	C:\Users\you\kiss	~/kiss

%APPDATA% is C:\Users\you\AppData\Roaming , and %LOCALAPPDATA% is C:\Users\you\AppData\Local . You can type either into the File Explorer address bar.

Inside the work folder (C:\Users\you\kiss or ~/kiss):

Item	What it holds
projects\	One folder per chat, named <date>-<project name>--<id> , for example 2026-10-01-Alptal-snow-example--<id> . Archived chats are in projects_archived\ .
One folder per installed model, in lower case	The default install place for each model, for example C:\Users\you\kiss\fsm2 .
_install-locations.json	A record of any model you installed in another folder.
sessions\ and other small folders	GeoForge's own bookkeeping, including where to find projects you created elsewhere. Leave them alone.

You can create a project somewhere else (in the **New project chat** dialog) and install a model in another folder (on the **Agent setup** page). GeoForge records those places and finds them again later. On Windows, choose short paths without spaces, for example D:\GeoForge-Manual\models\fsm2 , on a drive with several GB free. Avoid OneDrive and iCloud folders.

Caution: settings.json holds your API keys as plain, unencrypted text. It sits in your own user folder, but other programs running under your account, and administrators of the computer, can read it. Do not share, sync or attach this file, or the folder that contains it.

Tip: For advanced users only: the app has no setting to move the whole work folder. You can make a shortcut whose target is "C:\Users\you\AppData\Local\Programs\GeoForge Desktop\GeoForge Desktop.exe" app --workroot D:\GeoForge-Manual\work . While you use that shortcut, chats from the default work folder do not appear in the sidebar; their files stay on disk.

Upgrade GeoForge

Your projects, chats, model installs, API keys, proxy settings, default AI and Database token are kept when you upgrade. Nothing needs to be moved.

Tip: If you used an older Windows build on Chinese-language Windows and saw missing planning cards, run records that failed to save, or AI agents that would not start under a Chinese user name, upgrade to 0.6.54. These problems are fixed in this build.

Windows, installed copy

1. Wait for running work to finish, or accept that it will be stopped.
2. Right-click the tray icon → **Exit GeoForge**.
3. Download the new installer and check its SHA-256 (see [Check the download](#)).
4. Run the new installer. It installs into the same folder, so it may skip the folder page.
5. Click **Finish** with **Launch GeoForge Desktop** ticked.

If the installer shows **Preparing to Install** and lists GeoForge among the applications using files, GeoForge is still running. Exit it from the tray, then click **Next**.

Windows, portable copy

1. Right-click the tray icon → **Exit GeoForge**.
2. Extract the new zip into a **new** folder.
3. Start the new `GeoForge Desktop.exe`.
4. When the new copy works, delete the old folder. Your data is not in it.

macOS

1. Close the GeoForge window to quit.
2. Unzip the new download.
3. Drag the new `GeoForge Desktop.app` into **Applications**.
4. When Finder asks, click **Replace**.
5. Allow the new app to open again, as in [Install on macOS](#).

Uninstall GeoForge

Windows, installed copy

1. Right-click the tray icon → **Exit GeoForge**.
2. Open Windows **Settings** → **Apps** → **Installed apps** (Windows 10: **Apps & features**).
3. Find **GeoForge Desktop 0.6.54** in the list.
4. Click ... next to it (Windows 10: click the entry).
5. Choose **Uninstall**.
6. Confirm the uninstall in the dialogs that follow.

Windows, portable copy

1. Right-click the tray icon → **Exit GeoForge**.
2. Delete the extracted folder.

macOS

1. Close the GeoForge window to quit.
2. Drag `GeoForge Desktop.app` from **Applications** to the Trash.

What uninstalling keeps

Uninstalling removes only the program. Everything below stays on your computer until you delete it yourself.

Kept	Windows	macOS	Delete it if...
Settings and API keys	<code>%APPDATA%\KISS</code>	<code>~/Library/Application Support/KISS</code>	you want to remove your keys and settings
All projects and installed models	<code>C:\Users\you\kiss</code>	<code>~/kiss</code>	you no longer need any result or model install
Projects and models in folders you chose	Where you put them	Where you put them	as above
Database token	Credential Manager entry <code>com.geoforge.desktop:observation-activation-token</code>	Inside <code>~/Library/Application Support/KISS</code>	you want to remove the token

Caution: Deleting the work folder (`C:\Users\you\kiss` or `~/kiss`) permanently removes every project, result and installed model in it. Copy the results you need first. This cannot be undone.

On macOS, open the `Library` folder with Finder → **Go** → **Go to Folder...** (Shift-Command-G) and type `~/Library/Application Support/KISS`.

If something goes wrong

What you see	What to do
The browser says the download "isn't commonly downloaded" or holds it back	This is the browser's check for unsigned files. Choose to keep the file (in Edge: ... → Keep , then Show more → Keep anyway). Then check the SHA-256.
The SHA-256 does not match	Delete the file and download it again from the release page. Do not run it.
SmartScreen shows no Run anyway button	Click More info first. If the button is still missing, your organisation blocks unknown apps; ask your IT support.
Windows says Smart App Control blocked the app	Windows offers no override for this. Turning Smart App Control off is a security decision for you or your IT support.
Nothing seems to happen after you start GeoForge (Windows)	Wait 10–20 seconds. Look for the tray icon (click ^) and double-click it. Do not start GeoForge again until you have checked the tray, or you get a second copy.
The page says the site cannot be reached	The tab belongs to an earlier launch, or GeoForge has quit. Close the tab. Start GeoForge, or double-click its tray icon, and use the new tab.
GeoForge could not start. with a Reload button	Click Reload . If it appears again, quit with tray → Exit GeoForge and start GeoForge once more.
Two GeoForge icons in the tray	GeoForge was started twice. Right-click each icon → Exit GeoForge , then start it once.
The language, theme or Local API choice changed back	Expected after a restart, because the local address changes. Set it again. See Chapter 8.
Connect an AI to begin.	Normal until an AI is connected. See Chapter 2.
"No local agent CLI is installed." although you use an API key	Click API under AI CONNECTION .
On a Mac, the <code>.zip</code> is gone from Downloads and only <code>GeoForge Desktop.app</code> is there	Safari unpacked it for you, so the zip cannot be checked. Download it again with another browser, or turn off Open "safe" files after downloading in Safari's settings first.
macOS says the app cannot be opened, or that it is damaged	Run the <code>xattr</code> command from Allow the app to open with Terminal , then open the app again.
On macOS, a running setup or model run stopped	Closing the GeoForge window quits the app and stops its work. Open GeoForge again and continue in the same chat.
After uninstalling, the program folder is still there	Delete <code>%LOCALAPPDATA%\Programs\GeoForge Desktop</code> by hand. Your projects are not in it.

For other problems, see Chapter 9, Troubleshooting and known issues.

2 Connect an AI

In this chapter you will connect GeoForge to an AI, choose which AI a chat uses, set up a proxy if you need one, and check the connection before you start a project.

GeoForge cannot plan or run anything until at least one AI is connected. Until then the top of the sidebar reads **AI setup needed** and a banner says "Connect an AI to begin."

A working AI connection only means GeoForge can talk to the AI. It says nothing about whether a model run is correct. GeoForge's run records show what was actually run; judging whether the results are scientifically right stays your job. Later chapters explain both.

Two ways to connect an AI

	API key	Local agent CLI
What it is	A key from an AI company. GeoForge talks to the service directly.	A command-line AI program that is installed and signed in on this computer. GeoForge runs it for you under your user account.
Services in 0.6.54	DeepSeek (API), OpenAI (API), OpenRouter (API), Claude (API) (Anthropic)	Claude Code, OpenAI Codex, Kimi Code, Gemini CLI, Qwen Code
Extra software	None	Node.js for most of them; Git for Windows for Claude Code on Windows
Who bills you	The AI company, per use, on your API account	Your account or subscription for that CLI
Taken through a full project in the Windows 0.6.54 test	DeepSeek (API) , with the FSM2 snow model example	None yet

If you are unsure, start with **DeepSeek (API)**. It needs no other software, it usually works from mainland China without a proxy, and it is the only AI that has been tested through a full project on Windows in 0.6.54.

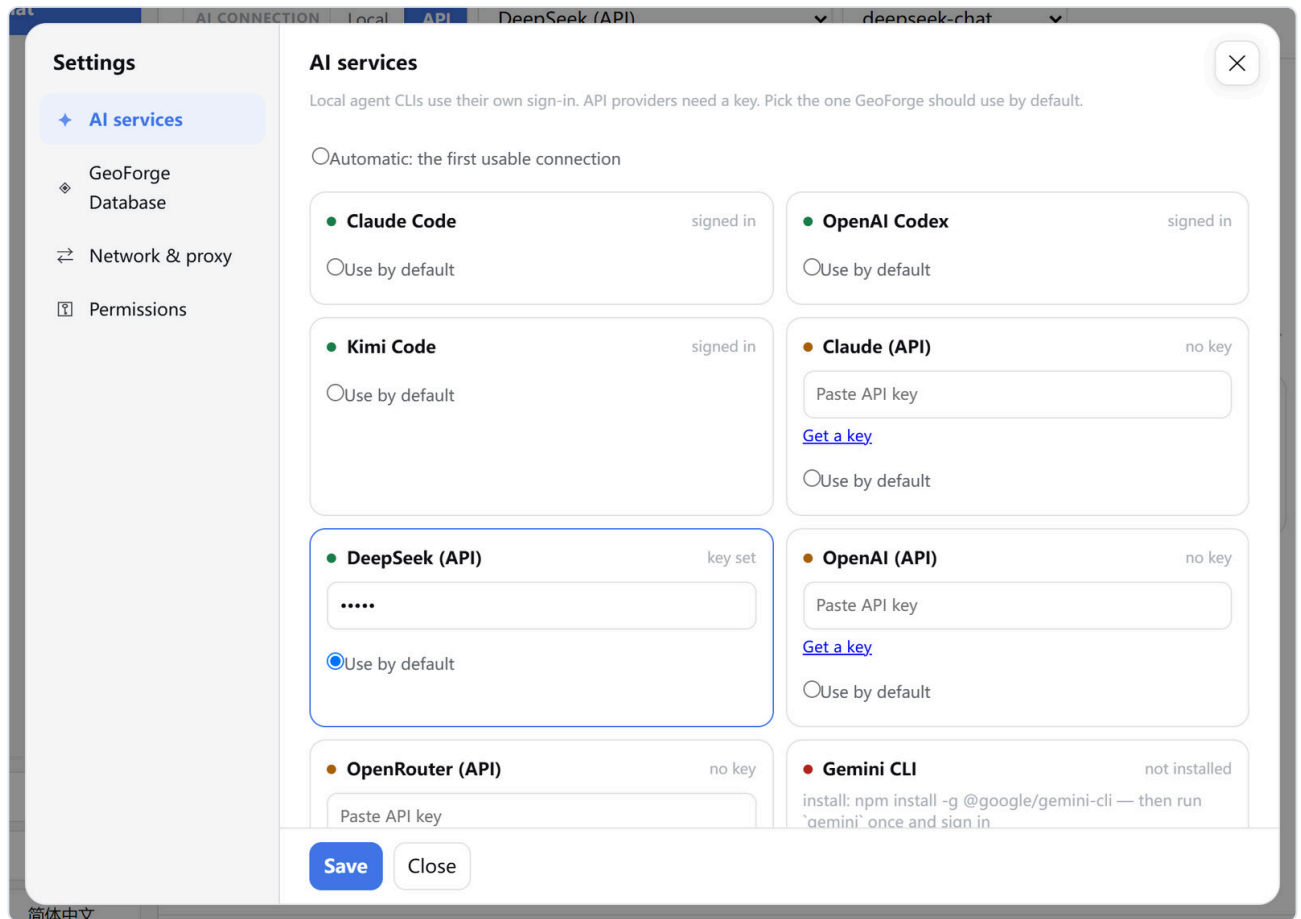
You can connect several AIs. Each chat uses one of them.

Open Settings

All AI settings are in one window.

1. Click **Settings** at the bottom of the left sidebar.

The **Settings** window opens. The first time, it shows the **AI services** page. After that it reopens on the page you used last, until the GeoForge page is reloaded or GeoForge restarts.
2. If another page is showing, click **AI services** on the left.



Check that the left side lists four pages, and that **Save** and **Close** are at the bottom.

3. To switch pages, click a page name on the left:

Page	What you set there
AI services	API keys, local agent CLIs, the default AI, and the connection test
GeoForge Database	The optional data catalogue (see Chapter 3)
Network & proxy	Whether GeoForge uses a proxy, and for which services
Permissions	Kimi Code file access, and MCP servers for the open chat

4. Click **Save** when you have finished. One **Save** keeps the changes on all four pages. "saved" appears for a moment next to **Close**.

Caution: Changes are kept only when you click **Save**, **Test AI & GitHub** or **Save & test database** (both of these save first). **Close**, **X**, the Esc key or a click outside the window all close it, and the next time you open **Settings** your unsaved changes are gone, on every page.

Tip: The KI Library page has its own small ⚙ settings box. It is an older, simpler version without **Permissions**, **Recheck local CLIs** or **Test AI & GitHub**. Use the **Settings** button in the main sidebar instead.

Connect an AI with an API key

Each API service has a card on the **AI services** page. These are the services and the models GeoForge offers for them, in the order the model list shows them:

Card	Models in the chat header	Where Get a key leads
DeepSeek (API)	deepseek-chat (preselected), deepseek-reasoner	https://platform.deepseek.com/api_keys
OpenAI (API)	gpt-4o (preselected), gpt-4o-mini	https://platform.openai.com/api-keys
OpenRouter (API)	claude-sonnet-4-5 (preselected), deepseek-chat (preselected)	https://openrouter.ai/keys
Claude (API)	claude-sonnet-4-5 (preselected), claude-opus-4-1	https://console.anthropic.com/settings/keys

The steps use DeepSeek. The other cards work the same way.

1. Click **Settings**.
2. If another page is showing, click **AI services**.
3. Find the **DeepSeek (API)** card. Without a key it shows **no key** and an amber dot.
4. Click **Get a key** on the card. The provider's key page opens in your browser.
5. On that page, sign in to your DeepSeek account, create an API key and copy it. You do this on the provider's site, not in GeoForge.
6. Back in GeoForge, click in the **Paste API key** field on the card.
7. Paste the key.
8. Select **Use by default** on the same card if this should be your usual AI. (See "Choose the default AI" below.)
9. Click **Save**.

Compare with the **AI services** screenshot in "Open Settings" above. The **DeepSeek (API)** card reads **key set**, the dot is green, the key field shows only dots, and **Use by default** is selected (the card gets a blue border). The **Get a key** link is gone.

10. Click **Test AI & GitHub** to prove the key works (see "Check the connection with Test AI & GitHub" below). A key that is saved is not yet a key that works.
11. Click **Close**.

The top of the sidebar, next to **GeoForge**, now reads for example **1 AI ready** instead of **AI setup needed**.

Replace or remove a key

After you save a key, GeoForge never shows it again. When you reopen **Settings**, the field holds only a placeholder: "..." and the key's last four characters, displayed as five dots.

To replace a key:

1. Click in the key field on the card.
2. Press Ctrl+A (macOS: Cmd+A), then Delete. The field is now empty.
3. Paste the new key.
4. Click **Save**.

To remove a key, do steps 1, 2 and 4 only.

Caution: If you paste a new key without first emptying the field, the new key is added to the placeholder, and GeoForge silently keeps the old key. Always empty the field first.

Where the key is stored

GeoForge saves keys in a settings file inside your own user profile:

Platform	File
Windows	%APPDATA%\KISS\settings.json (usually C:\Users\you\AppData\Roaming\KISS\settings.json)
macOS	~/Library/Application Support/KISS/settings.json (readable only by your account)

The file is not encrypted. Do not share, upload or sync that folder, and do not send it when you ask for help.

Keys set as environment variables

GeoForge also reads these environment variables from your computer: DEEPSEEK_API_KEY , OPENAI_API_KEY , OPENROUTER_API_KEY and ANTHROPIC_API_KEY . (An environment variable is a named setting that Windows or macOS passes to every program.)

- If one is set, the matching card shows **key set** even when you never pasted a key.
- If you also save a key in **Settings**, the saved key is used until GeoForge restarts. From the next start, the environment variable wins.

So if a key you saved is rejected after a restart, look for such a variable:

- **Windows:** Start menu, search "environment variables", open **Edit environment variables for your account**. Remove the variable, or give it the same key. Then restart GeoForge.
- **macOS:** look for a line such as export DEEPSEEK_API_KEY=... in ~/.zshrc . Remove it, or give it the same key. Then quit and reopen GeoForge.

Tip: A saved **Claude (API)** key is also visible to the Claude Code CLI. Claude Code then counts as signed in, and it may bill that API key instead of your Claude subscription. If you want Claude Code to use your subscription, leave the **Claude (API)** key empty.

Connect a local agent CLI

A local agent CLI is an AI program that runs in PowerShell (Windows) or Terminal (macOS). GeoForge starts it for you, under your user account, with your own sign-in. You install it and sign in once, outside GeoForge.

What you need first

- **Node.js (LTS version)** from https://nodejs.org . It provides npm , the installer that Claude Code, OpenAI Codex, Gemini CLI and Qwen Code use.
- **Git for Windows** from https://git-scm.com , if you want Claude Code on Windows. Claude Code runs its commands through Git Bash, which comes with it.
- An account with the AI company that makes the CLI.

Install and sign-in commands

GeoForge shows these same commands under each card that is not ready.

Card	Install command	Sign-in command
Claude Code	<code>npm install -g @anthropic-ai/claude-code</code>	<code>claude auth login</code>
OpenAI Codex	<code>npm install -g @openai/codex</code> (version 0.144.0 or newer)	<code>codex login</code>
Gemini CLI	<code>npm install -g @google/gemini-cli</code>	Run <code>gemini</code> once and sign in
Qwen Code	<code>npm install -g @qwen-code/qwen-code</code>	Run <code>qwen</code> once and sign in
Kimi Code	Windows: no installer is published; the card points to <code>https://code.kimi.com</code> . macOS: see below.	<code>kimi login</code>

On macOS, Kimi Code installs with this Terminal command:

```
curl -fsSL https://code.kimi.com/kimi-code/install.sh | bash
```

Steps

1. Install Node.js.
2. **Windows, Claude Code only:** install Git for Windows.
3. Open a **new** PowerShell window (Windows) or Terminal window (macOS). A window opened before Node.js was installed cannot find `npm`.
4. Type the install command from the table and press Enter.
5. Type the sign-in command and press Enter.
6. Finish the sign-in in the browser page the CLI opens.
7. **Windows only:** right-click the GeoForge icon in the system tray and choose **Exit GeoForge**.

Caution: **Exit GeoForge** also stops any agent, download or model run that is still working. Exit only when nothing important is running.

8. **Windows only:** open **GeoForge Desktop** from the Start menu.

GeoForge reads the list of program folders only when it starts, so a CLI installed while it was running can stay invisible until a restart. If you only signed in to a CLI that GeoForge already found, skip steps 7 and 8.

9. Click **Settings**.

10. If another page is showing, click **AI services**.

11. Scroll below the cards and click **Recheck local CLIs**.

The note beside the button shows "checking...", then "checked".

12. Read the card of your CLI (see the table below).

Tip: GeoForge also rechecks installed CLIs that are not ready yet each time you come back to its tab or window. **Recheck local CLIs** forces it.

What the card tells you

Card shows	Dot	Meaning	What to do
signed in	green	Installed and signed in.	Nothing. It is ready.
signed in	amber	Installed, but GeoForge cannot confirm the sign-in. This is always the case for Gemini CLI and Qwen Code .	You can use it. The first message in a chat shows whether the sign-in works.
installed; sign-in required	amber	Installed, not signed in.	Run the sign-in command shown under the card, then click Recheck local CLIs .
... update required (0.130.0 < 0.144.0)	amber	OpenAI Codex is too old for GeoForge. The numbers are your version and the minimum.	Run <code>npm install -g @openai/codex</code> again, then click Recheck local CLIs .
not installed	red	GeoForge cannot find the program.	Use the install command shown under the card.

The words "installed", "signed in" and "works" are three separate checks. Only a real message proves the last one; **Test AI & GitHub** sends one for you.

Caution: On Windows 0.6.54, none of the local agent CLIs has been tested through a full project yet. If you want the tested path, use **DeepSeek (API)**.

Kimi Code on Windows: allow full computer access

On macOS, GeoForge keeps Kimi Code inside the project folder with a macOS security feature. Windows has no such feature yet, so on Windows Kimi Code stays off until you allow it full access to your files.

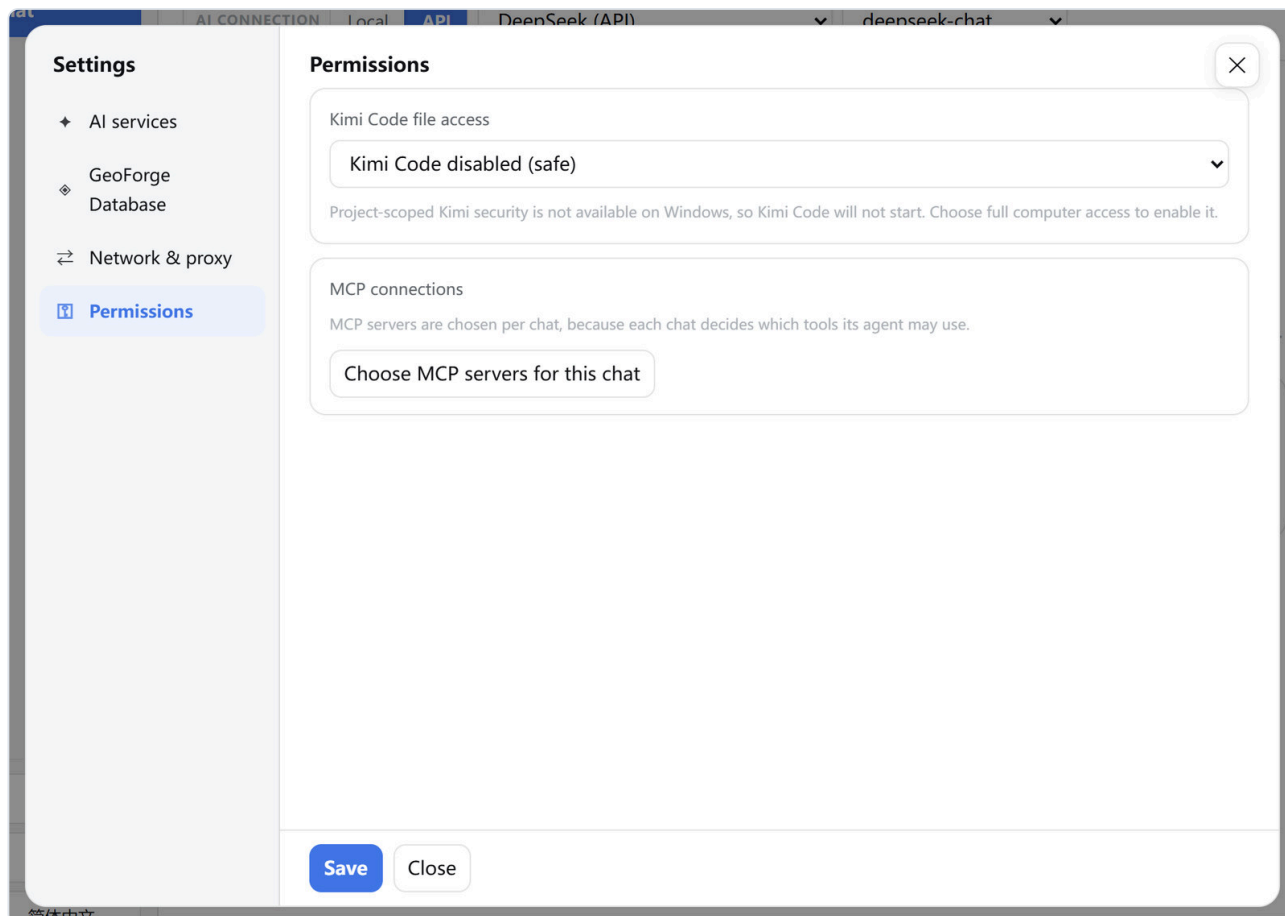
Until you do, the **Kimi Code** card can still say **signed in**, and you can even pick Kimi in a chat. But the reply is only this message:

"[Kimi Code was not started: project-scoped security could not be applied (project-scoped Kimi is currently available on macOS only). Choose Full computer access in AI Settings only if you accept that risk.]"

("AI Settings" in that message means **Settings** → **Permissions**.)

To allow Kimi Code:

1. Click **Settings**.
2. Click **Permissions** on the left.



Check that **Kimi Code file access** shows **Kimi Code disabled (safe)**, the default. The note under it says Kimi Code will not start on Windows.

3. In the **Kimi Code file access** list, choose **Enable Kimi with full computer access**.

A browser question appears: "Kimi Code will be able to read and change every file accessible to this Windows account. Accept this risk and enable it?"

4. Click **OK** only if you accept that risk. **Cancel** sets the list back to **Kimi Code disabled (safe)**.

After **OK**, the note under the list reads "Kimi Code can read and change every file accessible to this Windows account. Enable only if you accept this risk."

5. Click **Save**.

6. Send your message to Kimi again.

On macOS the same list has two other choices:

macOS choice	What Kimi Code may use
Project scoped (recommended)	The project, its KI, approved data and its own sign-in. When it needs another folder, the chat shows a card "Kimi needs access to one folder" with Allow once and Always for this project .
Full computer access	Every file your macOS account can open. Use it only for a task that cannot run in project-scoped mode.

The **MCP connections** part of the **Permissions** page is for advanced users. (MCP servers are add-on tools an agent can be given.) **Choose MCP servers for this chat** does nothing unless a chat is open.

Choose the default AI

The default AI is the one GeoForge picks for you when you do not choose.

1. Click **Settings**.
2. If another page is showing, click **AI services**.
3. Choose one:
 - **Use by default** on one card, for a fixed default.
 - **Automatic: the first usable connection** at the top of the page, to let GeoForge take the first AI that is ready.
4. Click **Save**.

The default is used in three places:

- **New chats:** it is preselected in the chat header, as long as the **Local | API** switch is on its kind. With **DeepSeek (API)** as default, the header must show **API**.
- **Agent setup** (model installation, Chapter 4): it is preselected there.
- **Test AI & GitHub:** step [6/6] tests the default AI.

Changing the default never changes a chat that has already started.

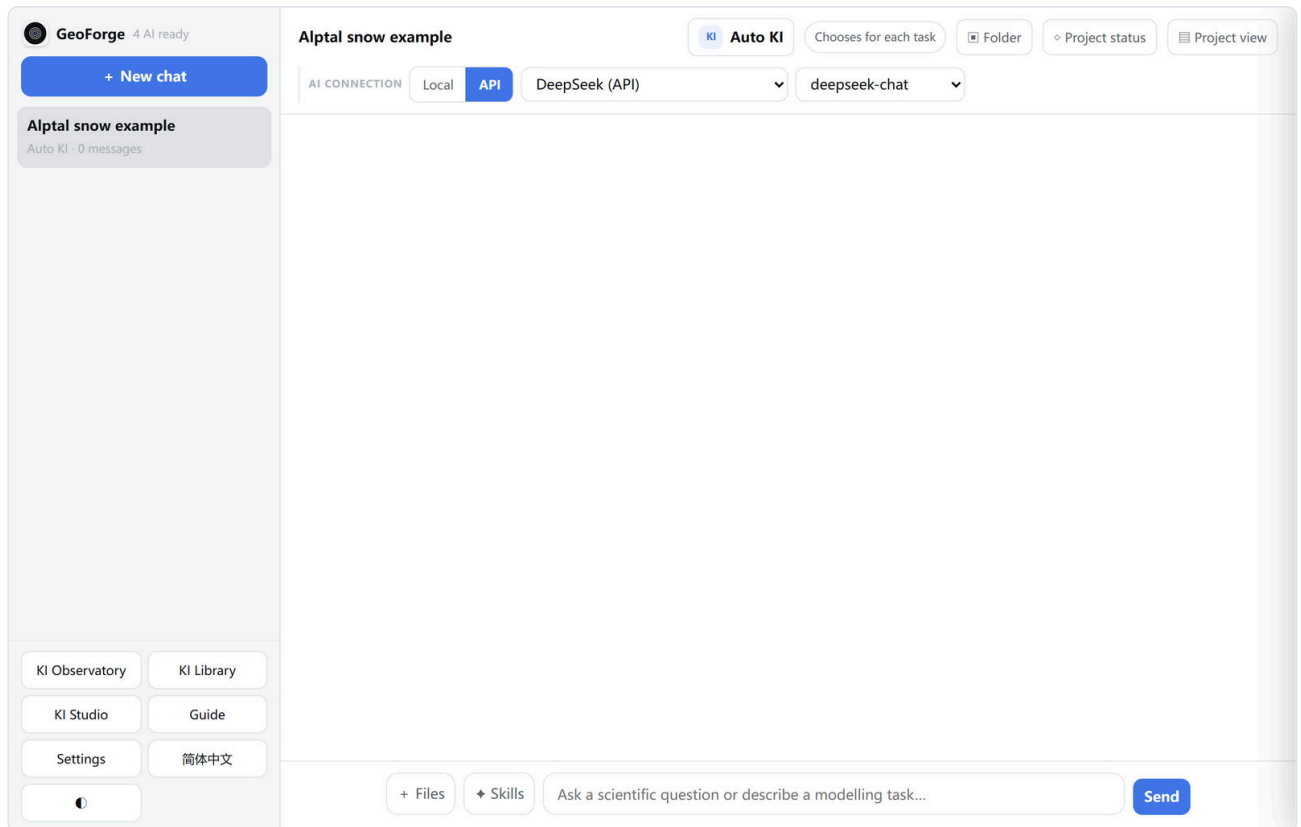
Choose the AI for one chat

Each chat has its own AI. You choose it in the chat header, under **AI CONNECTION**, before you send the first message.

1. Start a new chat with **+ New chat** (Chapter 6 explains the dialog).
2. Before typing, look at **AI CONNECTION** at the top of the chat.
3. Click **Local** for an installed agent CLI, or **API** for a key-based service.
4. In the first list, choose the AI.

Entries that are not ready are greyed out with a reason after a dash, for example "Claude Code — sign-in needed". An API service without a key shows "— not installed", which here means "no key saved". "— login checked on first use" (Gemini CLI, Qwen Code) is not an error: you can choose it.

5. In the second list, choose the model. For a local CLI, **CLI default** uses whatever model the CLI itself is set to.



Check the header before sending: **API** is highlighted, the lists show **DeepSeek (API)** and **deepseek-chat**, and the chat in the sidebar says "0 messages". **Send** is blue only when the chosen AI is ready.

6. Type your first message.

7. Click **Send**.

Tip: Hover over **Local** or **API** to see how many AIs of that kind are ready.

The AI is fixed after the first message

Once the first message is sent, the **Local** | **API** switch, both lists and the KI button (**Auto KI** or the name of the pinned KI) are greyed out for that chat. Hovering over the AI list shows "Fixed for this chat — start a new chat to switch AI"; the model list shows "Fixed for this chat — start a new chat to switch model".

The reason: the conversation lives inside that AI's own session, and another AI could not read it. Switching would lose the history.

To use a different AI, model or KI, start a new chat and choose before sending. Since 0.6.54, a chat keeps its AI after GeoForge restarts.

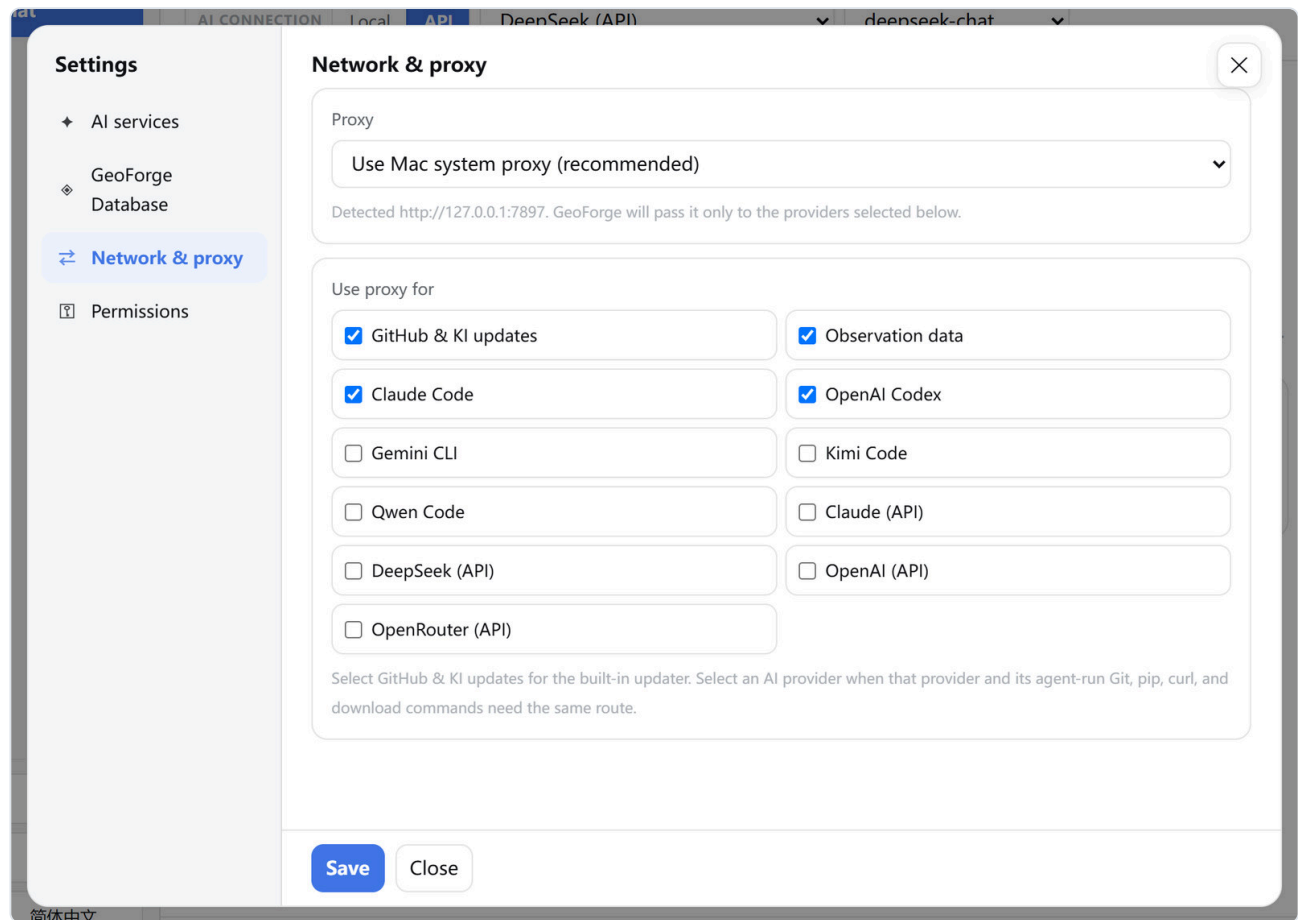
Tip (Windows): GeoForge opens at a new local address each time it starts, so your browser forgets whether you last chose **Local** or **API**, and a new chat can start on **Local**. If you use only an API key and a new chat shows **Local** with the banner "No local agent CLI is installed...", click **API**.

Network and proxy

You need this section only if some services are blocked or slow on your network. In mainland China, GitHub, Claude, OpenAI, OpenRouter and Gemini usually need a proxy; DeepSeek, Kimi and Qwen usually do not.

GeoForge does not provide a proxy. It can use one you already run, such as a local proxy app.

- 1. Click **Settings**.
- 2. Click **Network & proxy** on the left.



Check the note under **Proxy**. Here it reads "Detected http://127.0.0.1:7897. GeoForge will pass it only to the providers selected below." The four ticked boxes (**GitHub & KI updates**, **Observation data**, **Claude Code**, **OpenAI Codex**) are the defaults.

- 3. Under **Proxy**, choose one option:

Option	What it does
Use Mac system proxy (recommended)	Uses your computer's system proxy. On Windows too, despite the word "Mac".
Enter proxy manually	Uses the address you type, for example <code>http://127.0.0.1:7897</code> .
Do not use a proxy	Connects directly everywhere, and removes proxy settings from GeoForge's AI connections.

- 4. If you chose **Enter proxy manually**, type the address and port your proxy app shows. It must start with `http://` , `https://` , `socks5://` or `socks5h://` . A user name or password in the address is refused; let your local proxy app handle the login.
- 5. Under **Use proxy for**, tick each service that needs the proxy and untick the others (see the tables below).
- 6. Click **Save**.
- 7. Click **AI services** on the left.

8. Click **Test AI & GitHub**. Step [4/6] tests GitHub through the **GitHub & KI updates** tick; step [6/6] tests the AI through that AI's own tick.

Caution: On Windows the first option still reads **Use Mac system proxy (recommended)**. It means the Windows system proxy: the one set in Windows proxy settings, or by your proxy app's "system proxy" mode. If none is set, the note says "No system proxy detected. GeoForge will use the normal network connection."

What each tick covers

Tick	What goes through the proxy
GitHub & KI updates (on by default)	GeoForge's own GitHub check in Test AI & GitHub , and KI library updates
Observation data (on by default)	Connections to the GeoForge Database (Chapter 3)
Each AI, for example Claude Code (on by default), OpenAI Codex (on by default), DeepSeek (API)	That AI's own connection, and the downloads its agent makes with Git, pip or curl, for example fetching a model's source code from GitHub during setup

An unticked service always connects directly, even when a proxy is set.

A typical choice in mainland China:

Tick	Leave unticked
GitHub & KI updates , Observation data , Claude Code , OpenAI Codex , Gemini CLI , Claude (API) , OpenAI (API) , OpenRouter (API)	DeepSeek (API) , Kimi Code , Qwen Code

Tip: GitHub & KI updates does not cover downloads made by the AI's agent. If you use DeepSeek, Kimi or Qwen without the proxy and the agent cannot download model source code from GitHub during setup, tick that AI as well, click **Save**, and ask the agent to try again.

Check the connection with Test AI & GitHub

Run this test after any change on these pages, and before your first project.

1. Click **Settings**.
2. If another page is showing, click **AI services**.
3. Choose the AI to test: select **Use by default** on its card. With **Automatic: the first usable connection**, the test uses the AI currently shown in the chat header.
4. Scroll below the cards and click **Test AI & GitHub**.

"testing a real connection..." appears next to **Close**. GeoForge saves all your settings first, then a grey box fills with a six-step report, line by line. The report is in English only.
5. Wait for the last line, "done.", and for "test complete" next to **Close**.
6. Read the report with the table below.

Step	What it checks	Good result	Notes
[1/6] KI harness contract	GeoForge's built-in rules for running models are intact	OK	A FAIL here is not something you can fix in Settings. Copy the report and ask for help.
[2/6] Python 3 for model environments	Whether a Python is installed on the computer	OK and a path	See the caution below for Windows.
[3/6] Agent CLIs on this machine	Each installed CLI, its location and sign-in state	OK for the CLIs you use	"FAIL none found — install one" is fine if you use an API key.
[4/6] GitHub model-source connection	Whether GitHub can be reached, and which proxy was used	"OK GitHub answered (HTTP 200)"	On FAIL, follow the "Fix:" line: tick GitHub & KI updates or correct the proxy address.
[5/6] API keys in the environment	Which API keys GeoForge has	"OK DeepSeek (API) (DEEPSEEK_API_KEY is set)"	"none set (fine if you use an agent CLI)"
[6/6] Agent sign-in — running the agent for real	Sends one short real message to the AI being tested	API: "OK ... answered (HTTP 200) — key is valid". CLI: "OK ... answered — signed in and working"	On FAIL, read the lines under it and the "Fix:" line, if there is one.

OK means the step passed. **FAIL** lines are followed by the raw error and, for most steps, a line starting with "Fix:".

Caution (Windows): Some advice in the report is written for macOS. [2/6] often shows FAIL on Windows and suggests `brew install python` or `xcode-select --install`; those commands do not exist on Windows. This line does not stop you connecting an AI or chatting; Python matters only for models whose setup needs it (Chapter 4). Under a [6/6] failure, read "Terminal" as PowerShell, ignore the `echo $HOME` advice, and read "switch the menu bar to API" as "click **API** in the chat header".

Caution: Step [6/6] is a real call. With an API key, the provider charges you for a very small request; with a CLI, it uses a little of your plan.

Caution (Windows): [6/6] runs Kimi Code directly. It can report Kimi as working while **Permissions** is still on **Kimi Code disabled (safe)**, and chats will still refuse to start Kimi until you enable it.

When you ask for help, copy the whole report. GeoForge does not print your keys in it, but a provider's error message may quote part of a key, and folder paths show your user name. Read it before you share it.

If something goes wrong

What you see	Cause	What to do
Banner "Connect an AI to begin. Open AI Settings to add a key or configure a local agent." and AI setup needed in the sidebar	No AI is ready.	Add a key or sign in to a CLI as described above. "AI Settings" is the old name of Settings .
Banner "No API connection yet. Open AI Settings and add a key."	The header is on API , but no key is saved.	Add a key, or click Local if you use a CLI.
Banner "No local agent CLI is installed. Open AI Settings for setup details." with Recheck	The header is on Local , but no CLI was found.	Click API if you use a key. If you just installed a CLI on Windows, restart GeoForge, then click Recheck .
Banner "Local CLIs are installed but not ready — OpenAI Codex: sign-in needed. run <code>codex login</code> in PowerShell." with Recheck	A CLI is installed but not signed in.	Run the command in the banner, then click Recheck .
Send stays grey	The AI chosen in the header is not ready, or the chat is still busy.	Pick an AI that is not greyed out in the list, or fix the one you want.
PowerShell says "npm.ps1 cannot be loaded because running scripts is disabled on this system"	Windows blocks the PowerShell version of <code>npm</code> .	Type <code>npm.cmd</code> instead of <code>npm</code> , for example <code>npm.cmd install -g @openai/codex</code> .
A CLI still shows not installed after you installed it	GeoForge was running during the install, or the install failed.	Windows: exit GeoForge from the tray and start it again. macOS: quit GeoForge and open it again. Then, in a new PowerShell or Terminal window, type <code>claude --version</code> (or <code>codex --version</code> , <code>gemini --version</code>): if the command is not found, repeat the install.
A CLI shows "installed; sign-in required" although you signed in	The sign-in was done for a different account, or did not finish.	In PowerShell or Terminal, run <code>claude auth status</code> or <code>codex login status</code> to see what the CLI reports. Sign in again, then click Recheck local CLIs .
"update required (... < 0.144.0)" on OpenAI Codex	The Codex CLI is too old.	Run <code>npm install -g @openai/codex</code> , then click Recheck local CLIs .
Test [6/6]: "FAIL DeepSeek (API): HTTP 401 — key rejected" (or 403)	The provider refused the key.	Check the key on the provider's site. Empty the field fully, paste the key again, click Save , and test again. Check for an environment variable with the same name.
Test [6/6]: "FAIL cannot reach DeepSeek (API): ..."	The network or the proxy route is wrong for that AI.	On Network & proxy , tick or untick that AI, click Save , and test again.
Test [4/6]: "FAIL cannot reach GitHub: ..."	GitHub is blocked or the proxy address is wrong.	Tick GitHub & KI updates , check the proxy address, click Save , and test again.
You pasted a new key, but the old one is still used	The new key was added to the placeholder instead of replacing it.	Empty the field completely, paste again, click Save .

What you see	Cause	What to do
Save fails with "proxy address is required in manual mode"	Enter proxy manually is chosen with an empty address.	Type the address, or choose another option.
Save fails with "proxy must start with http://, https://, socks5://, or socks5h://"	The address has no scheme or a wrong one.	Write it in full, for example <code>http://127.0.0.1:7897</code> .
Save fails with "proxy credentials are not stored here; use a local authenticated proxy"	The address contains a user name or password.	Remove them and let your proxy app handle the login.
Kimi replies only "[Kimi Code was not started: project-scoped security could not be applied ...]"	Windows: Kimi is off by default.	Settings → Permissions → Enable Kimi with full computer access , OK , Save , then send again.
The AI or model list is greyed out, with "Fixed for this chat — start a new chat to switch AI"	The chat has already started.	Start a new chat and choose the AI before the first message.
Your changes are gone when you reopen Settings	The window was closed without Save .	Make the changes again and click Save .

3 GeoForge Database

In this chapter you will connect GeoForge Desktop to the GeoForge Database, check that the connection works, browse what the database holds, and follow approved data from the plan into your project folder.

3.1 What the GeoForge Database is

The GeoForge Database is a curated online catalogue of about 1,100 datasets for environmental modelling. It holds gauge and station observations, meteorological forcing and reanalysis, gridded products, soil maps, terrain data and similar inputs.

When the database is connected, the agent can search it while it plans your study. It writes the exact dataset IDs into the plan. The agent never downloads anything itself: after you approve the plan, GeoForge fetches the approved data (or gives you a download link for very large files), checks it, and records it in your project.

The database is optional. It is separate from any GeoForge web account and needs its own activation token.

You need the database when...	You can work without it when...
you want GeoForge to find and fetch observation, forcing or soil data for your study area	your plan uses a public source the KI already knows (for example NASA POWER)
you want a large gridded product cut to your study area on the server instead of downloading the whole file	you provide your own files
you want every downloaded file checked and recorded in the project	you run a KI's bundled example case (the FSM2 Alptal example needs no database)

3.2 Get an activation token

Each person gets their own token from the GeoForge Database owner. You cannot sign up for one inside the app.

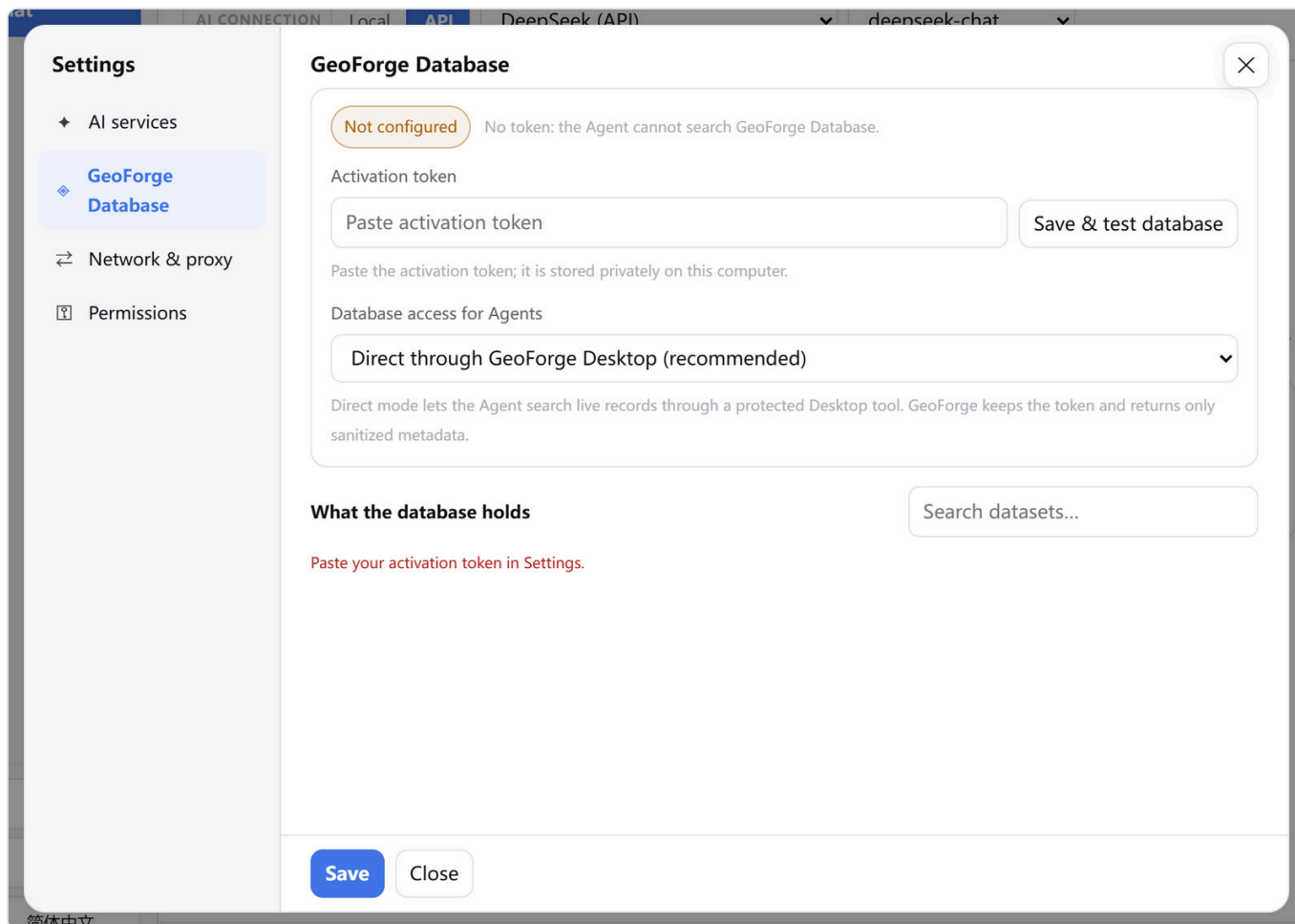
- The token is a long text that starts with `gfd_`.
- It is valid for 6 months. After that, ask for a new one.
- The owner can revoke one person's token without affecting anyone else.
- Each token has a daily limit of 20 GB or 2,000 requests. The limit resets at midnight server time.

Request a token from the person who manages your GeoForge Database deployment. This build has no public self-service token-issuance form; the desktop app cannot create a token.

Caution: Treat the token like a password. Do not paste it into a chat message, an e-mail to a colleague, or a file in your project. Every Baidu Pan link GeoForge shows you is logged against your token, so a shared token means shared responsibility.

3.3 Enter the token in Settings

1. Click **Settings** at the bottom of the left sidebar.
2. In the Settings menu on the left, click **GeoForge Database**.



Check: the pill at the top reads **Not configured**, the line beside it reads "No token: the Agent cannot search GeoForge Database.", and **What the database holds** shows "Paste your activation token in Settings." in red. This is the normal starting point.

3. Click in the **Activation token** box (it shows "Paste activation token").
4. Paste your token. The characters appear as dots.
5. Leave **Database access for Agents** on **Direct through GeoForge Desktop (recommended)**.
6. Click **Save & test database**. The note under the token box changes to "Saving and testing data access...".
7. Wait for the result. On success the note reads "Connected: N catalogue records available.", where N is about 1,100.
8. Click **Close**.
9. Click **Settings** again.
10. Click **GeoForge Database**. The pill and the dataset list refresh only when you open this page again.

Check: the pill reads **Connected** in green, and the status line shows a record count and a recent sync time ("synced just now" or "synced N min ago").

What Save & test database does. It saves every Settings page, not only this one. It then sends one test request to the database. If the test passes, GeoForge starts downloading a local copy of the catalogue in the background. The copy holds descriptions only: no data files, links or passwords.

The background download normally finishes within a minute. If you reopen the page before it finishes, the pill may show **Not synced** (or, on a computer where GeoForge has already run without a token, **Unavailable**), and the list may say "The catalogue is being fetched; try again in a moment." Wait a minute, close Settings and open the page again.

Tip: Save at the bottom of the window also stores the token, but it does not test it or download the catalogue. Use **Save & test database** whenever you paste a new token.

Replace or remove the token

Once a token is saved, the box shows only dots. GeoForge never shows the saved token again; the dots only mean "a token is stored". If you leave the dots as they are, the saved token is kept.

To replace the token:

1. Click in the **Activation token** box.
2. Select everything in it (Windows: Ctrl+A; macOS: Cmd+A).
3. Press Delete.
4. Paste the new token.
5. Click **Save & test database**.

Caution: Delete the dots before you paste. If you paste after the dots, GeoForge treats the whole box as "unchanged" and keeps the old token.

To remove the token:

1. Click in the **Activation token** box.
2. Select everything in it (Windows: Ctrl+A; macOS: Cmd+A).
3. Press Delete, so the box is completely empty.
4. Click **Save**.

GeoForge deletes the stored token. When you next open the page, the pill reads **Not configured**.

Tip: The **KI Library** page has its own small settings dialog (⚙) with the same token field. Both save to the same place. This manual uses the main **Settings** window.

3.4 Read the database status

The pill at the top of **Settings** → **GeoForge Database** shows one of five states. **Connected** has two colours.

Pill	What it means	What to do
Not configured	No token is stored, or GeoForge could not read it from the system password store (section 3.10).	Enter a token (section 3.3).
Off	You set Database access for Agents to Do not provide database access .	If you want the database, switch back to Direct through GeoForge Desktop (recommended) and click Save .
Unavailable	A token is stored, but the server rejected it (wrong, expired or revoked). It can also show for a few seconds right after you first save a token.	Read the line beside the pill, then see section 3.10.
Connected (green)	The token works and the local catalogue copy is up to date.	Nothing.
Connected (amber)	The token works, but the last refresh failed, usually because you were offline. GeoForge keeps using the older copy.	Check your network (section 3.10).
Not synced	The token works, but the catalogue copy has not been downloaded yet.	Wait a minute, then reopen the page.

The line beside the pill gives details:

- With a working token: "local catalogue: N records (M direct download) · synced N min ago". If the last refresh failed, the line ends with "· stale:" and the reason.
- Before the copy exists: "catalogue not downloaded", sometimes followed by the reason.
- With a rejected token: "Access unavailable: the token is not authorized, so the Agent cannot search GeoForge Database.", followed by the server's message.
- With access switched off: "Disabled: the Agent will not search the database."

GeoForge refreshes the catalogue copy when it starts and then every 6 hours.

The **GeoForge Database** card inside **Project status** → **Details** has its own pill with different words:

Project status pill	Matches the Settings state
Not set up	Not configured
Off	Off
Unavailable	Unavailable
Stale	Connected (amber)
Local copy	Connected (green) or Not synced

When the database counts as "activated"

GeoForge uses the database for a project only when all three conditions hold:

1. **Database access for Agents** is not set to **Do not provide database access**.
2. A token is stored.
3. The server has not rejected that token.

A network outage does not deactivate the database. Searches keep working from the local copy, which is marked stale.

When the database is not activated:

- The agent gets no database tools and is told that database access is disabled.
- GeoForge does not suggest database datasets for your study.
- A plan that still names a database dataset is sent back to the agent to fix.
- The approval card shows no **Data sources, your pick** list.
- Plans that use only public sources or your own files still work.

If you approve a plan that needs the database after access was switched off or the token was rejected, nothing is approved or downloaded. The chat says: "GeoForge Database access is off or not activated, but this plan still needs it for ... Nothing was approved or downloaded. Activate the Database in Settings and approve again, or choose "Modify the plan" to use public sources or your own files. Your saved choices are retained."

3.5 Browse the catalogue

You can look through the catalogue yourself before or during a project. This helps you check what exists for your region before you write a request.

In Settings

1. Open **Settings** → **GeoForge Database**.
2. Scroll to **What the database holds**. The list loads from the local copy on your computer, so it works offline once the copy exists.

3. To filter, click a chip under **By domain**, **By delivery** or **By kind**. Each chip shows how many datasets it covers. Click the same chip again to clear it.
4. To search, type in **Search datasets....** The text is matched against the dataset ID, name, type and variable names.
5. Click a row to open its details.
6. Click **Hide** to close the details.

Check: the line above the rows reads "N of M datasets". The list shows at most 120 rows; when there are more, the line adds "showing the first 120 — narrow with search".

The **By delivery** chips tell you how a dataset reaches you:

Delivery tag	What it means
direct download	GeoForge can download the file itself (files up to 100 MB).
manual (Baidu Pan)	The file is larger. GeoForge either asks the server to clip it to your study area, or gives you a Baidu Pan link to download it yourself (section 3.7).

The details panel lists **Variables**, **Period**, **Extent**, **Resolution**, **Format**, **Size**, **Delivery** and **Domains**. A field is left out when the catalogue has no value for it.

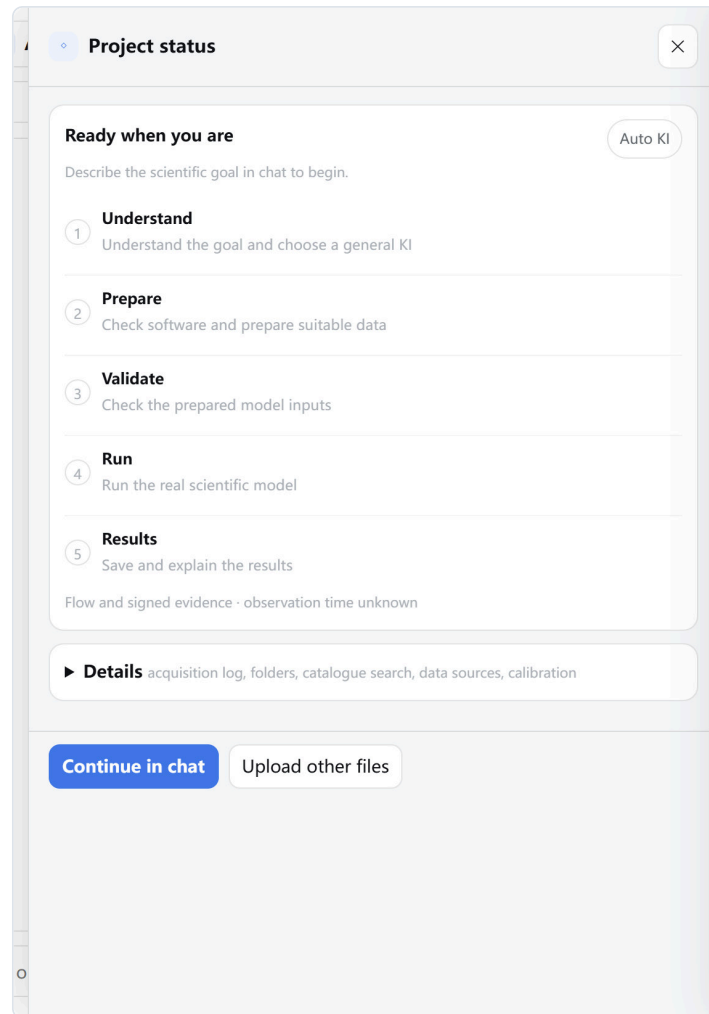
Check: the panel names the dataset you clicked and shows its ID under the name.

Things to know when browsing:

- **Search in English.** Catalogue text is English. Use English names, station numbers or dataset IDs, for example `huai` , `cmfd` or `51080` . Chinese place names do not match.
- **Search one word at a time here.** The Settings search box looks for the text exactly as you type it, so `huai prec` finds only records containing that exact phrase.
- **Sizes are whole-product sizes.** The size in the list is the size of the entire product, not what your project will download. Section 3.6 explains where to find the real download size.
- **Only main products are listed.** Per-variable and per-year sub-files are hidden, so the total here can be smaller than the record count in the status line.

From inside a project

1. In an open project chat, click ◇ **Project status** at the top right.



Check: **Details** is the last card, with the hint "acquisition log, folders, catalogue search, data sources, calibration".

2. Click **Details** to expand it.
3. Scroll to the **GeoForge Database** card inside **Details**.
4. Type in the search box ("variable, place, period, or dataset ID"). Here every word you type must appear in a record, so `huai prec works`.
5. Click **Search** (or press Enter).

Check: the card shows "N found; showing M." and one entry per dataset with its delivery tag (**direct** or **manual Baidu**), ID, kind, format, period, variables, coverage, domains and size.

6. To ask the agent about a dataset, click **Check with Agent** under it.

Check with Agent closes **Project status** and sends the agent a prepared message. The message asks it to compare the dataset with the KI's input requirements and, during planning, only to write the exact ID into the plan. If the chat is still working, the message waits in the message box for you to send.

If no token is stored or the token was rejected, a search shows the error and an **Open AI settings** button. That button opens the **Settings** window; click **GeoForge Database** in its menu. (The hint on this card still says "AI settings"; it means **Settings** → **GeoForge Database**.)

3.6 How planning uses the database

You do not need to search the catalogue yourself. When the database is activated, the agent does it during planning.

1. **Your request is matched to the catalogue.** GeoForge compares the words in your request (place names, station numbers, dataset names) with the local copy and gives the agent a short list of candidates. Station numbers, basin names and river names help most. Some major Chinese place names, such as 淮河 and 蚌埠, are recognised here.
2. **The agent checks candidates.** While it works, the activity line above the message box may read **Searching GeoForge Database**. The work details show steps such as **Searched GeoForge Database**, **Read dataset schema** (the real variable names and units in the files) and **Estimated a server clip**. The agent may request a few clip estimates while it is understanding the task. These are read-only and download nothing.
3. **The plan records exact IDs.** Each input in the plan's data list names one dataset ID. Nothing is downloaded before you approve.
4. **GeoForge checks the plan.** When the agent submits the plan, GeoForge looks up every ID in the catalogue and fills in delivery, size, coverage and period. Unknown IDs go back to the agent. A whole national or global product of hundreds of gigabytes is never accepted as one download; the plan must use a server clip, a regional subset or smaller per-variable or per-year files.

A catalogue match only shows that the data exists. It does not prove that the data suits your model. The agent must still check units, resolution and coverage, and so should you.

What the approval card shows

The **Data** section of the approval card ("Approve the plan?") sorts every input into groups. Database inputs appear in two of them:

Group on the card	Tag on the row	What happens after you approve
GeoForge fetches after approval	direct download	GeoForge downloads the file.
GeoForge fetches after approval	server clip	GeoForge asks the server to cut the product to your study area, period and variables, then downloads the clip.
You	Baidu download, link shown after approval	You download the files from Baidu Pan yourself (section 3.7).

Each row reads `input ← dataset ID · size`. If the agent found more than one suitable dataset for an input, the card also shows **Data sources, your pick**: one radio button per candidate, with the recommended one already selected. If you pick another one and approve, GeoForge switches the plan to it and shows the updated card for you to approve once more.

Check: the row tagged **server clip** shows a size in kilobytes or megabytes, not the whole-product size.

Which size counts

There are two different sizes, and only one of them is your download:

- **Whole-product size**: shown in the catalogue browser, in search results, and sometimes quoted by the agent in chat. A national forcing product can be hundreds of gigabytes.
- **Clip size**: shown on the approval card row tagged **server clip**. This is what your project will download. In one test, a soil map that was 1.7 GB as a Baidu download became a 14 KB server clip.

Trust the approval card, not sizes quoted in chat. GeoForge estimates the clip size again just before it shows the card and once more when you click **Approve and start**. If the clip has grown noticeably, the source version changed or the estimate failed, the card is shown again with the reason, and you approve again. Approving the plan also approves its server clips; there is no separate data approval.

Caution: Known issue in 0.6.54. The **Data sources, your pick** list can disagree with the dataset in the **Data** section, and sources outside the catalogue (for example NASA POWER) are left out of the list. A line under **Waiting on you** may still say "defaults to NASA POWER unless you pick another". Before you approve, compare the dataset ID after the ← in the **Data** section with what you expect. If they disagree, choose **Modify the plan** and name the source you want.

3.7 How approved data arrives

After you approve the plan, GeoForge gathers every approved input before any model step runs. There are three kinds of delivery.

Delivery	Who does it	Size limit	How it is checked
Direct download	GeoForge	up to 100 MB per file	The file is compared with the server's SHA-256 checksum (a fingerprint of the file's exact contents) and downloaded again once if it does not match. ZIP files are unpacked safely; archives with unsafe paths are refused.
Server clip	GeoForge	up to 2 GB per clip, and enough free disk space	Every part is checked against the job's list of sizes and checksums. A clip much larger than its estimate is refused.
Baidu Pan manual delivery	You	none	GeoForge records the files you placed and their fingerprints. Baidu files have no server checksum.

Automatic downloads keep running when **Project status** is closed or you switch to another chat, as long as GeoForge stays open. Waiting for a Baidu file does not hold them up.

Caution: Known limit of the Windows 0.6.54 release: real GeoForge Database downloads (direct, server clip and Baidu Pan) had not been tested end to end on Windows when it was published. If something in this section behaves differently on Windows, report it.

Follow progress in the chat

While data is being gathered, a banner appears above the message box. Its title tells you what kind of delivery is running:

Banner title	Meaning
Data preparation · fetching approved data	Only automatic downloads are running.
Data preparation · manual download needed	Only Baidu Pan downloads are waiting for you.
Data preparation · automatic and manual downloads are independent	Both kinds are running side by side.
Approved data acquired · run not started	Everything is in. The model has not started.
Approved data acquired · software setup needed	Everything is in, but the model software must be set up first.

Each automatic input has its own line, for example "Automatic: dataset ID — Server subsetting". Typical states are **Server queued**, **Server subsetting**, **Subset ready to download**, **Downloading and checking files**, **Files acquired** and **Acquisition failed**. Each Baidu input has a line "Manual download needed: dataset ID".

The banner's **Project status** button opens the panel.

When the last input arrives during a chat turn (for example after you click **Files are in place, continue**), the run continues in that turn. When it arrives in the background, the title changes to **Approved data acquired · run not started** or **Approved data acquired · software setup needed**. The model then waits until you click **Start the approved run**.

Check: the banner shows one line per input, and the automatic line changes state while the manual line waits.

Server clip jobs are also listed under **Project status** → **Details** → **GeoForge Database acquisitions**, with states such as **Server queued**, **Server processing**, **Ready to download**, **Downloading and verifying** and **Files acquired; scientific checks pending**. GeoForge downloads a ready clip by itself. If a job shows **Download failed; retry available**, click **Download and check all files** under it.

Download from Baidu Pan yourself

Some datasets are too large to serve directly and cannot be clipped for your study. For these, **Project status** shows a card headed **One thing needs you**, titled "Download 1 dataset from Baidu Pan" (or "Download N datasets from Baidu Pan"). The chat only points you to **Project status**: the link and code are never shown in the chat.

Each row of the card shows:

- the dataset name and its size;
- **Extraction code**: the code Baidu Pan asks for;
- **Inside the share, download only**: shown when you need just one file or folder from a larger share;
- **Place at**: the exact project folder for the files, with a **Copy path** button;
- **Open download link**: opens the Baidu Pan share.

Check: the **Place at** path ends in your dataset's ID.

1. Click **Open download link**. The share opens in your web browser.
2. Enter the code shown after **Extraction code**.
3. If the card shows **Inside the share, download only**, download only that file or folder. Otherwise download the whole share.
4. Wait until the download has completely finished.
5. Click **Copy path** next to **Place at**. The button changes to **Copied**.
6. Open the **Place at** folder:
 - **Windows**: paste the path into the File Explorer address bar and press Enter.
 - **macOS**: in Finder, choose **Go** → **Go to Folder...**, paste the path and press Return.
7. If the folder does not exist yet, create it with exactly the name at the end of the path (the dataset ID).
8. Move or copy the downloaded files into that folder. Do not leave them in the Baidu Pan download folder.
9. Wait until the copy has finished.
10. In **Project status**, click **Files are in place, continue**.

GeoForge closes **Project status** and sends "I placed ... at ... Please continue." in the chat. (If the chat is still working, the message waits in the message box for you to send.) GeoForge then fingerprints the files in the folder and signs a run record for them. If no usable files are there yet, the card stays open.

Caution: GeoForge ignores files that are still being written (names ending in `.part`, `.crdownload`, `.tmp`, `.bc!` and similar), files whose names start with a dot, and symbolic links. Click **Files are in place, continue** only after the download and the copy have both finished.

If the download is not practical, click **Change the plan**. The message box fills with "Please change the plan: ". Finish the sentence, for example by asking for a public source or by offering your own file, and send it. The revised plan needs a new approval.

Tip: Baidu Pan is often slow or unusable outside mainland China. If you cannot use it, ask the GeoForge Database owner for another way to receive the files, or click **Change the plan**.

If an approved download fails

GeoForge stops and shows a card titled "Some approved data could not be fetched" that lists each failure. The chat says "Some approved data could not be fetched. Use the card in the chat to retry or modify the plan."

1. Choose one option on the card:
- **Retry the missing data:** fetches only the failed items. Files already checked are kept.

◦ **Modify the plan:** sends the plan back to the agent with your note.
2. Click **Continue with this choice**.

3.8 Where the data goes, and what the run record proves

Everything is saved inside your project folder. In these examples the project folder is `D:\GeoForge-Manual\projects\2026-10-01-huai-test--...`.

What	Where in the project
Direct downloads	<code>inputs\observations\<dataset ID>\</code>
Baidu Pan files you placed	the Place at folder, normally <code>inputs\observations\<dataset ID>\</code>
Server clips	<code>inputs\geoforge_subsets\<acquisition ID>\</code>
Original ZIP archives of direct downloads	<code>.geoforge\downloads\</code>
Signed run records for data	<code>.geoforge\receipts\data-receipts\</code>

The `.geoforge` folder belongs to GeoForge. On macOS Finder hides it; on Windows File Explorer shows it. Do not edit or delete anything in it.

Every input GeoForge brings in gets a signed run record (the app calls it a receipt). **Data in this plan** in **Project status** reads its states from these records and the files on disk, not from what the agent says:

State	Meaning
Pending	Not fetched or prepared yet.
Waiting for you	A Baidu Pan download or a file you must provide is still missing.
Acquired; scientific checks pending	The data arrived and has a signed run record.
Present; not verified	Files are in the folder, but no run record covers them.
Fetch failed	GeoForge could not fetch the item.
No step uses it	The plan lists the item, but no step uses it, so nothing fetches it.

Check: every database input reads **Acquired; scientific checks pending**.

If you delete or replace a downloaded file, it stops counting as acquired. Checked downloads are reused when the plan is changed but the same dataset request is kept, so the same data is not fetched twice.

What a data run record proves, and what it does not:

Delivery	The record proves	The record does not prove
Direct download	The file matches the server's checksum and came from the approved request.	That the values are correct or suit your model.
Server clip	Every part matches the server job's sizes and checksums.	That the clip covers your study area as intended; check the extent yourself.
Baidu Pan	Which files you placed, and their fingerprints.	That the files match the original; Baidu Pan files have no server checksum.

Downloaded data is not checked science. The KI still prepares and checks the inputs (units, extent, values) before the model runs. What GeoForge guarantees is that every approved step has a signed run record before the project is marked **Completed**. **Completed** does not mean the results are scientifically validated; judging them is still your job.

3.9 Privacy

Your token stays on your computer. It is never written to the settings file, your project folders, chat transcripts or logs, and it is never sent to the agent. The Settings page only ever shows dots.

	Windows	macOS
Where the token is stored	Credential Manager, generic credential <code>com.geoforge.desktop:observation-activation-token</code>	a private file readable only by your user account: <code>~/Library/Application Support/KISS/secrets/com.geoforge.desktop.observation-activation-token</code>
Where the catalogue copy is stored	<code>%APPDATA%\KISS\database\catalogue.json</code>	<code>~/Library/Application Support/KISS/database/catalogue.json</code>

On Windows you can see or delete the stored token in **Control Panel → Credential Manager → Windows Credentials**, under **Generic Credentials**. If you delete it there, restart GeoForge (tray icon → **Exit GeoForge**, then start it again), because a running GeoForge keeps using the token it has already read.

The agent sees search results, not the token. API agents receive cleaned-up dataset descriptions. CLI agents (for example Claude Code, OpenAI Codex or Kimi Code) receive a temporary, read-only pass to GeoForge's own local service. It allows only catalogue search, schema reading and clip estimates, and it stops working when GeoForge closes.

Baidu Pan links are for you only.

- Links and extraction codes are shown only to you in **Project status**, never to the agent or in the chat.
- The server logs every link it reveals against your token. A revealed link cannot be withdrawn.
- So that the card survives a restart, GeoForge keeps the links in the project file `.geoforge\manual-download-details.json`.

Caution: Do not share or publish a project folder that still contains `.geoforge\manual-download-details.json`. Remove that file from the copy you share.

3.10 If something goes wrong

Messages from the database server appear in English. Results of **Save & test database** appear in the note under the token box.

What you see	What it means	What to do
"Paste your activation token in Settings."	No token is stored.	Enter your token (section 3.3).
"That token isn't right — check for a copy/paste miss."	The server does not recognise the token.	Copy the whole token again, with no spaces or line breaks. Replace it as in section 3.3 and click Save & test database .
"Your token expired — ask the group owner for a new one."	The token is older than 6 months.	Request a new token (section 3.2).
"This token was revoked — ask the group owner."	The owner withdrew the token.	Contact the owner (section 3.2).
Pill Unavailable , "Access unavailable: the token is not authorized..."	The server rejected the stored token at the last refresh.	Paste a valid token and click Save & test database . If you saved a token only seconds ago, wait a minute and reopen the page first.
"Daily download limit reached; resets at midnight server time."	Your token used 20 GB or 2,000 requests today.	Wait until after midnight server time, then click Retry the missing data .
"GeoForge could not reach GeoForge Database." or pill Connected in amber with "... stale: ..."	GeoForge is offline, or the proxy route is wrong.	Searches still work from the local copy. Reading dataset schemas, clip estimates and downloads wait for a connection. Check your network and the proxy settings below, then click Save & test database to refresh.
A message starting "GeoForge Database server or gateway" (for example "...returned HTTP 502" or "...is unavailable")	The problem is on the server side.	Try again later. Do not switch the plan to a manual download because of it.
A message naming Credential Manager (Windows), for example "Credential Manager write failed (...)", or "the private token file" (macOS)	GeoForge could not read or save the token in the system password store. The pill may show Not configured .	Restart GeoForge and click Save & test database again. On Windows, check that Credential Manager opens from Control Panel.
"The catalogue is being fetched; try again in a moment."	The first catalogue download is still running.	Wait a minute and reopen Settings → GeoForge Database .
The note stays on "Saving and testing data access..."	Another Settings page has an invalid value, so nothing was saved. The error appears at the bottom of the window, next to Save (for example "proxy address is required in manual mode").	Fix that page (for example, type a proxy address in Network & proxy), then click Save & test database again.
The pill or the dataset list does not change after a successful test	The page is not refreshed automatically.	Click Close , open Settings and click GeoForge Database . If the list is still empty, reload. Windows: press F5 in the GeoForge browser tab. macOS: quit and reopen GeoForge.
Search finds nothing you expected	Catalogue text is English, and the Settings box matches your text as typed.	Search for one English word, a station number or a dataset ID.
Approve is refused with "GeoForge Database access is off or not activated..."	The plan needs the database, but it is not activated.	Activate it (section 3.3) and approve again, or choose Modify the plan to use public sources or your own files.

What you see	What it means	What to do
The Baidu Pan card stays open after Files are in place, continue	GeoForge found no usable files in the Place at folder.	Check that the files are in exactly that folder, that the download and copy are finished, and that the names do not start with a dot. Then click the button again.
"... already holds files with no receipt ..."	The target folder already contains files GeoForge cannot account for. Nothing was overwritten.	Move those files out of the folder and click Retry the missing data , or choose Modify the plan to use them as your own input.
"The downloaded file failed integrity verification twice."	The file did not match the server's checksum.	Click Retry the missing data . If it happens again, report it to the database owner.
"The downloaded archive contains an unsafe path."	GeoForge refused a ZIP file for safety.	Report it to the database owner and choose Modify the plan .
"Insufficient local disk space for the complete verified acquisition"	Not enough free space for a server clip.	Free disk space on the drive that holds the project, then click Retry the missing data .
An entry under Clip estimates on file (in GeoForge Database acquisitions) shows Estimate failed; acquisition unverified	The server could not estimate the clip. No job was created and nothing was downloaded.	Ask the agent in chat to estimate the clip again. If the message names an unknown source variable, ask it to read the dataset schema and estimate again with the exact variable names. Do not ask it to remove the variable filter.
The agent says a dataset is "unavailable" because it is a Baidu Pan download	Manual delivery is a normal path, not a missing dataset.	Tell the agent to keep the dataset if it suits the study, or choose another source.

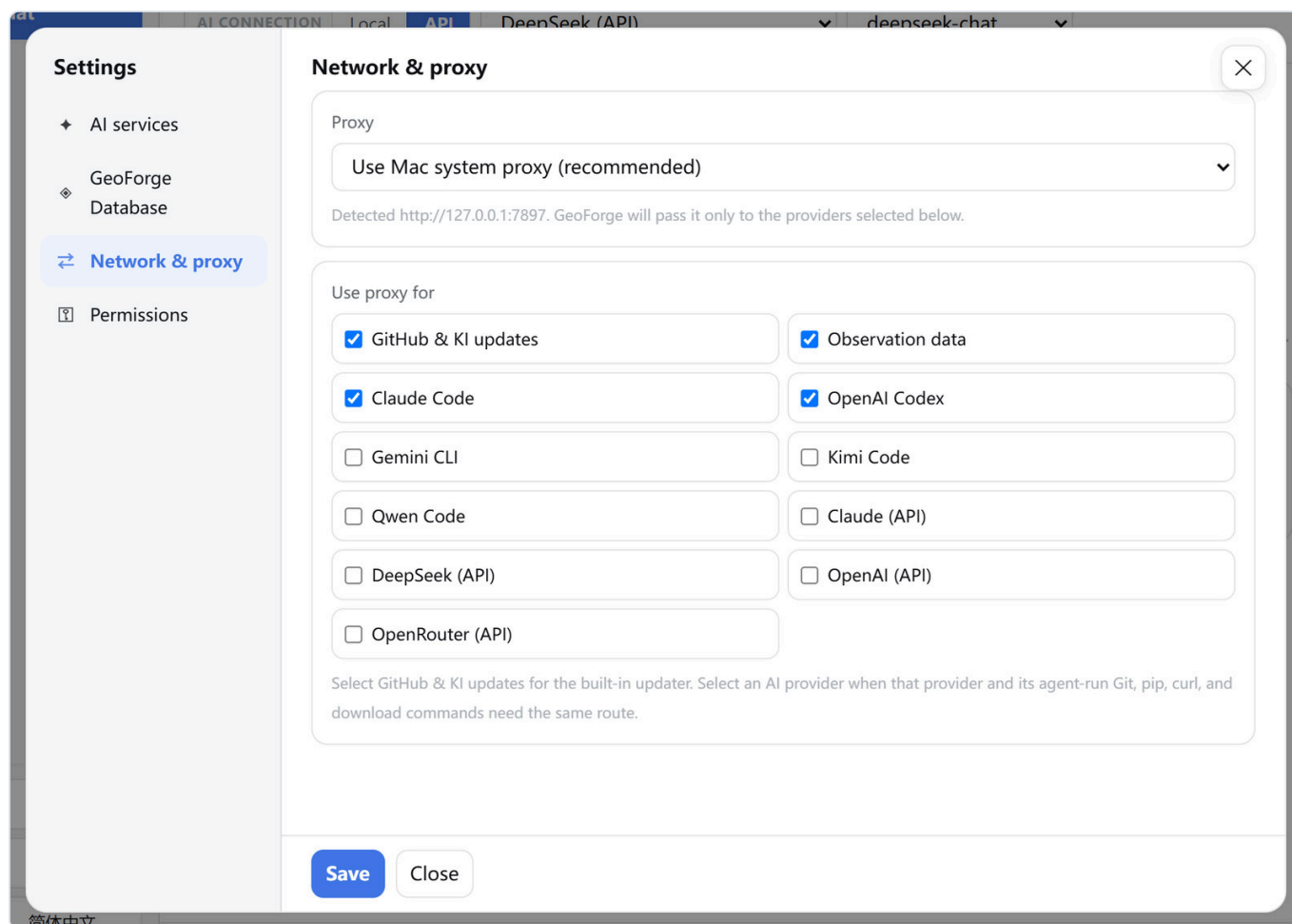
Caution: If you replace a working token with a wrong one, the test shows an error, but the pill can stay **Connected** for up to 6 hours, until the next catalogue refresh. Trust the note under the token box, and paste the correct token again.

Tip: The stale warning the agent may quote says "Refresh explicitly in Settings". There is no separate refresh button: **Save & test database** is the way to force a refresh.

Proxy settings and the database

Database traffic follows **Settings** → **Network & proxy**.

1. Click **Settings** at the bottom of the left sidebar.
2. Click **Network & proxy**.



Check: under **Use proxy for**, **Observation data** is ticked. It is ticked by default.

- **Observation data** ticked: database requests use the proxy chosen under **Proxy**.
- **Observation data** not ticked: database requests go out directly and ignore every proxy.
- **Enter proxy manually**: type the address, for example `http://127.0.0.1:7897`. Proxy passwords are not stored.
- **Do not use a proxy**: no request uses a proxy.

	Windows	macOS
First proxy option	Reads Use Mac system proxy (recommended) , but uses the Windows system proxy.	Use Mac system proxy (recommended) uses the macOS system proxy.

If the database works without your VPN but fails with it on:

1. On **Network & proxy**, untick **Observation data**.
2. Click **Save**.
3. Click **GeoForge Database** in the Settings menu.
4. Click **Save & test database**.

Baidu Pan downloads happen in your own browser or Baidu client, so GeoForge's proxy settings do not affect them.

4 Install a scientific model

In this chapter you will get a model's software running on your computer, either inside a project after you approve a plan or ahead of time on the **Agent setup** page, read what **Verified on this machine** means, and choose the KI for a chat.

The examples use FSM2, the snow model of the worked example in Chapter 5, with **DeepSeek (API)**. That is the combination tested end to end on Windows in 0.6.54.

KIs and model software

GeoForge keeps two things apart for every model:

	The KI	The model software
What it is	GeoForge's package of checked knowledge about one model: how to install it, prepare its inputs, run it and check it	The scientific program itself, for example <code>FSM2.exe</code>
Where it comes from	Ships with GeoForge (127 KIs in 0.6.54), or you import it (Chapter 7)	An agent downloads or builds it on your computer, from the official source the KI names
Size, for FSM2	Under 1 MB	About 1.1 GB in the Windows test, almost all of it the private compiler used to build it

The FSM2 KI holds instructions for the agent, four tools (to convert forcing data and soil parameters, to run FSM2 and to read its output), a list of known errors with their fixes, Windows installation notes and a Windows recipe, and a start-up check. It does not hold the FSM2 program. Installing GeoForge installs no model software.

A KI that came with GeoForge, or with an official KI library update, carries the badge **Bundled KI** in the **KI Library**.

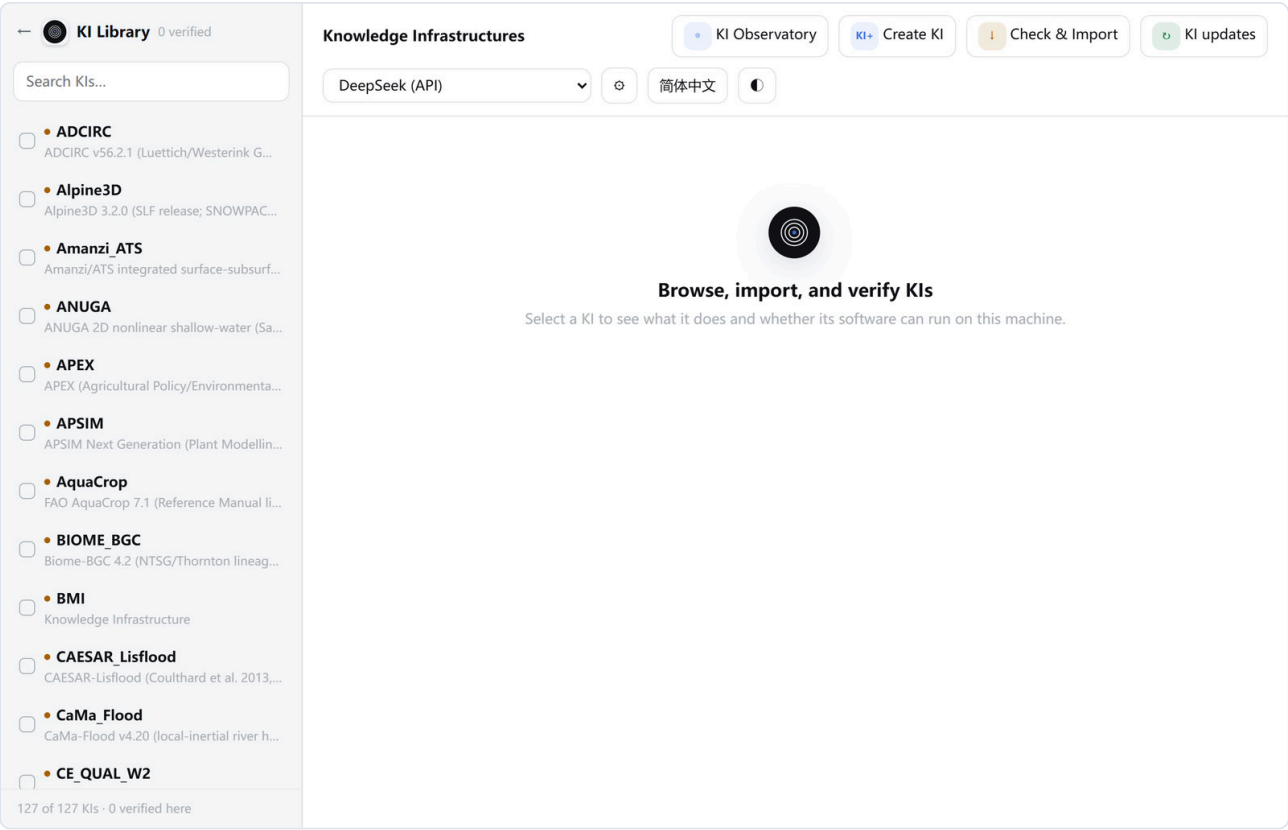
The software state of a KI

Each KI shows the state of its software on this computer. You see it as a coloured dot before the name in the **KI Library**, as a label in the chat's KI picker, and as a pill next to the KI name in the chat header.

Label	Dot	Meaning
Verified on this machine	green	The model software passed the KI's final check on this computer.
Setup needed	orange	Not verified here yet. The agent works out the installation from the KI.
Ready to set up	orange	Not verified here yet. The KI carries an installation recipe that has been seen to work.
Manual setup	orange	Not verified here yet. GeoForge cannot fetch the software by itself, so expect to supply some of it yourself.
Verification failed	red	The last check on this computer failed. Set the software up again.

For APEX only, **Recheck APEX0806** means an earlier check tested a different APEX version. Set it up again.

1. Click **KI Library** at the bottom of the sidebar.



Check the line at the bottom of the list, here **127 of 127 KIs · 0 verified here**, and the orange dot before every name. On a new installation no model software is verified yet. That is normal.

Verification belongs to one computer. GeoForge saves the result with the installed software, in that model's installation folder. Another computer, another Windows user account or another installation folder starts again at **Setup needed**. On the other hand, once a model is verified, every project on this computer uses the same installation: you set each model up once.

Two ways to set up the software

	Inside a project	Ahead of time
When	After you click Approve and start on a plan	Whenever you like, before any project
How it starts	GeoForge starts it after your approval, once the approved data are in	KI Library → Set up with agent , or the chat banner's Let the agent set it up
Which AI does the work	The chat's AI	The AI you choose on the Agent setup page
Installation folder	The KI's recorded folder; by default <code>C:\Users\you\kiss\fsm2</code>	You can choose the folder, or point to software you already have
Where you follow it	The chat, the activity bar and ◇ Project status	The log and the Needs you box on the Agent setup page
How to stop it	Stop in the activity bar	The page has no Stop button; exit GeoForge
Afterwards	GeoForge starts your approved run	The KI shows Verified on this machine for all later projects

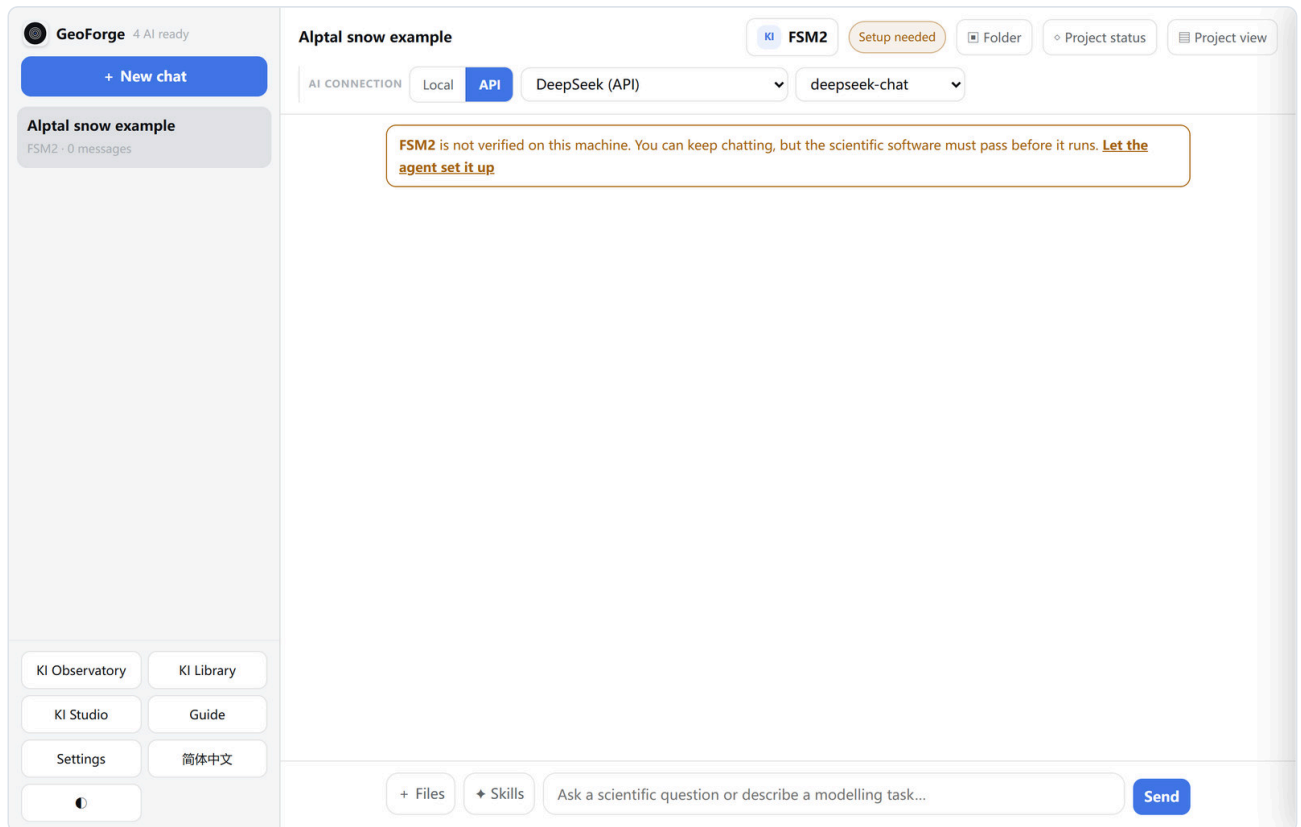
Both ways use the same installation folder, run the same final check and save the same record. Setting up costs AI usage on the chosen AI, like any other agent work.

Set up ahead of time when you want another folder or drive, want GeoForge to use software you already have, want a different AI for the installation, or simply want to know that the model runs here before you plan a study.

Let a project set up the software

Nothing is installed during planning. The software is set up only after you approve a plan. Chapter 5 follows the whole FSM2 project; this section covers only the setup part.

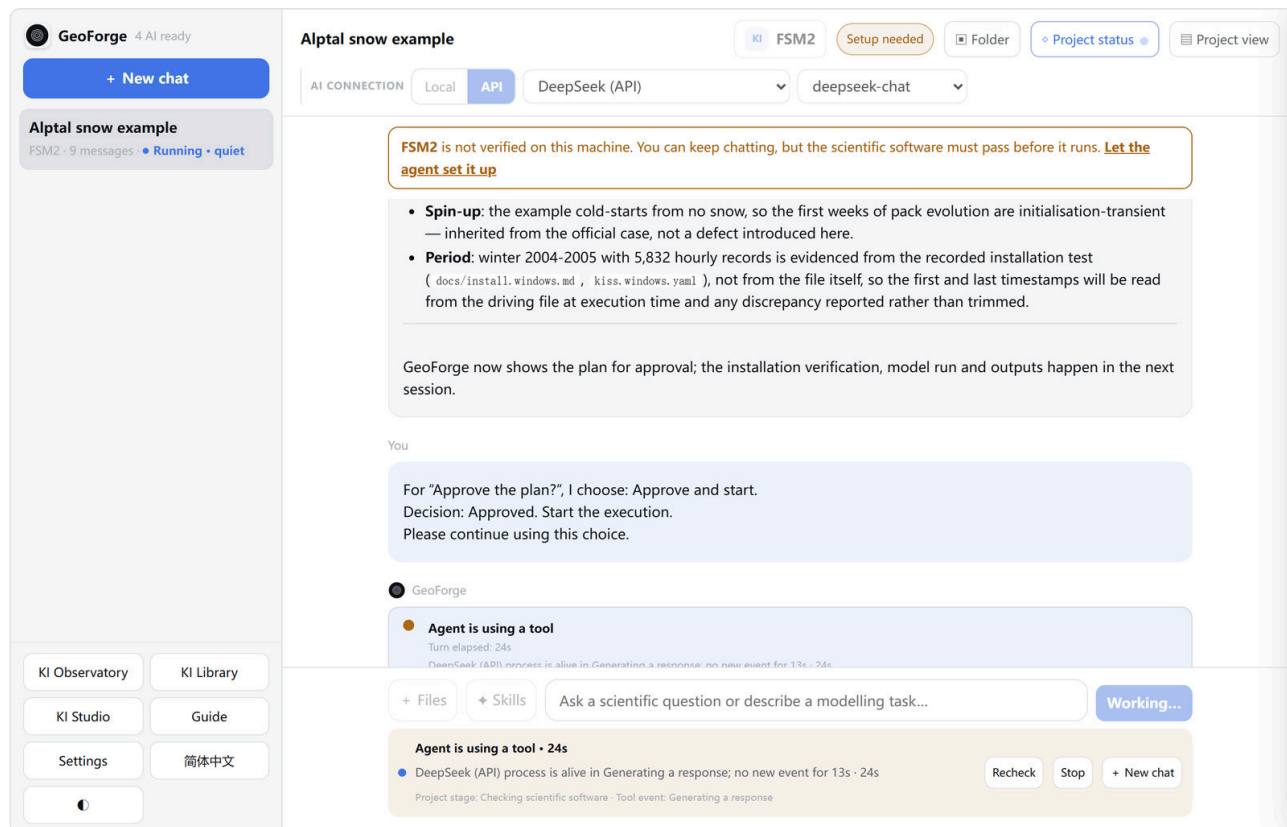
1. Start a new chat and choose the KI (see "Use a KI in a chat" below), or leave **Auto KI**.



Check the header: **FSM2** with the pill **Setup needed**, and the banner "**FSM2** is not verified on this machine. You can keep chatting, but the scientific software must pass before it runs." You can ignore the banner's link **Let the agent set it up** if you want the project to set the software up.

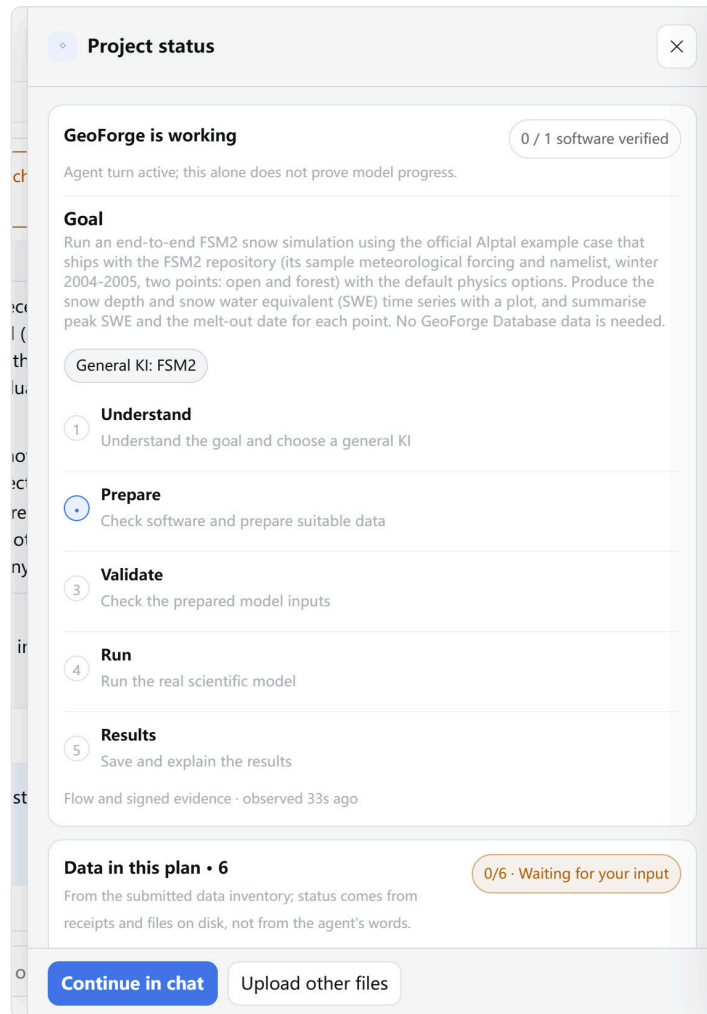
2. Send your request (Chapter 5).
3. Answer the planning questions.
4. Read the plan on the approval card.
5. Click **Approve and start**.

GeoForge first fetches the approved data, if the plan needs any. Then, because the software is not verified, the chat's AI sets it up in the KI's installation folder.



Check that the pill in the header still reads **Setup needed** and that **Stop** is available in the activity bar under the message box. The pill changes to **Verified on this machine** only after the final check passes.

6. Click **Project status** in the chat header to see the stage.



Check that **Prepare** ("Check software and prepare suitable data") is the active stage and that the pill on the right reads **0 / 1 software verified**.

7. Wait for the final check. When it passes, the chat shows "**Software verified. Starting your approved plan in a new session...**" and GeoForge starts the approved steps. Chapter 5 continues from there.

If the approved data finished downloading in the background, the banner above the message box reads **Approved data acquired · software setup needed**. Click **Start the approved run**: GeoForge sets up the software first, then runs (Chapter 3).

To stop the setup, click **Stop** in the activity bar. GeoForge does not run the final check after a Stop. Files already downloaded or built stay in the installation folder. Send a message when you want to continue.

If the turn ends without a passing check, send a message in the same chat, for example "Please continue setting up FSM2." The agent tries again, and the files of the first attempt are still there. Finish a setup that a project has started in that project's chat.

Current Windows behaviour: API software setup installs and verifies the model environment. Run the scientific example through a separately reviewed project plan so its outputs receive passing run records.

Tip: To choose the installation folder, or to use software you already have, open the **Agent setup** page before you approve the plan. A project always uses the folder recorded for the KI.

Set up the software ahead of time

Open the Agent setup page

1. Click **KI Library** at the bottom of the sidebar.
2. Type **FSM2** in **Search KIs...**
3. Click **FSM2** in the list.

The screenshot shows the KI Library interface. On the left sidebar, there's a 'KI Library' header with '0 verified' and a search bar containing 'FSM2'. Below the search bar, 'FSM2' is listed with a description: 'FSM2 (Flexible Snow Model) 2.1.x linea...'. The main content area shows the details for 'FSM2'. At the top, it says 'FSM2 FSM2 (Flexible Snow Model) 2.1.x lineage; Essery et al. 2025 (GMD 18, 3583-3605) forest-canopy extension of FSM 1.0 (Essery 2015)'. Below this are buttons for 'KI Observatory', 'KI+ Create KI', 'Check & Import', 'KI updates', and a dropdown menu set to 'DeepSeek (API)'. There are also buttons for '简体中文' and a moon icon. The main content area has a header 'KI FSM2' followed by the same description. Below this are two badges: 'Bundled KI' and 'Setup needed'. There are four cards: 'SCIENTIFIC SOFTWARE' with 'fsm2 2.1.2', 'LAST CHECKED ON THIS MACHINE' with 'Not yet', 'LANGUAGE & LICENCE' with 'fortran · Not declared', and 'SOURCE' with 'Open project website'. At the bottom of these cards are two buttons: 'Use in a new chat' and 'Set up with agent'. Below these is a 'Run data' section with a 'Software setup needed' badge. It shows a flow from 'INPUTS' (62 declared) to 'KI FSM2 Setup needed' to 'OUTPUTS' (17 declared). The inputs are 'Forcing 8 · Parameters 22 · Initial conditions 14 · Boundary & controls 18'. The outputs are 'snw · snd · svg +14 more'. At the bottom, there's a warning: 'Local data check · Dataset locations not declared' and a note: 'Local files are not machine-checkable yet. This KI has no dataset paths in its setup manifest. The declared input groups below still show the model interface, but GeoForge will not guess whether your files satisfy it.'

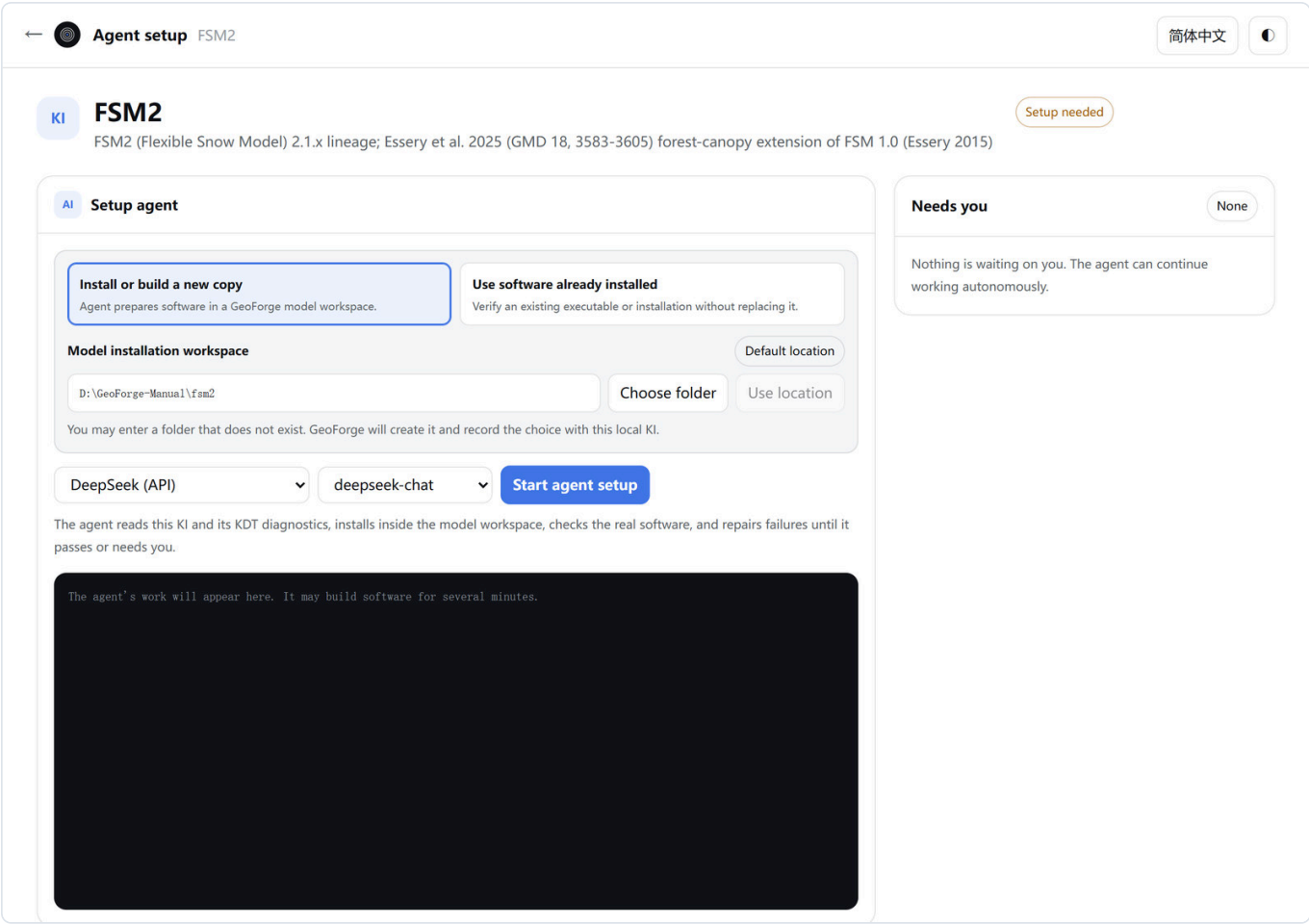
Check the two badges under the name, **Bundled KI** and **Setup needed**, the card **Last checked on this machine** with **Not yet**, and the buttons **Use in a new chat** and **Set up with agent**.

4. In the AI list at the top of the **KI Library** (here **DeepSeek (API)**), choose the AI that should do the setup. The **Agent setup** page starts with this AI; you can still change it there.
5. Click **Set up with agent**. The **Agent setup** page opens in the same tab.

From a chat, the banner link **Let the agent set it up** opens the same page for that KI. The page then starts with your default AI (Chapter 2), not necessarily the chat's AI. When a chat has several KIs that are not verified, the banner link reads **Open KI Library** instead.

To go back, click ← at the top left. It leads to the **KI Library**; its own ← leads to the main window.

The parts of the page



Check that **Install or build a new copy** is selected, that the badge beside **Model installation workspace** reads **Default location**, that the lists show **DeepSeek (API)** and **deepseek-chat**, that **Needs you** shows **None**, and that the badge at the top right reads **Setup needed**.

In this screenshot GeoForge's work folder is `D:\GeoForge-Manual` , so the default folder reads `D:\GeoForge-Manual\fsm2` . With a normal installation it is `C:\Users\you\kiss\fsm2` .

Part	What it shows
Header	Agent setup , the KI name, ←, the language button and 🌙
Title	The KI name, its reference and, at the right, the software state badge
Setup agent	Where the software goes, which AI does the work, the start button, the progress card and the log
Needs you	The only place where the agent asks you for something during this setup
Verification	The result of the final check, one line per step
How this works	A four-point summary of the setup

Model installation workspace

Default location

D:\GeoForge-Manual\fsm2

Choose folder

Use location

You may enter a folder that does not exist. GeoForge will create it and record the choice with this local KI.

DeepSeek (API)

deepseek-chat

Start agent setup

The agent reads this KI and its KDT diagnostics, installs inside the model workspace, checks the real software, and repairs failures until it passes or needs you.

The agent's work will appear here. It may build software for several minutes.

Verification

Not checked yet. The agent will run the KI preflight.

How this works

1. The agent reads the KI before touching the software.

2. It tries, checks the actual error, repairs, and retries.

3. Licence gates, protected downloads, login, and system privileges come back to you here.

4. "Verified" appears only after the KI's real preflight passes.

Check that **Verification** reads "Not checked yet. The agent will run the KI preflight." "Preflight" is the KI's own start-up check.

Choose where the software goes

With **Install or build a new copy** selected, the agent installs a new copy in the folder shown under **Model installation workspace**. To keep the default folder, skip to "Choose the AI and start".

To use another folder:

1. Click in the path field under **Model installation workspace**.

2. Press Ctrl+A (macOS: Cmd+A) to select the path.

3. Type or paste the full path, for example `D:\GeoForge-Manual\model1s\fsm2` . The folder does not need to exist yet.

4. Click **Use location**. The button becomes clickable once you change the path.

GeoForge creates the folder and records it. The badge changes to **Custom location**, and the note reads "Recorded in ... and this local KI's .geoforge-install.json."

Platform	Choose folder
Windows	Only shows the message "Type the full path directly in the installation field." Copy the path from File Explorer's address bar and paste it into the field.
macOS	Opens the system folder picker. Click Use location after you pick a folder.

GeoForge refuses a path without a drive letter, a whole drive, your home folder, a folder already given to another KI, and a folder you cannot write to. The reason appears in a message box (see "If something goes wrong").

On Windows, choose a short path without spaces, on a drive with several GB free, outside OneDrive. Windows limits most paths to 260 characters, and build tools create deep folders inside the installation folder.

Caution: Changing the folder does not move software that is already installed. GeoForge then looks for the software in the new folder, so the KI shows **Setup needed** again until you set it up there. The old folder stays on disk.

Use software that is already installed

If the model is already installed on this computer, for example by your lab, GeoForge can check that copy instead of building a new one. GeoForge gives the agent only read and run access to that software, and does not overwrite it. GeoForge's own files for the KI (its working copy, the log and the verification record) still go in the KI's installation folder.

- 1. Click **Use software already installed**. The panel **Existing installation or executable** appears, with the badge **Not selected**.
- 2. Click in the path field of that panel.
- 3. Paste the full path of the program file, for example `D:\GeoForge-Manual\lab-models\FSM2\FSM2.exe` , or of its folder.
- 4. Click **Use path**.

The badge reads **Found**, and the note reads "Will verify ... The agent may read and run this software but will not overwrite it." The start button now reads **Verify existing software**.

- 5. Continue with "Choose the AI and start" below.

Platform	Choose folder and Choose file
Windows	Only show the message "Type the full existing software path directly." Paste the path instead.
macOS	Open the system folder or file picker.

If you do not know where the software is, choose the AI first, then click **Let Agent find and verify it** instead of steps 2 to 5. This starts the agent at once.

GeoForge looks for programs named after the model on the Windows program search path, and for folders or programs with that name directly inside `C:\Program Files` , `C:\Program Files (x86)` and `C:\Users\you\AppData\Local` . The agent then checks what was found.

If nothing was found, the log says "No likely installation was found automatically; the agent will ask for a path instead of reinstalling." The agent is told not to install a new copy unless you switch back to **Install or build a new copy**.

Choose the AI and start

- 1. In the first list below the folder box, choose the AI. AIs that are not ready are greyed out with "— unavailable".
- 2. In the second list, choose the model. For a local agent CLI, **Agent default** uses the CLI's own setting.
- 3. Click the start button.

The start button's label tells you what it will do:

Label	When it appears
Start agent setup	First setup of a new copy
Continue agent	An earlier setup already prepared this folder
Verify existing software	Use software already installed is selected
Re-check with agent	The KI is already verified

GeoForge records the folder if you changed it, prepares the folder (its working copy of the KI and written instructions for the agent) and runs the KI's check once. If the check already passes, the log says "FSM2 is already verified and ready to use." and the AI is not used. Otherwise the agent starts from the check's failure.

Unlike a chat, this page does not fix the AI. You can choose another AI for the next attempt.

Follow the progress

While the agent works, a progress card appears under the start button:

- Its title says what the agent is doing, for example "Running setup commands" or "Reading the KI and diagnostics".
- The line below reads "Agent process connected · signal 3s ago" while the page keeps receiving signals, even during a long compile. If no signal has arrived for more than 14 seconds, it reads "Command still running · awaiting the next agent signal (...s)".
- The timer at the right counts minutes and seconds.
- Three phases light up in turn: **1 · Read KI**, **2 · Install & repair**, **3 · Verify**. GeoForge estimates the phase from the log, so treat it as a guide.

The black log below shows the agent's messages and command output. Runs of tool calls are shortened to lines such as "↳ Running setup commands · 4 steps". GeoForge also saves the log as `setup-agent.log` in the installation folder.

In the Windows installation test of September 2026, FSM2 took about three minutes; in the run for this manual, setup inside the project took about five. Most models took a few minutes; the slowest model that installed took about 17 minutes. A slow network makes the first build longer, because the agent downloads the source and often a compiler.

Browser disconnection or closing a tab no longer ends the active turn in 0.6.55. Reopen GeoForge and the chat to inspect its saved progress; use **Stop** to stop work.

The page has no Stop button. To stop a setup started here:

- **Windows:** right-click the GeoForge tray icon and choose **Exit GeoForge**. It stops running agents and commands before it quits.
- **macOS:** quit GeoForge or close its window.

Files already downloaded or built stay in the installation folder, and the next setup can reuse them.

When the agent needs you

The agent stops only for something it should not do by itself: a licence, a protected download, a login, a change to the whole computer, or a scientific choice. Then the badge on **Needs you** reads **Needs you now**, and the log ends with "Paused for the user. The required action is shown beside this log."

The request card can contain:

Part	What it is for
A label: DOWNLOAD, LICENCE, LOGIN, PERMISSION, CHOICE or OTHER	The kind of help needed
A title and a short explanation	What to do and why. A long technical report folds under View the full technical report .
Open the official page ↗	The official page for the download, licence or login
"Needed at: ..."	Where a file has to go
A command and Copy	A command you can copy and run yourself, for example in PowerShell. Read it first; GeoForge does not run it for you.
A file box and Give file to agent	DOWNLOAD and LICENCE only: hand the agent a file you downloaded
Optional note for the agent	A short note sent to the agent with your answer

To hand over a file you downloaded:

1. Click **Open the official page** ↗.
2. Download the file from that page. If the page asks you to sign in or to accept a licence, you do that yourself.
3. Back on the **Agent setup** page, click the file box above **Give file to agent**.
4. Choose the downloaded file.
5. Click **Give file to agent**.

GeoForge saves the file in `user-files\` in the installation folder and restarts the agent at once. The file can be up to 300 MB.

For any other request:

1. Do what the card asks.
2. Type a note in **Optional note for the agent** if it helps. For a CHOICE request, write your answer here.
3. Click **I've done this — continue**. The agent continues.

If you cannot do it, click **I can't do this — help me**. GeoForge opens a new chat with this KI and puts a description of the problem in the message box. Read it, then click **Send** to discuss alternatives with the AI. The setup stays paused on the **Agent setup** page until you answer the card there.

Caution: Never type passwords, licence keys or tokens into the note. The note is saved in the installation folder and sent to the AI.

The other badges on **Needs you: None** means nothing waits for you. **Action needed** means the agent stopped before the check passed (see "If something goes wrong"). **Ready to resume** means you answered and the agent has not picked up yet. Files you handed over are listed under "Files supplied".

When the setup finishes

When the final check passes, the log ends with "GeoForge final check: PASS" and "FSM2 is verified and ready to use.", and the badge at the top right reads **Verified on this machine**.

←

Agent setup

FSM2

简体中文

KI

FSM2

FSM2 (Flexible Snow Model) 2.1.x lineage; Essery et al. 2025 (GMD 18, 3583-3605) forest-canopy extension of FSM 1.0 (Essery 2015)

Verified on this machine

AI

Setup agent

Install or build a new copy

Agent prepares software in a GeoForge model workspace.

Use software already installed

Verify an existing executable or installation without replacing it.

Model installation workspace

D:\GeoForge-Manual\fsm2

Choose folder

Use location

Default location

Recorded in D:\GeoForge-Manual\fsm2\kiss.toml and this local KI's .geoforge-install.json.

DeepSeek (API)

deepseek-chat

Re-check with agent

Use KI in chat

The agent reads this KI and its KDT diagnostics, installs inside the model workspace, checks the real software, and repairs failures until it passes or needs you.

The agent's work will appear here. It may build software for several minutes.

Needs you

None

✓

No action needed

The software passed its check.

Check that the badge reads **Verified on this machine**, that **Needs you** shows ✓ **No action needed** with "The software passed its check.", and that the buttons **Re-check with agent** and **Use KI in chat** are shown. Here FSM2 was set up inside the project of Chapter 5, so the log on this page shows only its placeholder text and the note under the folder reads "Recorded in D:\GeoForge-Manual\fsm2\kiss.toml and this local KI's .geoforge-install.json."



Verification

- ✓ **materialise**
▶ details
- ✓ **python-env**
▶ details
- ✓ **ki-tools-common**
▶ details
- ✓ **system-deps**
▶ details
- ✓ **python-deps**
▶ details
- ✓ **acquire[build]**
▶ details
- ✓ **data**
▶ details
- ✓ **preflight**
▶ details

How this works

1. The agent reads the KI before touching the software.
2. It tries, checks the actual error, repairs, and retries.
3. Licence gates, protected downloads, login, and system privileges come back to you here.
4. "Verified" appears only after the KI's real preflight passes.

Check that every line under **Verification** has a ✓. For FSM2 these are **materialise**, **python-env**, **ki-tools-common**, **system-deps**, **python-deps**, **acquire[build]**, **data** and **preflight**.

Each line under **Verification** has a mark:

Mark	Meaning
✓	Passed
↷ with "· repaired"	Failed earlier, then passed after the agent's repair
◇ with "· handled in each project"	A data step. Data belong to each project, so this does not block the software.
–	Skipped
✗	Failed

Click **details** under a line to read the check's output. For FSM2, the **preflight** details list each check as OK, WARN or FAIL. A WARN line, such as "WARN binary: gfortran", is advice and does not block verification.

After a successful setup:

- **Use KI in chat** creates a new chat with this KI chosen and the AI from this page. It skips the **New project chat** dialog (Chapter 6).
- **Re-check with agent** runs the check again later, for example after a Windows update. If the check passes, the AI is not used.

Back in the **KI Library**, the KI now looks like this:

FSM2 FSM2 (Flexible Snow Model) 2.1.x lineage; Essery et al. 2025 (GMD 18, 3583-3605) forest-canopy extension of FSM 1.0 (Essery 2015)

• KI Observatory • **KI** Create KI • Check & Import • KI updates • DeepSeek (API)

简体中文

KI FSM2
FSM2 (Flexible Snow Model) 2.1.x lineage; Essery et al. 2025 (GMD 18, 3583-3605) forest-canopy extension of FSM 1.0 (Essery 2015)

Bundled KI **Verified on this machine**

SCIENTIFIC SOFTWARE
fsm2 2.1.2

LAST CHECKED ON THIS MACHINE
2026/10/1 10:34:32

LANGUAGE & LICENCE
fortran · Not declared

SOURCE
[Open project website](#)

Use in a new chat **Open setup & verification**

Run data
Built from this KI's declarations and the files visible on this machine—not invented by AI. [Review declared inputs](#)

INPUTS
62 declared
Forcing 8 · Parameters 22 · Initial conditions 14 · Boundary & controls 18

KI
FSM2
Verified on this machine

OUTPUTS
17 declared
snw · snd · svg +14 more

Local data check • Dataset locations not declared

Local files are not machine-checkable yet. This KI has no dataset paths in its setup manifest. The declared input groups below still show the model interface, but GeoForge will not guess whether your files satisfy it.

1 of 127 KIs · 1 verified here

Check the badges **Bundled KI** and **Verified on this machine**, a date and time under **Last checked on this machine**, and the button **Open setup & verification** in place of **Set up with agent**. On a computer where only FSM2 is set up, the line under the list ends with **1 verified here**.

What "verified" means

Verified on this machine means one thing: after setup, the KI's own final check passed on this computer. GeoForge runs that check itself when the agent has finished. The agent saying "it works" is not enough.

For FSM2, the check confirms that:

- the four FSM2 tools of the KI are present;
- `FSM2.exe` exists and starts: with no input it reaches the point where FSM2 reads its settings, and stops there as expected;
- the FSM2 source folder, its compile script and the official Alptal example files are present;
- Python can load the numpy and pandas packages the tools need.

The check does not run a simulation. Other KIs check other things; read the **preflight** details to see what was checked.

Verified on this machine does not mean that your data are ready, that a plan is right, or that the model gives correct results for your site. What GeoForge guarantees for a project is that every approved step has a signed run record before it shows **Completed** (Chapter 5). Even a completed run of an official example is a forward run, not a validation against observations. Judging the science stays your job.

GeoForge saves the result as `status.json` in the installation folder and shows its time under **Last checked on this machine**.

Windows notes

What a setup may need, and where it comes from

Need	How it is handled	Example
The model's source code	The agent downloads it from the official source named in the KI, usually with Git. GeoForge does not include Git. Install Git for Windows (https://git-scm.com) before you set up a model that is built from source.	FSM2: a fixed version from GitHub
A compiler (Fortran, C, C++)	The agent downloads a private copy into the installation folder, as the KI's Windows notes describe. It needs no administrator rights and installs nothing for the whole computer.	FSM2: the WinLibs (MinGW-w64) compiler with gfortran
Python and Python packages	GeoForge looks for a Python 3 on your computer (step [2/6] of Test AI & GitHub , Chapter 2) and records the Python a model uses in <code>kiss.toml</code> in the installation folder. The agent installs the packages the KI needs. If GeoForge finds no Python, the agent has to provide one, for example a private copy in the installation folder, or ask you.	FSM2's tools need numpy and pandas. In the Windows test, the computer's own Python 3.13 was used.
Other build helpers	Private copies of tools named in the KI's Windows notes	micromamba for conda-forge packages (SUMMA, LISFLOOD and others), 7-Zip to unpack installers (APEX, Alpine3D and others), winflexbison (RHESSys)
Tools for the whole computer that the agent may not install	The agent asks you in Needs you	"Install the required build libraries"

All 127 KIs carry Windows installation notes from earlier test installations. 23 also carry a Windows install recipe: APEX, Alpine3D, CAESAR_Lisflood, CE_QUAL_W2, CRHM, Cell2Fire, DSSAT, Daisy, ESMF, FSM2, HYPE, HexWatershed, LISFLOOD, MARRMoT, RAPID, RHESSys, SUMMA, SWAP, SWAT_Plus, SimFire, TELEMAC_MASCARET, TRIGRS and TopoFlow.

Downloads by the agent, including Git, go through your proxy only if that AI is ticked under **Settings** → **Network & proxy** (Chapter 2).

GeoForge reads Windows' program search path only when it starts. After you install Git for Windows or another tool, exit GeoForge from the tray, start it again, and then continue the setup.

What the setup agent may change

GeoForge tells the setup agent to keep every file it writes inside the installation folder and to leave software outside it unchanged. With an API provider such as **DeepSeek (API)**, GeoForge also enforces this: it runs only known installation tools, and refuses a command whose file paths lead outside the installation folder. Anything that needs administrator rights or a change to the whole computer comes back to you as a request (on the **Agent setup** page, in **Needs you**).

GeoForge starts setup commands without console windows, so a setup normally opens no black windows. A command that waits for keyboard input fails instead of waiting forever.

The installation folder and disk space

Item	What it holds
binaries\	The model software. For FSM2: binaries\FSM2\source\repo\FSM2.exe
ki\	GeoForge's working copy of the KI for this computer
ki_tools_common\	GeoForge's shared helper tools, used by every KI
kiss.toml	The folders the KI uses and the Python the model uses
status.json	The verification record
.geoforge-install.json	The record of where this installation is
setup-agent.log	The log of setups started on the Agent setup page
user-files\	Files you handed to the agent
setup-request.json	The current Needs you request; earlier ones are kept with a number in the name
CLAUDE.md , AGENTS.md and similar files	The setup instructions for the agent

Private compilers and other tools are also stored in this folder. FSM2 took about 1.1 GB in the Windows test, almost all of it the compiler. Larger models can need several GB.

Leave this folder alone. Deleting it removes the software and its verification record, and the KI shows **Setup needed** again.

Windows verification scope in 0.6.55

The historical FSM2 walkthrough remains useful for the setup controls. Windows 0.6.55 also includes assisted installation and actual approved native runs of SHAW, VIC and CRHM; Chapter 11 records their inputs, versions and scientific output checks. These tests required repairs to compiler/runtime or KI integration details. They do not establish unattended installation or scientific validation of all KIs.

API-based software setup is confined to installation and preflight verification. Run scientific examples in a separate approved project workflow. Browser disconnection no longer ends the active turn; reopen GeoForge and the chat to inspect saved progress, and use **Stop** to stop work.

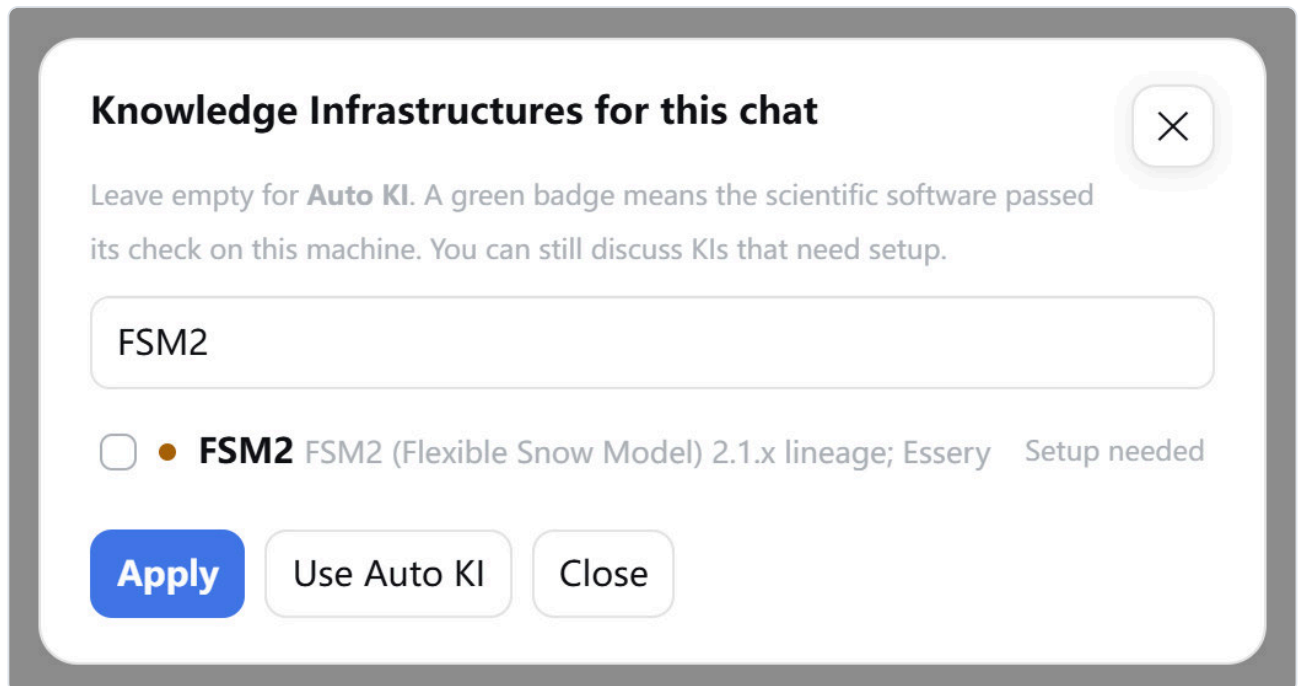
A September installation-only test found 91 of 127 KIs installable and able to start, 35 requiring a person, and one failure. That historical startup test is neither a local verification on your computer nor a check of scientific results. Protected downloads, logins, licences and large models designed for Linux can still require work from you.

Use a KI in a chat

Choose the KI for a chat

Choose the KI before the first message. After the first message, the KI is fixed for the chat, like its AI (Chapter 2).

1. Click **+ New chat** and create the chat (Chapter 6).
2. Click **Auto KI** in the chat header. The window **Knowledge Infrastructures for this chat** opens.
3. Type `FSM2` in **Search KIs....**



Check the label at the right of the FSM2 row. Here it reads **Setup needed**: you can still choose the KI and plan with it.

4. Click the FSM2 row to tick it.

5. Click **Apply**.

The header now shows **FSM2** and its state, as in the screenshot in "Let a project set up the software". If the software is not verified, the banner "... is not verified on this machine" appears above the chat.

To go back to automatic choice before the first message:

1. Click the KI button in the chat header. It now shows the KI name.
2. Click **Use Auto KI**. This only clears the ticks.
3. Click **Apply**.

You can tick several KIs for a coupled study. The header then shows the number of KIs and, for example, **1/2 verified on this machine**.

Tip: If a chat uses several KIs, set each one up from the **KI Library** before you approve the plan.

Let GeoForge choose: Auto KI

If you tick nothing, the header shows **Auto KI** and the pill **Chooses for each task**. GeoForge reads your first request, chooses the KI and announces it in the chat, for example "**Task understood · KI: FSM2**. Building the plan...".

If the chosen KI is not verified, its software is set up after you approve the plan, as described above. If GeoForge chose the wrong model, start a new chat and choose the KI yourself.

Other ways to start a chat with a KI

- **Use in a new chat** in the **KI Library**, with the AI chosen at the top of the **KI Library**.
- **Use KI in chat** on the **Agent setup** page, once the KI is verified.

Both skip the **New project chat** dialog (Chapter 6).

Keep KIs up to date

A KI library update replaces KI definitions: instructions, tools, checks and installation notes. It never changes installed model software, verification records, your projects or KIs you imported. A verified KI stays verified after an update.

Platform	How updates work
Windows	Only when you ask: KI Library → KI updates → Check again . In 0.6.54 the result is normally "GeoForge kept the current KI library because the repository version would remove Windows installation guidance.", and the button then reads Library kept . This is expected and safe.
macOS	Checked automatically at each launch

Chapter 7 describes the **KI library updates** window in full.

If something goes wrong

What you see	Cause	What to do
Message box "Type the full path directly in the installation field." after Choose folder	On Windows, GeoForge runs in your browser, which cannot open a folder picker for it.	Copy the path from File Explorer's address bar, paste it into the field and click Use location .
Message box "Type the full existing software path directly."	The same, for Use software already installed .	Paste the path and click Use path .
"the installation folder must be an absolute path"	The path has no drive letter.	Write the full path, for example <code>D:\GeoForge-Manual\models\fsm2</code> .
"choose a dedicated model folder, not the disk or home folder"	You entered a whole drive or <code>C:\Users\you</code> .	Use a folder of its own for this model.
"that folder is already assigned to ..."	Another KI uses that folder.	Give each KI its own folder.
"the installation folder is not writable: ...", or a message with "[WinError 5]"	Your account cannot write to that folder, for example one inside <code>C:\Program Files</code> .	Choose a folder on your own drive or in your user folder.
"the existing installation was not found: ..."	The path of the existing software is wrong.	Check the path in File Explorer, paste it again and click Use path .
"Choose the existing installation, or let the Agent find it."	Use software already installed is selected without a path.	Enter the path, or click Let Agent find and verify it .
The start button stays grey	Needs you shows a request, or no AI is ready ("— unavailable" in the list).	Answer the request first. If no AI is ready, connect one (Chapter 2).
Needs you shows Action needed with "Setup is not finished" or "Agent connection stopped"	The agent stopped before the check passed: "The agent stopped before verification passed. Continue the repair; GeoForge will show a specific request here if an external action is needed."	Click Continue repair . If it stops again in the same way, click Ask what went wrong .
The log ends with "GeoForge final check: FAIL" and "The check still fails. Continue the agent for another repair pass."	The software is not working yet.	Open details under the × line in Verification and read the FAIL lines. Click Continue repair .
The log ends with "setup agent failed: ..."	GeoForge hit an error while running the agent.	Click Continue repair . If it happens again, report it with <code>setup-agent.log</code> (Chapter 9, section 9.18).
"The live display was interrupted. The agent will continue in the background; this page will refresh from its saved log."	The page lost its connection: the tab was reloaded or closed, or GeoForge could not start the agent.	Watch the log for a minute. If it stops growing, click Continue repair . On Windows, keep the tab open during setup.
Needs you request "Install the required build libraries"	The build needs tools for the whole computer that are missing, for example	If the card shows a <code>pacman -S ...</code> command (it does when MSYS2 is installed), run it in PowerShell, then click I've done this — continue . Otherwise you can ask the

What you see	Cause	What to do
	gfortran . The agent may not install them itself.	agent to use a private copy instead: write that in Optional note for the agent and click I've done this — continue .
Needs you request "... cannot connect" (LOGIN)	The AI used for setup could not reach its service. The model installation itself has not failed. The text mentions "this Mac" and "AI Settings" even on Windows.	Check your network. If that AI needs a proxy, tick it in Settings → Network & proxy and click Save (Chapter 2). Then click I've done this — continue . To switch AI instead, choose another one in the AI list on the page first, then click I've done this — continue .
Message box "file larger than 300 MB" after Give file to agent	Files handed over this way are limited to 300 MB.	Copy the file yourself to the folder shown after "Needed at:", then click I've done this — continue .
A KI that was verified shows Setup needed again	You changed the installation folder, started GeoForge with another work folder, moved or deleted the installation folder, or you are on another computer or user account.	Set it up again with Set up with agent . To use the old copy, choose Use software already installed and point to it.
The KI shows Verification failed	The last check on this computer failed.	Click Set up with agent in the KI Library . On the Agent setup page, read the X line under Verification , then click Continue repair in Needs you (or the start button, which then reads Continue agent).
In a chat: "[could not prepare the FSM2 setup workspace: ...]"	GeoForge could not prepare the KI's installation folder.	Open the KI's Agent setup page and check the folder under Model installation workspace . Choose a folder you can write to, click Use location , then send your message again.
In a chat, setup ended without "Software verified. Starting your approved plan in a new session..."	The final check did not pass, or you clicked Stop .	Send a message in the same chat, for example "Please continue setting up FSM2."
Setup inside a project ran the model's example and took long	Historical 0.6.54 setup behaviour; current API setup is limited to installation and preflight.	Nothing to fix if the results stayed in the installation folder. If they landed in the project and block Completed , see Chapter 5, "If the project does not reach Completed".
A build program keeps running after Stop or Exit GeoForge (Windows)	Known issue 2 (Chapter 9, 9.16): a job started through Git Bash escaped.	End it in Task Manager (Chapter 9, section 9.10).
KI updates reports that the current KI library was kept	Expected on Windows in 0.6.54.	Nothing. Your KIs and your verified software are unchanged.

5 Your first project

In this chapter you will run one complete project, from a written request to **Completed**: the official FSM2 snow example for Alptal, Switzerland, winter 2004–2005. You will see every card GeoForge shows on the way, what to check on each, and how to read the results.

The screenshots come from a real run on Windows on 1 October 2026, with **DeepSeek (API)** and the model **deepseek-chat**. Your AI will word its questions and plan differently. The steps and the checks stay the same.

Before you start

You need	Where to set it up
An AI that is ready, here DeepSeek (API)	Chapter 2
The FSM2 KI. It ships with GeoForge.	Nothing to do
The FSM2 software verified on this computer	Optional: if it is not, GeoForge sets it up after you approve the plan (Chapter 4). That adds about five minutes the first time.
GeoForge Database access	Not needed: the example uses the forcing data that ships with FSM2.

How long it takes, in the run shown here (FSM2 already verified):

Moment	Minutes after Send
First planning question appears	1
Approval card appears	3
Completed	4

Each planning turn and the run use your AI. With DeepSeek this example costs little, but the number of turns varies with the AI.

Historical screenshot note: this FSM2 walkthrough was captured in 0.6.54. Its Windows browser-reload interruption was fixed in 0.6.55; browser reconnection no longer cancels the running turn.

Step 1: Create the project chat

1. Click **+ New chat** at the top of the sidebar.
2. In **Project name**, type `Alptal snow example` .
3. In **Create the project inside**, keep the default folder or type another one, for example `D:\GeoForge-Manual\projects` .

New project chat

Each chat has its own project folder for memory, inputs, model runs, outputs, and plots.

Project name

Alptal snow example

Create the project inside

D:\GeoForge-Manual\projects Choose...

Project folder (created automatically)

D:\GeoForge-Manual\projects\2026-10-01-Alptal-snow-example--{id}

Edit the suggested name or keep it. GeoForge creates the folder automatically, including missing parent folders. {id} becomes a unique ID so projects with the same name stay separate. Existing files are not changed.

Create chat Use default location Cancel

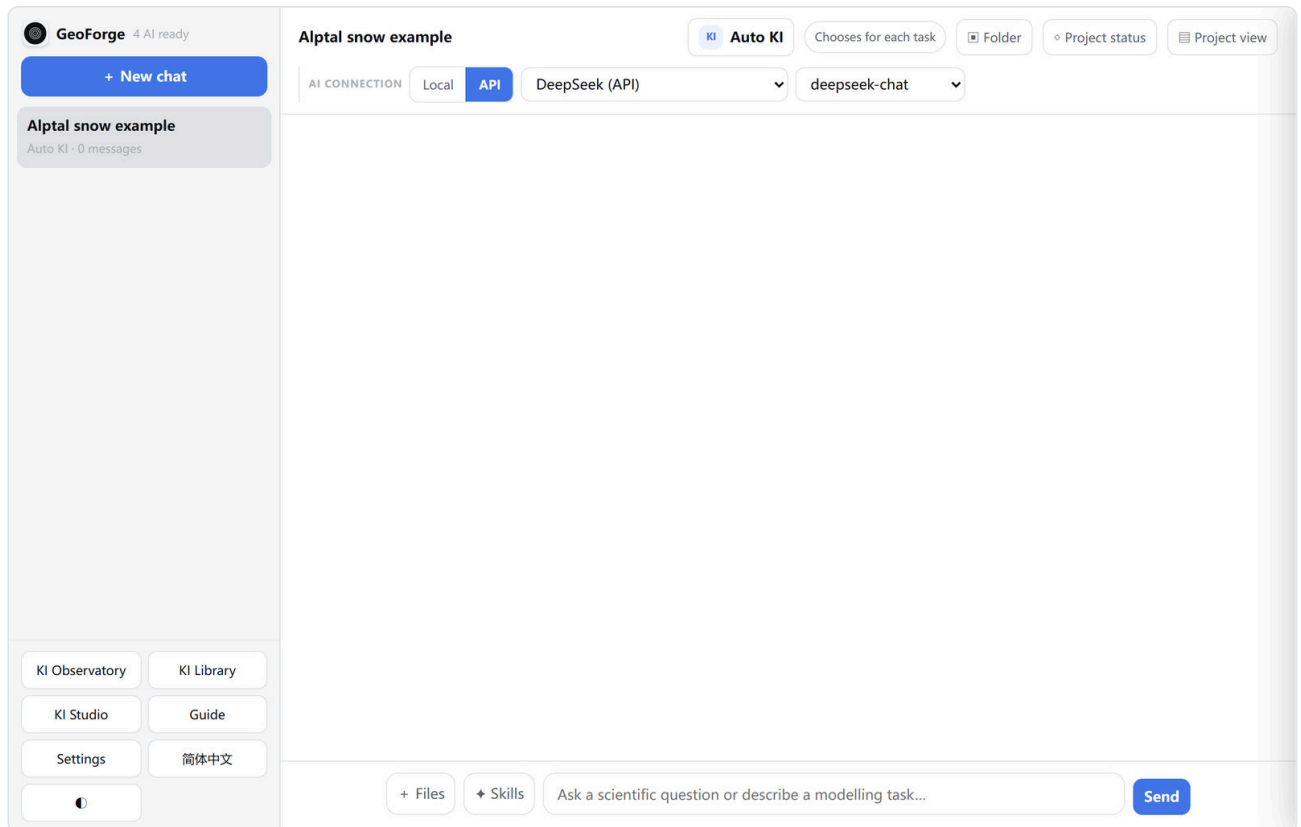
Check that **Project folder (created automatically)** shows where the project will be created.

4. Click **Create chat**.

Chapter 6 explains the dialog and the project folder in detail.

Step 2: Choose the AI for this chat

1. Under **AI connection**, click **API**.
2. In the first list, choose **DeepSeek (API)**.
3. In the second list, choose **deepseek-chat**.



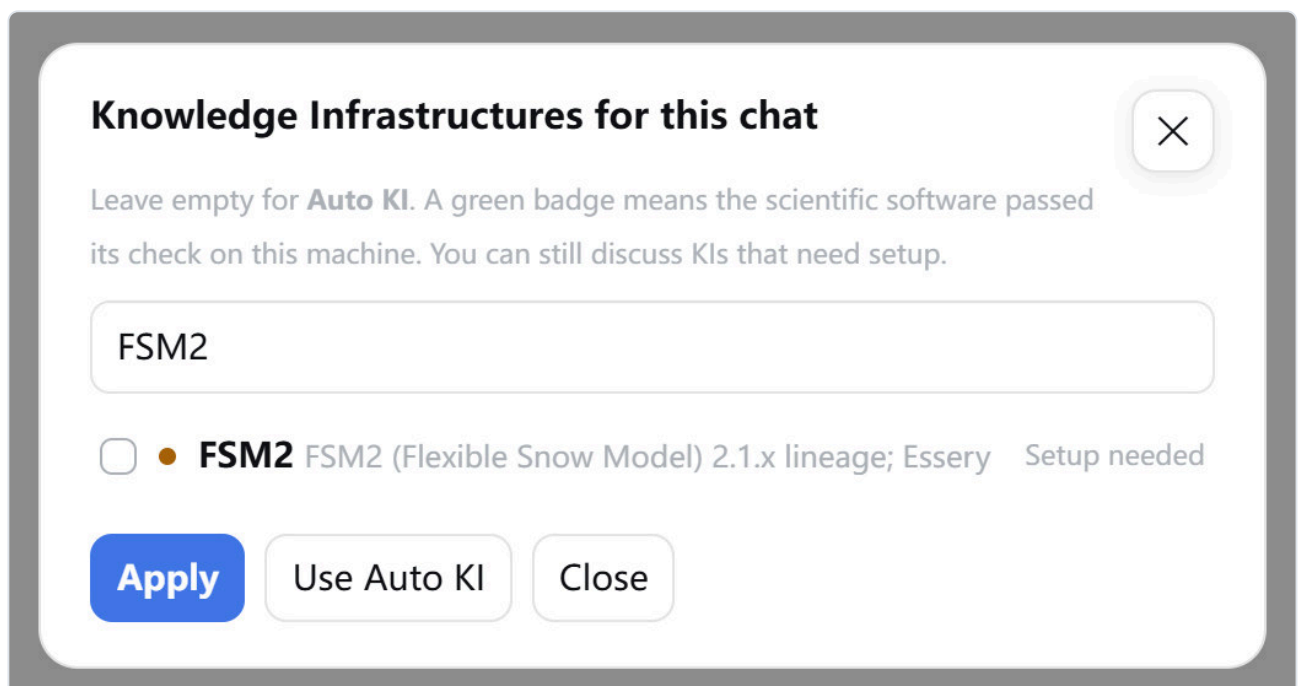
Check that the chat name appears at the top and in the sidebar, with **Auto KI · 0 messages** under it.

These choices are fixed once you send the first message. To use another AI later, start a new chat.

Step 3: Choose the KI

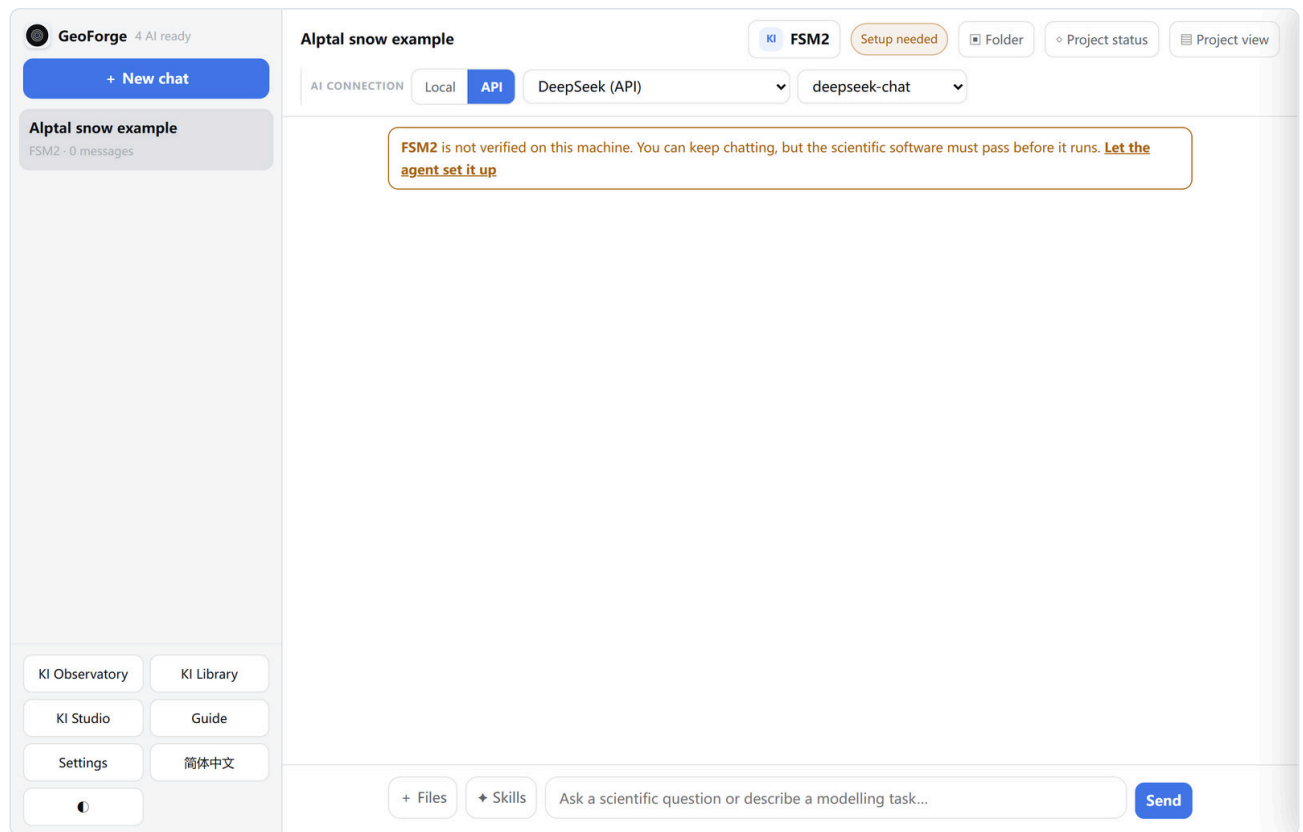
You can leave **Auto KI**, and GeoForge chooses the model from your request. Choosing it yourself is more predictable:

1. Click **Auto KI** in the chat header.
2. Type `FSM2` in **Search KIs....**



3. Click the FSM2 row to tick it.

4. Click **Apply**.

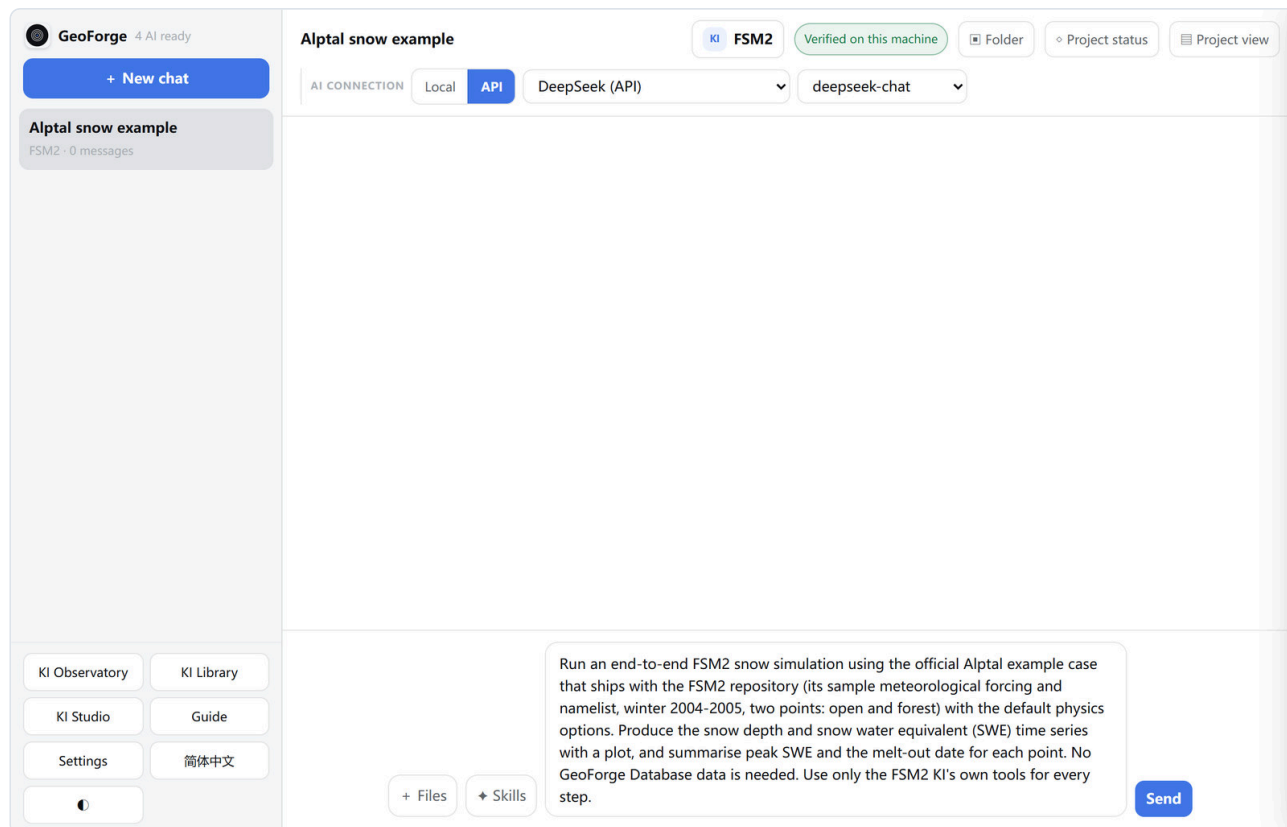


Check that the header shows **FSM2** and its state. On a new computer it reads **Setup needed**, with the banner "FSM2 is not verified on this machine. You can keep chatting, but the scientific software must pass before it runs." That is fine: GeoForge sets the software up after you approve the plan.

Step 4: Write the request

1. Click in the message box (**Ask a scientific question or describe a modelling task...**).
2. Type or paste the request:

Run an end-to-end FSM2 snow simulation using the official Alptal example case that ships with the FSM2 repository (its sample meteorological forcing and namelist, winter 2004-2005, two points: open and forest) with the default physics options. Produce the snow depth and snow water equivalent (SWE) time series with a plot, and summarise peak SWE and the melt-out date for each point. No GeoForge Database data is needed. Use only the FSM2 KI's own tools for every step.



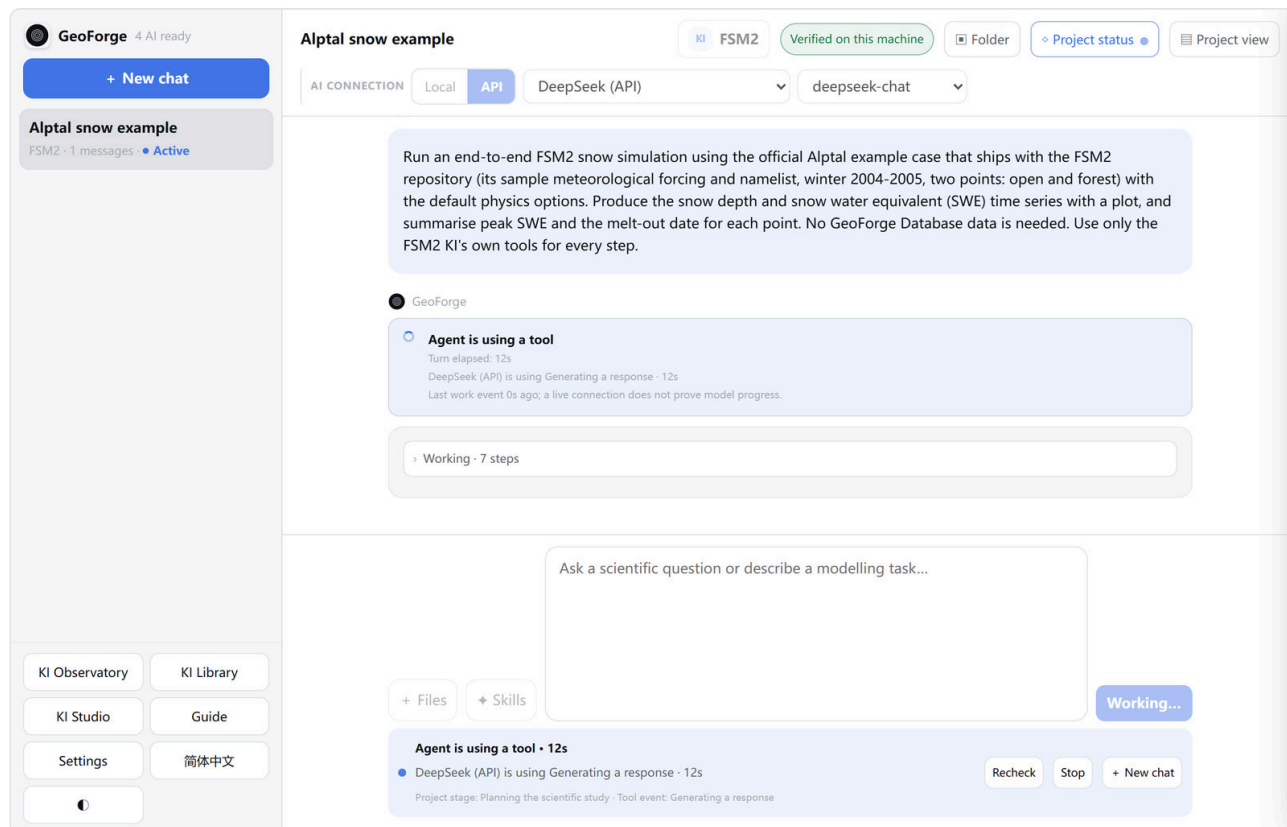
A good request names:

What	In the example
The model and the case	FSM2, the official Alptal example
Place and period	Alptal, winter 2004–2005, an open and a forest point
Settings that matter to you	the default physics options
The outputs you want	snow depth and SWE series, a plot, peak SWE and melt-out date per point
Data you have, or do not want	the shipped forcing; no GeoForge Database data
How the work should be done	only the KI's own tools

Tip: Ask for steps that use the KI's own tools. In the first run for this manual, the plan included a step "written by the agent" (no tool), and the project could not reach **Completed** (see "If the project does not reach Completed" below).

Step 5: Send, and let the AI plan

1. Click **Send**.



Check that **Send** reads **Working...**, and that the activity bar under the message box shows what the AI is doing, for example **Agent is using a tool** with the elapsed time and the line "Project stage: Planning the scientific study".

While the AI plans, GeoForge lets it read the KI and your project, and write a draft plan. It cannot download, install or run anything yet. The activity bar has three buttons:

Button	What it does
Recheck	Refreshes what the bar shows about the AI's activity
Stop	Ends the current turn. Send a message to continue later.
+ New chat	Starts another chat. This one keeps working.

Step 6: Answer the planning questions

When the AI needs a decision from you, a card opens with the label **Needs you**. In this run there was one question: which FSM2 physics options to use.

?

FSM2 physics options for the Alptal example run

Needs you

×

The run period (winter 2004-2005), the two points (open + forest) and the data source (the official example forcing ``met_Alptal_0405.txt`` shipped beside the installed binary) are all settled by your request and by the KI — no further decision needed there.

One choice remains: which compile-time physics option set the run should use. FSM2's physics switches are compile-time ``#define``'s, so this determines which binary we run.

EVIDENCE FOR THE RECOMMENDED OPTION

``kiss.windows.yaml`` (agent_hint) records the exact option set written into ``src/OPTS.h`` when the machine's VERIFIED ``FSM2.exe`` was built, and states it matches the upstream ``compil.sh``:

ALBEDO=2, CANINT=1, CANMOD=1, CANRAD=1, CANUNL=1, CONDUCT=1, DENSITY=1, EXCHNG=1, HYDROL=1, SGRAIN=1, SNFRAC=1, DRIV1D=1, SWPART=0, ZOFFST=0, PROFNC=0.

``tools/run_fsm2.py`` ``DEFAULT_OPTIONS`` declares the identical set, and ``docs/03_compilation.md`` lists ALBEDO=2, CONDUCT=1, DENSITY=1, EXCHNG=1, HYDROL=1 as "Recommended". This is the configuration the KI's ``example_validated`` Alptal result and its recorded 5,832-row execution were produced with — so it is both "the default physics options" you asked for AND the only set that is machine-verified here.

WHAT EACH OPTION WOULD COST

- Option A needs no compilation and no gfortran: it uses the already-built, hash-verified binary. Fastest, and matches the KI's validated example.
- Option B recompiles from the pinned source (commit f0c86be1, gfortran 16.2.0) and produces a NEW binary. ``SKILL.md`` warns that ``run_fsm2.compile_fsm2`` overwrites ``src/OPTS.h`` in the source tree — so a recompile alters the shared installation's physics unless done in a copied

Check the title, here "FSM2 physics options for the Alptal example run", and the explanation under it. The AI explains the evidence it used and what each choice would cost.

1. Read the question and its explanation. Scroll down in the card to see all of it.
2. Click the option you want. Here: **A — Use the verified pre-built FSM2.exe with its default physics options (recommended).**

×

The option set above rests on `kiss.windows.yaml` + `install.windows.md` + the tool default, not on reading `OPTS.h` from disk. I will confirm this at execution. I have also not yet confirmed that `met\_Alptal\_0405.txt` and `nlst\_Alptal.txt` are physically present at the installed repo path — that is a file existence check for the acquisition step, and if they are absent the fallback is re-running the pinned build step, not fabricating forcing.

This run is the KI's official example case; per `docs/validation\_convention.yaml` and SKILL.md §11 it is the source of the KI's `example\_validated` status, not a validation against observations for your site. A separate decision about observational validation can be asked later if you want it — it is deliberately not bundled here.

☒
A — Use the verified pre-built FSM2.exe with its default physics options (recommended)

Run the already-built, machine-verified binary. Physics = the official default set recorded in kiss.windows.yaml / run_fsm2.py DEFAULT_OPTIONS (ALBEDO=2, CANINT=1, CANMOD=1, CANRAD=1, CANUNL=1, CONDUCT=1, DENSTY=1, EXCHNG=1, HYDROL=1, SGRAIN=1, SNFRAC=1, DRIV1D=1, SWPART=0, ZOFFST=0, PROFNC=0). No compilation, no toolchain needed; reproduces the KI's recorded 5,832-row Alptal example.

☐
B — Recompile from the pinned source with the same default options, in a copied work directory

Build a fresh binary from commit f0c86be1 with the same official option set, using the staged gfortran 16.2.0. Adds a compile step and copies source/repo/src first so the shared installation's OPTS.h and binary are not modified. Physically identical results to A.

☐
C — Recompile with a different physics option set (I will specify which switches)

Choose non-default switches, e.g. SWPART=1 (solar-position diffuse/direct split, needs a correct lon-derived `noon` in &drive), CANMOD=2 (two canopy layers), or SNFRAC=2/3. This is no longer the KI's default configuration and its output cannot be compared directly to the KI's recorded example validation.

Check that the option you chose is highlighted.

3. Scroll to the bottom of the card and click **Continue with this choice**.

Question cards end with these controls. The approval card (Step 7) has the same ones except **Use my own answer / file**.

Control	Use it to
Use my own answer / file	Answer in your own words, or give a file location. Write the answer in Optional note for the agent .
Optional note for the agent	Add context to your choice
Continue with this choice	Send your answer. The AI continues planning.
Ask about these choices	Ask the AI to explain before you decide
Not now or ✕	Close the card without answering. Reopen it with ◇ Project status → Respond .

Answer on the card, not by typing in the chat: a typed reply is not recorded as your answer.

Other runs may ask more questions, for example which evaluation you want or where outputs should go. Each is answered the same way.

Step 7: Read the plan

When planning is finished, the card **Approve the plan?** opens. Its title and options stay in English in Chinese mode.

?
Approve the plan?
Needs you
X

WHAT I UNDERSTOOD

Run an end-to-end FSM2 snow simulation using the official Alptal example case that ships with the FSM2 repository (its sample meteorological forcing and namelist, winter 2004-2005, two points: open and forest) with the default physics options. Produce the snow depth and snow water equivalent (SWE) time series with a plot, and summarise peak SWE and the melt-out date for each point. No GeoForge Database data is needed. Use only the FSM2 KI's own tools for every step.

KI · FSM2

area · Alptal, Switzerland (47.05 N, ~8.7 E) — the FSM2 upstream example site, one open point and one forested point

period · 2004-12-01 00:00 to 2005-09-01 00:00 (winter 2004-2005; 5832 hourly records)

DATA • 10

► The run prepares itself (10) already on disk · made by a step

STEPS • 3

- FSM2:s6_run_official_example run · run_fsm2.py env 4
- FSM2:s7_parse_official_example postprocess · parse_fsm2_output.py env 5
- FSM2:postflight_check check · run by GeoForge (preflight)

SCIENTIFIC DECISIONS

- physics_options: A: verified pre-built FSM2.exe with the upstream default physics options (ALBEDO=2, CANINT=1, CANMOD=1, CANRAD=1, CANUNL=1, CONDUCT=1, DENSTY=1, EXCHNG=1, HYDROL=1, SGRAIN=1, SNFRAC=1, DRIV1D=1, SWPART=0, ZOFFST=0, PROFNC=0) / B: recompile from source with a different switch combination (factorial experiment) → **Use the verified pre-built FSM2.exe with default physics options; no recompilation.** decided

From your answer: FSM2 physics options for the Alptal example run → A — Use the verified pre-built

Read it from the top:

Section	What to check
What I understood	It matches your request. The chips below show the KI, and the area and period when the AI states them.
Data	Every input. You (...) lists inputs you must provide. The run prepares itself (...) lists inputs that are already on disk or made by a step: click it to see them. Here all 10 are prepared by the run.
Steps	Each step, its kind (run , postprocess , check , ...) and the KI tool it uses, here <code>run_fsm2.py</code> and <code>parse_fsm2_output.py</code> . run by GeoForge (preflight) means GeoForge runs the KI's own check.
Scientific decisions	The choices behind the plan, with "From your answer: ..." under the ones you made
Waiting on you	Choices that will take a default unless you pick another. Read them.
Not ready to execute yet	Steps that cannot run as planned, for example "step 's3_config' has no tool — not ready to execute".

STEPS • 3

1. FSM2:s6_run_official_example

run

• run_fsm2.py

env 4

2. FSM2:s7_parse_official_example

postprocess

• parse_fsm2_output.py

env 5

3. FSM2:postflight_check

check

• run by GeoForge (preflight)

SCIENTIFIC DECISIONS

physics_options: A: verified pre-built FSM2.exe with the upstream default physics options (ALBEDO=2, CANINT=1, CANMOD=1, CANRAD=1, CANUNL=1, CONDUCT=1, DENSTY=1, EXCHNG=1, HYDROL=1, SGRAIN=1, SNFRAC=1, DRIV1D=1, SWPART=0, ZOFFST=0, PROFNC=0) / B: recompile from source with a different switch combination (factorial experiment) → Use the verified pre-built FSM2.exe with default physics options; no recompilation.

decided

From your answer: FSM2 physics options for the Alptal example run → A — Use the verified pre-built FSM2.exe with its default physics options (recommended)

Text summary and tool policy

Approve and start

Execution starts in a fresh session with the approved plan; every run and download gets a receipt.

Modify the plan

Tell me what to change in the note; I will revise the plan and ask again.

Optional note for the agent

Add context only if needed...

Continue with this choice

Ask about these choices

Not now

Check the lower part: **Text summary and tool policy** (click to open), the options **Approve and start** and **Modify the plan**, and **Optional note for the agent**.

Look out for two things in particular:

- A step marked **written by the agent** or **no tool — cannot execute**. Such a step can never get a run record. If it writes result files, the project cannot reach **Completed**.
- A **Not ready to execute yet** section. GeoForge still lets you approve, but the run is likely to stop. In the third run for this manual, approving such a plan ended in a failed validation.

In both cases choose **Modify the plan** (Step 8). The plan in the screenshots has three steps, all with a tool or run by GeoForge, and no **Not ready to execute yet** section: it is ready.

Caution: Also compare **Data** with any data-source choice before you approve (Chapter 9, 9.16, issue 1). It does not apply to this example, which downloads nothing.

Step 8: Approve, or ask for changes

To approve:

1. Click **Approve and start**.

STEPS • 3

1. FSM2:s6_run_official_example

run

• run_fsm2.py

env 4

2. FSM2:s7_parse_official_example

postprocess

• parse_fsm2_output.py

env 5

3. FSM2:postflight_check

check

• run by GeoForge (preflight)

SCIENTIFIC DECISIONS

• physics_options: A: verified pre-built FSM2.exe with the upstream default physics options (ALBEDO=2, CANINT=1, CANMOD=1, CANRAD=1, CANUNL=1, CONDCT=1, DENSTY=1, EXCHNG=1, HYDROL=1, SGRAIN=1, SNFRAC=1, DRIV1D=1, SWPART=0, ZOFFST=0, PROFNC=0) / B: recompile from source with a different switch combination (factorial experiment) → **Use the verified pre-built FSM2.exe with default physics options; no recompilation.**

decided

From your answer: FSM2 physics options for the Alptal example run → A — Use the verified pre-built FSM2.exe with its default physics options (recommended)

► Text summary and tool policy

☒

Approve and start

Execution starts in a fresh session with the approved plan; every run and download gets a receipt.

☐

Modify the plan

Tell me what to change in the note; I will revise the plan and ask again.

Optional note for the agent

Add context only if needed...

Continue with this choice

Ask about these choices

Not now

2. Click **Continue with this choice**.

GeoForge records your approval with a fingerprint of the plan. From now on the AI may run only the approved steps with the approved tools, and every run gets a signed run record.

To ask for changes instead:

1. Click **Modify the plan**.
2. In **Optional note for the agent**, write what to change. When the card shows **Not ready to execute yet**, you can write: "Some steps cannot run yet (see Not ready to execute yet). Please revise the plan so that every step uses one of the KI's own tools, and drop any step this example does not need."
3. Click **Continue with this choice**.

The AI revises the plan and shows a new **Approve the plan?** card. Read it again from the top.

Step 9: Software setup, if needed

If FSM2 is not yet verified on this computer, GeoForge sets it up first, in the same turn. The chat shows the setup work, and
 ◇ **Project status** shows **Prepare** as the active stage with **0 / 1 software verified**.

GeoForge 4 AI ready

+ New chat

Alptal snow example

FSM2 · 9 messages · Running · quiet

KI Observatory

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简体中文

Alptal snow example

AI CONNECTION

Local

API

DeepSeek (API)

deepseek-chat

FSM2

Setup needed

Folder

Project status

Project view

FSM2 is not verified on this machine. You can keep chatting, but the scientific software must pass before it runs. [Let the agent set it up](#)

- Spin-up:** the example cold-starts from no snow, so the first weeks of pack evolution are initialisation-transient — inherited from the official case, not a defect introduced here.
- Period:** winter 2004-2005 with 5,832 hourly records is evidenced from the recorded installation test (docs/install.windows.md , kiss.windows.yaml), not from the file itself, so the first and last timestamps will be read from the driving file at execution time and any discrepancy reported rather than trimmed.

GeoForge now shows the plan for approval; the installation verification, model run and outputs happen in the next session.

You

For "Approve the plan?", I choose: Approve and start.
Decision: Approved. Start the execution.
Please continue using this choice.

GeoForge

Agent is using a tool

Turn elapsed: 24s

DeepSeek (API) process is alive in Generating a response; no new event for 13s · 24s

+ Files

+ Skills

Ask a scientific question or describe a modelling task...

Working...

Agent is using a tool · 24s

DeepSeek (API) process is alive in Generating a response; no new event for 13s · 24s

Project stage: Checking scientific software · Tool event: Generating a response

Recheck

Stop

+ New chat

Project status

GeoForge is working

0 / 1 software verified

Agent turn active; this alone does not prove model progress.

Goal

Run an end-to-end FSM2 snow simulation using the official Alptal example case that ships with the FSM2 repository (its sample meteorological forcing and namelist, winter 2004-2005, two points: open and forest) with the default physics options. Produce the snow depth and snow water equivalent (SWE) time series with a plot, and summarise peak SWE and the melt-out date for each point. No GeoForge Database data is needed.

General KI: FSM2

1 Understand

Understand the goal and choose a general KI

2 Prepare

Check software and prepare suitable data

3 Validate

Check the prepared model inputs

4 Run

Run the real scientific model

5 Results

Save and explain the results

Flow and signed evidence · observed 33s ago

Data in this plan · 6

0/6 · Waiting for your input

From the submitted data inventory; status comes from receipts and files on disk, not from the agent's words.

Continue in chat

Upload other files

Check that **Prepare** is the active stage and that the pill reads **0 / 1 software verified**.

Wait until the chat says "**Software verified. Starting your approved plan in a new session...**". In the first run for this manual, setup took about five minutes. Chapter 4 describes setup in full.

In 0.6.55, API software setup is restricted to installation and bounded startup checks. The approved scientific execution follows a successful preflight. Unrecorded results from an older attempt still cannot establish completion.

Step 10: The run

GeoForge starts the approved steps in a fresh AI session that has only the approved plan.

The screenshot displays the GeoForge application interface. On the left sidebar, there's a 'New chat' button and a chat titled 'Alptal snow example' with 5 messages and an 'Active' status. The main chat area shows a 'Scope note on validation' and an 'Inventory summary' for the 'Alptal snow example'. The chat history shows a user message: 'For "Approve the plan?", I choose: Approve and start. Decision: Approved. Start the execution. Please continue using this choice.' and a GeoForge response: 'Agent is using a tool'. The bottom bar shows 'Project stage: KI execution: prepare inputs and run the model' and 'Tool event: Generating a response'. The 'Project status' button is highlighted with a red dot.

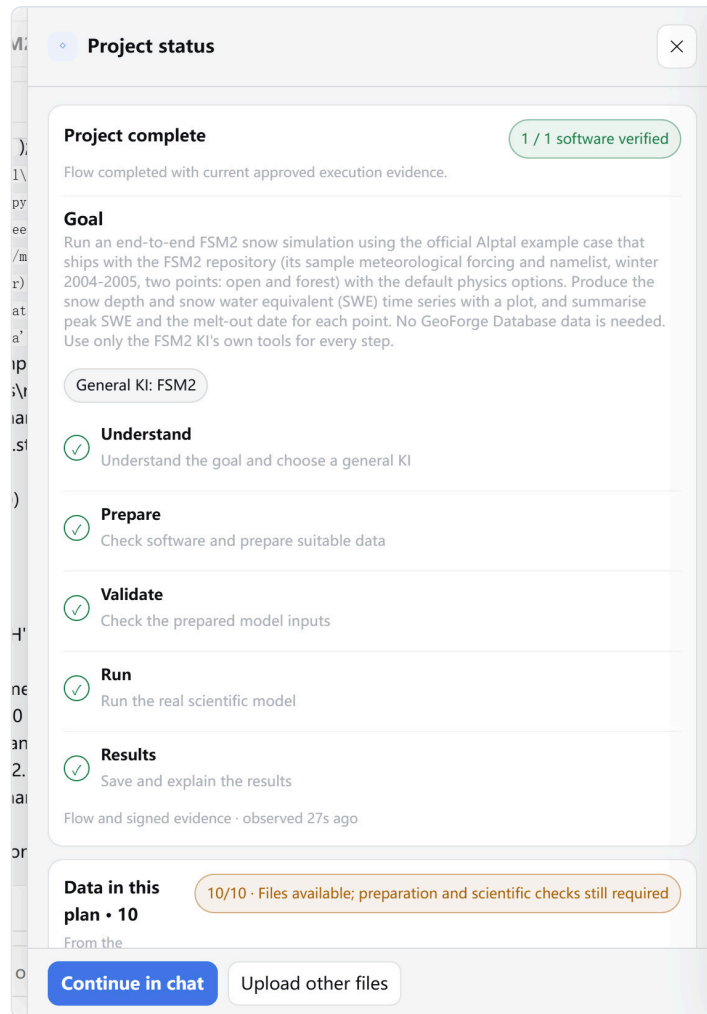
Check that the activity bar shows "Project stage: KI execution: prepare inputs and run the model". The **Project status** button may show a red dot for a moment if an attempt fails; a passing retry replaces it. What counts is the state at the end.

Browser disconnection or closing a tab no longer ends the active turn in 0.6.55. Reopen GeoForge and the chat to inspect its saved progress; use **Stop** to stop work.

Step 11: Check Completed

When the turn ends, the chat ends with the AI's summary and a line from GeoForge itself, starting with **Completed**.

1. Click **Project status** in the chat header.



Check that:

- the headline reads **Project complete**, with "Flow completed with current approved execution evidence.";
- all five stages, **Understand**, **Prepare**, **Validate**, **Run** and **Results**, have a tick;
- the pill on the right reads **1 / 1 software verified**.

2. Scroll down to **Data in this plan**. Here it reads "10/10 · Files available; preparation and scientific checks still required". GeoForge has the files; judging them scientifically is still your job.

3. Click X to close the panel.

If the **Completed**. line also says that earlier failed attempts were superseded by a passing retry, that is normal. In this run the AI first called the tools with `--help` and with malformed arguments; those attempts failed, the correct runs passed, and every attempt stays in the run history (`runs\evidence.json`).

Step 12: Read the results

In the chat

Scroll up to the AI's summary. It describes what ran, with the run record of each step:

GeoForge4 AI ready

+ New chat

Alptal snow example

FSM2 · 6 messages

KI Observatory

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Alptal snow example

FSM2

Verified on this machine

Folder

Project status

Project view · 1

AI CONNECTION

Local

API

DeepSeek (API)

deepseek-chat

What actually ran

Two KI tools, both by absolute path, both with signed receipts:
1. tools/run_fsm2.py --run-only with the verified pre-built FSM2.exe and the unmodified upstream nlst_Alptal.txt piped to stdin → FSM2 run completed successfully, validation passed. Receipt FSM2_20261001T110521_ddd0f5 .
2. tools/parse_fsm2_output.py --runid Alptal_ --npnts 2 --metrics --plot → validation passed. Receipt FSM2_20261001T110524_aacd6f .

Phase 1 of the plan's output-frequency choice was honoured: peak SWE and melt-out were computed on the model's native hourly records. --npnts 2 was mandatory — FSM2 writes multi-point columns variable-major. convert_forcing_to_fsm2.py and convert_soil_params.py were deliberately not used: the shipped forcing already provides Sf and Rf as two independent columns plus RH in percent (so triplets T19/T20 don't apply), and the official example does not consult HWSD.

The model wrote 5832 rows in each of Alptal_flux.txt , Alptal_stat.txt , Alptal_subc.txt , plus the restart Alptal_dump . Preflight passed 14/15 — the only FAIL is the optional gfortran compiler, which this run never needs since no compilation stage exists.

Results (from metrics.csv)

Point	Peak SWE [kg m ⁻²]	Peak SWE time	Peak depth [m]	Melt-out	Status
1 open	348.23	2005-03-17 08:00	1.138	2005-04-05 12:00	melted out
2 forest	155.68	2005-03-15 09:00	0.579	2005-03-27 14:00	melted out

Melt-out is the parser's own rule: first hourly depth == 0 after the SWE peak. Both points reached that, and both are 0.0 kg m⁻² / 0.0 m at the record end (2005-06-01 00:00), so no date was invented and no "snow remains" status was needed.

+ Files

+ Skills

Ask a scientific question or describe a modelling task...

Send

Then the results:

GeoForge4 AI ready

+ New chat

Alptal snow example

FSM2 · 6 messages

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Project view · 1

AI CONNECTION

Local

API

DeepSeek (API)

deepseek-chat

Results (from metrics.csv)

Point	Peak SWE [kg m ⁻²]	Peak SWE time	Peak depth [m]	Melt-out	Status
1 open	348.23	2005-03-17 08:00	1.138	2005-04-05 12:00	melted out
2 forest	155.68	2005-03-15 09:00	0.579	2005-03-27 14:00	melted out

Melt-out is the parser's own rule: first hourly depth == 0 after the SWE peak. Both points reached that, and both are 0.0 kg m⁻² / 0.0 m at the record end (2005-06-01 00:00), so no date was invented and no "snow remains" status was needed.

The forest point peaks at roughly 45 % of the open point's SWE and melts out 9 days earlier — the expected signature of canopy interception and reduced sub-canopy radiation at VAI 3.96. Peak depth leads peak SWE at both points, i.e. the pack is compacting while it is still gaining mass.

FSM2 snow depth and snow water equivalent

+ Files

+ Skills

Ask a scientific question or describe a modelling task...

Send

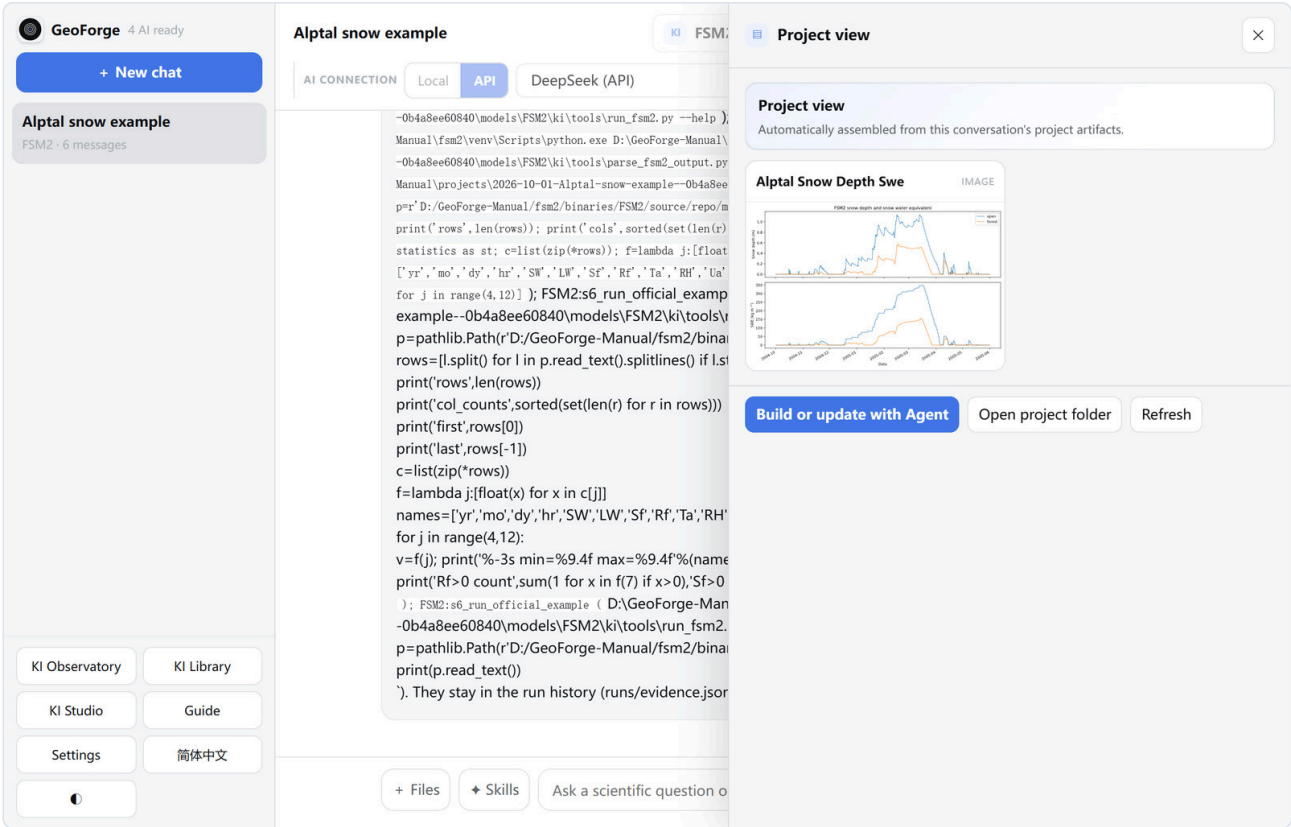
In this run:

Point	Peak SWE (kg m ⁻²)	Peak SWE time	Peak depth (m)	Melt-out
1 open	348.23	2005-03-17 08:00	1.138	2005-04-05 12:00
2 forest	155.68	2005-03-15 09:00	0.579	2005-03-27 14:00

The forest point holds about 45 % of the open point's peak SWE and melts out nine days earlier. The first run for this manual produced the same numbers, which is what a reproduction of an official example should do.

In Project view

- Click  **Project view** in the chat header. The number after it counts the panels.



Check the panel **Alptal Snow Depth SWE** with the two-part plot: snow depth above, SWE below, open and forest.

Button	What it does
Build or update with Agent	Asks the AI to add or update panels
Open project folder	Opens the project folder in File Explorer (Finder on a Mac)
Refresh	Reloads the panels from the project files

In the project folder

Click  **Folder** in the chat header. In this run the results are in:

File	What it holds
outputs\fsm2_alptal\Alptal_flux.txt , Alptal_stat.txt , Alptal_subc.txt	FSM2's own hourly output, 5,832 rows each
outputs\fsm2_alptal\Alptal_dump	FSM2's state at the end of the run
outputs\fsm2_alptal\series.csv	Snow depth, SWE and the other variables for both points, hourly
outputs\fsm2_alptal\metrics.csv	Peak SWE, peak depth and melt-out date per point
artifacts\alptal-snow-depth-swe.svg	The plot
runs\	The plan, your approval, the evidence summary and the run logs

Chapter 6 explains the folder in full.

What Completed means here

Completed means: every step of the plan you approved ran through GeoForge, each has a signed run record, the outputs passed GeoForge's automatic checks (they exist, are not empty, contain no NaN or infinite values), and no result file exists that no passing run produced.

It does not mean the model is right for Alptal. This was a forward run of FSM2's own example with its own forcing. No observations were compared, so no skill score exists, and the AI's summary says so. To evaluate the model, plan a new project that includes observations and an explicit comparison.

After **Completed** you can still ask the AI about the results in this chat. It can no longer run, download or change inputs here. For another period, another site or other physics options, start a new chat.

If the project does not reach Completed

The first project for this manual did not reach **Completed**, although the model ran correctly. Its story shows the three things to look for.

1. **Files without a run record.** The setup agent had run the example inside the project (known issue 3), and the plan contained a step "written by the agent" whose audit files had no run record. GeoForge listed five such files and kept the project open.

Project status

Ready to continue

1 / 1 software verified

Project unfinished; no active agent turn is observed.

Goal

Run an end-to-end FSM2 snow simulation using the official Alptal example case that ships with the FSM2 repository (its sample meteorological forcing and namelist, winter 2004-2005, two points: open and forest) with the default physics options. Produce the snow depth and snow water equivalent (SWE) time series with a plot, and summarise peak SWE and the melt-out date for each point. No GeoForge Database data is needed.

General KI: FSM2

1 Understand

Understand the goal and choose a general KI

2 Prepare

Check software and prepare suitable data

3 Validate

Check the prepared model inputs

4 Run

Run the real scientific model

5 Results

Save and explain the results

Flow and signed evidence · observed 12m 58s ago

Data in this plan · 6

2/6 · Waiting for your input

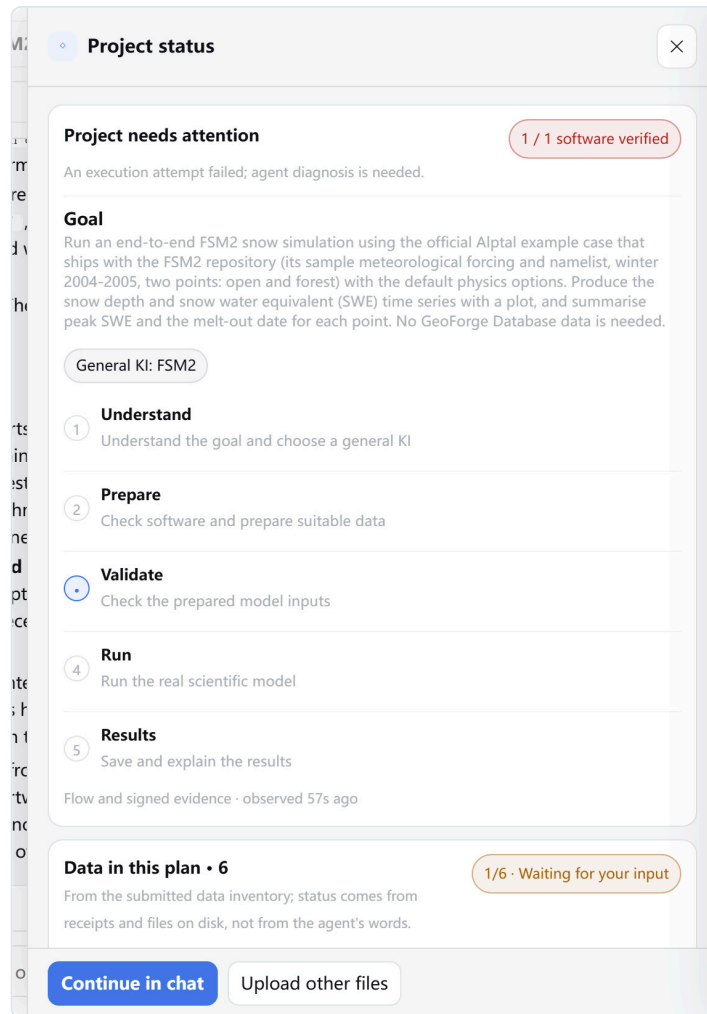
From the submitted data inventory; status comes from receipts and files on disk, not from the agent's words.

Continue in chat

Upload other files

Check the headline: **Ready to continue**, not **Project complete**. In 0.6.54 the chat did not say why, because the run came straight after setup (fixed in 0.6.55).

2. **A failed later attempt.** Asked to finish, the AI reran the model step in a scratch folder without naming its outputs. That attempt failed validation, and the latest attempt of each step is the one that counts.



Check the headline **Project needs attention** with "An execution attempt failed; agent diagnosis is needed."

3. **The AI's words are not evidence.** In its next reply the AI wrote that validation had passed. The run records said otherwise, and **Project status** kept showing the failure.

What to do in such a case:

1. Open ◇ **Project status** and read the headline and the status line. Chapter 9, 9.11 lists every message and its fix.
2. Send: "GeoForge says the project is not Completed yet. Please read runs/evidence.json, re-run every approved step whose latest attempt failed exactly as approved, and regenerate or remove any file that has no passing run record. Do not change the approved science or inputs."
3. After the reply, check ◇ **Project status** again rather than trusting the summary.
4. If files without a run record remain, and they are not your own inputs, move them out of `outputs\`, `artifacts\`, `inputs\` and `calibration\`, for example into a new folder `notes\` in the project, then send a message so GeoForge checks again.
5. If the project still does not complete, start a new chat with a clearer request (as in Step 4) and the software already verified. That is what produced the run in this chapter.

If something goes wrong

What you see	What to do
Send stays grey	Under AI connection , choose API or Local and a provider that is ready (Chapter 2).
No card appears after planning, and the chat ends with "Plan was not submitted for approval: - The provider failed or was interrupted. Your draft is preserved; retry planning."	Send "Please retry planning from your saved draft." On Windows, keep the tab open while the AI works.
"GeoForge could not finish this turn: ConnectionAbortedError ..."	Historical 0.6.54 Windows interruption when the browser disconnected; fixed in 0.6.55. Preserve the error if it recurs on a current build.
A question card or the approval card disappeared	Click ◇ Project status , then Respond .
The approval card shows Not ready to execute yet	Choose Modify the plan and ask for a plan whose steps all use the KI's tools (Step 8).
Setup fails after approval	Send "Please continue setting up FSM2." Chapter 4 lists setup problems.
The run ends without Completed	See "If the project does not reach Completed" above, and Chapter 9, 9.11.

6 Projects and files

In this chapter you will learn where GeoForge keeps each chat's work, how to give it your own files, how to read the **Project status** panel, and how to move, share or archive a project without losing what it proves.

6.1 One chat, one project folder

Every chat in the sidebar owns one folder on your disk. That folder holds the chat's history, your inputs, the downloaded data, the model runs, the outputs and plots, and the signed run records. GeoForge creates it when you click **Create chat**, before you send anything.

The installed model software is not inside the project. It stays in that model's install folder, by default inside GeoForge's work folder (for example `C:\Users\you\kiss\fsm2`), so all your projects share one installation of each model.

Tip: Use one chat for one scientific objective. A chat's AI and KI are fixed after its first message, so a new question, study area or experiment belongs in a new chat anyway.

6.2 Create a project folder

1. Click **+ New chat** at the top of the sidebar. The **New project chat** dialog opens.
2. Read **Project name**. GeoForge fills it in: with the first line of your message if you typed one before any chat was open, otherwise with a dated name such as `Project 2026-10-01 093015`. Change it if you like.
3. Check **Create the project inside**. This is the parent folder. Keep the default or type a full path, such as `D:\GeoForge-Manual\projects`.
4. Read **Project folder (created automatically)**. This is the exact folder GeoForge will create.
5. Click **Create chat** (or press Enter in **Project name**).

New project chat

×

Each chat has its own project folder for memory, inputs, model runs, outputs, and plots.

Project name

Alptal snow example

Create the project inside

D:\GeoForge-Manual\projects

Choose...

Project folder (created automatically)

D:\GeoForge-Manual\projects\2026-10-01-Alptal-snow-example--{id}

Edit the suggested name or keep it. GeoForge creates the folder automatically, including missing parent folders. {id} becomes a unique ID so projects with the same name stay separate. Existing files are not changed.

Create chat

Use default location

Cancel

Check that the preview ends in `<date>-<your name>--{id}`. GeoForge replaces `{id}` with a unique 12-character code, so two projects with the same name never share a folder.

The dialog also opens when you click **Send** or **+ Files** with no chat open. **Use default location** puts the default parent folder back. **Cancel** closes the dialog without creating anything.

Chats started with **Use in a new chat (KI Library)**, **Ask Agent (KI Observatory)** or **Use KI in chat** (a model's setup page) skip this dialog. Their folder is created in the default parent folder and is renamed after your first message.

GeoForge creates missing parent folders for you and does not change files that already exist there. The default parent is the `projects` folder in your GeoForge work folder:

	Windows	macOS
Default parent folder	<code>C:\Users\you\kiss\projects</code>	<code>~/kiss/projects</code>
Choose...	No folder picker opens, because GeoForge runs in your web browser. The dialog shows "Browser mode: type the folder path here. The desktop build opens the system folder picker." Type the path, or copy it from the File Explorer address bar and paste it.	Opens the macOS folder picker.

Name rules. A project name has 1 to 120 characters and must contain at least one letter or number. Chinese names work and give readable folder names. In the folder name, spaces and most punctuation become `-`, and only the first 56 characters are used.

Caution: Choose the final location now. A project's run records are tied to the folder's full path (see 6.9), so moving the folder later means GeoForge can no longer verify them. Pick a local drive with plenty of free space that you can write to.

6.3 What is inside a project folder

A project folder looks like this (example: `D:\GeoForge-Manual\projects\2026-10-01-Alptal-snow-example--<id>`). Some folders appear only when they are first needed.

Item	What it holds
<code>README.md</code>	A short description of this folder.
<code>session.json</code>	The chat's working state, used by the app.
<code>kiss.toml</code>	Path settings that GeoForge writes once a KI is prepared for this chat.
<code>memory\transcript.jsonl</code>	The full chat history, one message per line.
<code>inputs\</code>	Input data, in <code>forcing\</code> , <code>parameters\</code> , <code>initial_conditions\</code> , <code>boundary_conditions\</code> , <code>observations\</code> and <code>static\</code> . GeoForge Database downloads go to <code>observations\<dataset>\</code> or <code>geoforge_subsets\</code> .
<code>inputs\uploads\</code>	Files you add with + Files , drag and drop, or Upload other files .
<code>inputs\user\<input>\</code>	Files you upload for one specific plan input (see 6.6).
<code>outputs\<KI>\</code>	The model's own output files, for example <code>outputs\FSM2\alptal_0405\</code> .
<code>artifacts\</code>	Plots, tables, written run notes and the Project view file (<code>project-view.json</code>).
<code>runs\</code>	The plan (<code>plan.json</code> , <code>data-inventory.json</code>), your approval (<code>approval.json</code>), the progress state, the evidence summary (<code>evidence.json</code>), the event log and the run logs (<code>logs\</code>).
<code>models\<KI>\</code>	This project's copy of the KI instructions and tools (<code>ki\</code>) and its path settings. Not the model software.
<code>calibration\</code>	Calibration setup, cases and calibration runs for this project.
<code>references\papers\</code>	PDF papers you add for the agent to read.
<code>.geoforge\</code>	GeoForge's protected records: signed run records (<code>receipts\</code>), your saved planning answers (<code>user-answers.json</code>), download progress and manual download details.
<code>setup-request.json</code> , <code>setup-request-<number>.json</code>	The question or request card that waits for you now, and earlier cards, kept after you answered them.

On Windows, File Explorer shows `.geoforge`. On macOS, Finder hides folders whose names start with a dot; press Command-Shift-. (period) in Finder to show them.

6.4 What you may add, open or leave alone

You may	Where
Open and copy at any time	outputs\ , artifacts\ , memory\transcript.jsonl , README.md , and the files in runs\ (to read only)
Add files	inputs\user\<input>\ (before you approve), inputs\uploads\ , references\papers\
Never edit, move or delete	.geoforge\ , runs\approval.json , runs\plan.json , runs\data-inventory.json , runs\flow-state.json , session.json

Why the last row matters:

- The approval, the progress state and every run record are signed: sealed with a key that only GeoForge holds (see 6.9). A hand edit makes the signature fail, and GeoForge then treats the record as invalid.
- If plan.json or data-inventory.json changes after you approved, your approval no longer matches the plan and you must approve again.
- If .geoforge\user-answers.json cannot be read, approval stops until it is fixed.
- If you replace or delete a downloaded file, **Project status** stops showing it as available.

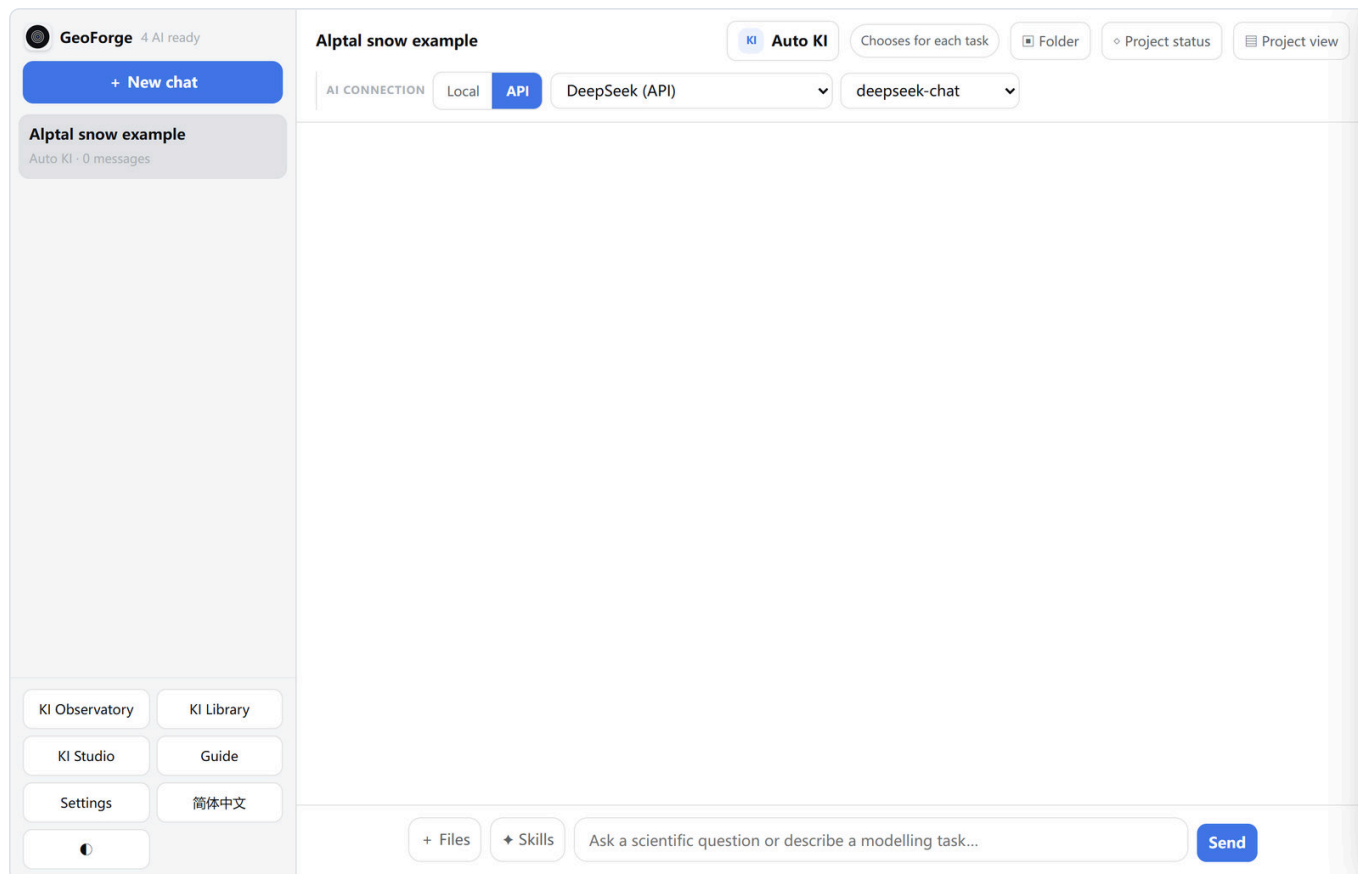
Caution: A data file that the approved plan does not list, and that no recorded run produced, keeps the project from reaching **Completed**. This applies to .csv , .txt , .dat , .out , .bin , .nc , .tif , .png and .json files in inputs\ , outputs\ , artifacts\ and calibration\ . Add files before you approve, so the plan can include them, and do not save your own notes or test files there. If the chat names such a file, see "If something goes wrong".

Tip: Close output files in Excel or other programs before a run starts or reruns. Windows does not let the model replace a file that another program has open.

6.5 Open the folder with the Folder button

1. Open the chat in the sidebar.
2. Click  **Folder** in the header.
3. Find the new window that opens at the project folder. If you do not see it, look on the taskbar or in the Dock.

	Windows	macOS
The folder opens in	File Explorer	Finder



Check that **Folder**, **Project status** and **Project view** are in the header. They are greyed out until a chat is open. After you click **Folder**, the line under the message box reads "Opened" followed by the folder path.

Other buttons open the same place. **Open project folder** at the bottom of the **Project view** panel opens the project folder. Once the chat has a KI, the **Data for this run** card under **Project status** → **Details** has **Open all inputs** and, inside each data group, **Open folder**; they open the input folders.

6.6 Give GeoForge your files

There are three main ways to add files. Only one of them ties a file to a specific input of the plan.

Button	Where it is	Files go to	Tied to a plan input?	Use it for
Upload files	◇ Project status → Data in this plan , on a row under You	inputs\user\ <input>\	Yes, when you approve the plan	Data the plan asks you to provide, such as your own site file or observations
Upload other files	Bottom of the Project status panel	inputs\uploads\	No	Loose files for the agent to inspect
+ Files , or drag files onto the message box	Left of the message box	inputs\uploads\	No	A file that belongs with the message you are writing

Two more buttons exist for special cases. **Add paper PDF** (in **Project status** → **Details** → **See technical data details**, in the **Optional model reading** card once the chat has a KI) saves PDF papers to references\papers\ , up to 100 MB each. **Add observations** (in the **Calibration** card under **Details**) works like **+ Files**.

What all three main ways have in common:

- The file is copied into the project at once and appears as a small label (chip) with its name and size above the message box. The agent learns about it only when you click **Send**.
- Uploading never sends, approves or checks anything.
- GeoForge never overwrites a file. If the name is taken, the new copy gets an added ending. Unusual characters in a file name become `_`.
- One file can be up to 300 MB. For larger data, see 6.6.3.

GeoForge's messages write paths with forward slashes, such as `inputs/user/site/`. This is the same folder as `inputs\user\site\` in File Explorer.

6.6.1 Attach files to a message

1. Click **+ Files**.
2. Choose one or more files in the file dialog and confirm.
3. Check the chips above the message box. The line under the message box reads, for example, "2 files ready to send".
4. To drop a file from this message, click **X** on its chip. The file stays in the project.
5. Type your message.
6. Click **Send**. The agent receives the project paths of the attached files. If you send no text, GeoForge sends "Please inspect the attached files."

Tip: Instead of steps 1 and 2, you can drag files from File Explorer or Finder onto the message box.

What you should see: one chip per file, and **Send** enabled.

You cannot attach files while the chat is working: **+ Files** is greyed out until the AI has finished its current reply (its "turn"). Wait for the reply to finish, or add the files in another chat.

6.6.2 Upload a file for a plan input

Use this when the plan says an input must come from you. On the **Approve the plan?** dialog, such an input is listed under **You** with "please provide". The dialog covers the window, so you close it first, upload, and then reopen it.

1. In the **Approve the plan?** dialog, click **Not now**. Nothing is approved or rejected.
2. Click **Project status**.
3. Under **Data in this plan**, find the group **You**. The row for your input shows **Waiting for you**, a note that starts "Provide it yourself:" (format, unit, rules and the target folder), and an **Upload files** button.
4. Click **Upload files**.
5. Choose the file or files in the file dialog and confirm.
6. Check the row. It now shows **Present; not verified**, and its **Upload files** button is gone.
7. Read the line at the bottom of the panel: "Saved and queued for your next message (not sent, approved, or verified): ...". GeoForge also puts a note in the message box that asks the agent to inspect the files. You can leave it: when you approve, your approval message replaces it, and the files go with that message.
8. Under **One thing needs you** at the top of the panel, click **Respond**. The **Approve the plan?** dialog opens again.
9. Click **Approve and start**.
10. Click **Continue with this choice**.
11. Wait for the reply in the chat: "Your file is now named as the input: ... The updated card is in the chat; approve it to start." GeoForge has linked each file in `inputs\user\<input>\` to that input and recorded its checksum (a fingerprint of the file's exact content). A new **Approve the plan?** dialog opens.
12. Check the **Waiting on you** section of the new dialog. It lists "your file is now named as the input: → inputs/user//". The input itself has moved to **The run prepares itself**, marked "already on disk".
13. Click **Approve and start**.

14. Click **Continue with this choice**. The signed approval now records that the input is your file.

What you should see: the row's status on the right, and the **Upload files** button under the note.

What you should see: your file's path in **Waiting on you**. If it names the wrong file, choose **Modify the plan** instead of approving.

Caution: If an input waits for your file and its folder is empty, approval stops with "Upload your file with the Upload button on the card". The **Upload files** button is in **Project status** → **Data in this plan**, not on the card.

At approval, GeoForge links every file in `inputs\user\<input>\` except files whose names start with a dot. To replace a file, delete the old one in File Explorer or Finder before you approve; otherwise both are linked. To use a different file after you approved, ask in the chat for a plan change, so the new file is linked at the next approval.

6.6.3 Large files

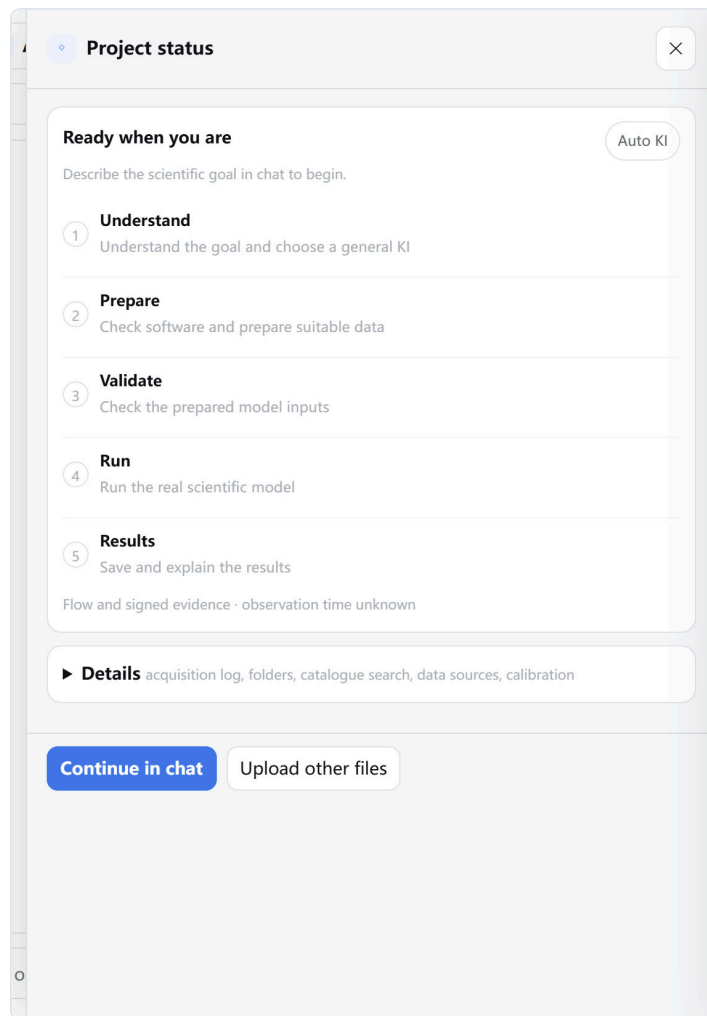
Uploads stop at 300 MB per file ("file larger than 300 MB"). For larger data:

1. Click  **Folder**. The project folder opens in File Explorer or Finder.
2. Open the target folder. For an input the plan asks you to provide, that is `inputs\user\<input>\`; create it if it does not exist, using exactly the input's name shown in **Data in this plan**. For any other file, use `inputs\uploads\`.
3. Copy the data into that folder.
4. Tell the agent in the chat where the files are. Do this before you approve, so the plan can name the files (see the caution in 6.4).

Files in `inputs\user\<input>\` are linked at approval however they got there. GeoForge computes the checksum of each file at every approval, which takes a while for very large files.

6.7 The Project status panel

1. Click  **Project status** in the header. The panel opens on the right.
2. Read it from top to bottom.
3. Click  to close it, or **Continue in chat** to close it and go back to the message box.



In a new chat, check that the card reads **Ready when you are** with **Auto KI** on the right, that none of the five stages is marked, and that **Details** is folded.

The panel has these blocks:

Block	What it shows
One thing needs you	Only when a card waits for you: a choice or question such as Approve the plan? (Respond reopens it), a manual Baidu Pan download (Files are in place, continue), or a licence, login or permission step (sometimes with Open official page).
Progress card	A headline: Ready when you are, Ready to continue, GeoForge is working, One thing needs you, Project complete, Project needs attention or Status unconfirmed ; just Progress while the One thing needs you card is shown above it. On the right, Auto KI or "N / M software verified". Then a one-line summary, the Goal , the KI chips (for example "General KI: FSM2") and the five stages. The last line says where the status comes from and how old it is.
Data in this plan	After a plan exists: every input in three groups, GeoForge fetches, You and The run prepares itself , each with its status.
Details	Folded; its label reads "acquisition log, folders, catalogue search, data sources, calibration". Inside: the status evidence, data download jobs, Data for this run (with Open all inputs once the chat has a KI), the GeoForge Database search, Data sources & download record (when there is one), Calibration and See technical data details .
Bottom buttons	Continue in chat and Upload other files .

The five stages:

Stage	Meaning
Understand	Understand the goal and choose a general KI
Prepare	Check software and prepare suitable data
Validate	Check the prepared model inputs
Run	Run the real scientific model
Results	Save and explain the results

The dot marks where the work is now. It is not proof that earlier checks passed, and earlier stages are not ticked as work moves on. All five get a tick only when the project is complete. During planning and data download the dot stays on **Understand**; read the summary line instead.

Statuses in **Data in this plan** come from signed records and files on disk, not from what the agent says:

Status	Meaning
Waiting for you	You must upload, download or choose something.
Pending	Not prepared yet.
Present; not verified	A file is in place; nobody has checked its content.
Acquired; scientific checks pending	GeoForge downloaded it and signed a record; its scientific fitness is not yet checked.
Produced; scientific checks pending	A recorded run step produced it.
Fetch failed	The download failed.
No step uses it	Listed in the plan, but no step uses it.

A completed project can still show "scientific checks pending". That is honest: a finished run is not a validated model.

The  **Project status** button keeps its name. Once the project has a goal, a coloured dot after it shows the state, and its tooltip starts with the state name:

Dot	Tooltip starts with	Meaning
Blue, pulsing	Working	GeoForge or the agent is working.
Amber	Needs you	Something waits for you.
Green	Complete	Every approved step has a passing, signed run record, and no unrecorded data file is left. It does not mean the results are scientifically valid.
Red	Needs attention	A step failed, work is blocked, or the records could not be read.
Grey	Ready or Unconfirmed	Nothing is running and nothing waits for you, or the state could not be confirmed.

6.8 Switch between chats

1. Click any chat in the sidebar to open it. The newest chat is at the top.
2. Read the line under each title: its KI (or **Auto KI**) and the number of messages. A working chat also shows **Active**, **Running · quiet**, **Check status** or **Connection issue**.

Work continues in a chat you leave. While one chat is running, you can open another chat or click **+ New chat**, either at the top of the sidebar or on the activity bar (the strip under the message box that appears while a chat works); the app is not frozen. When you return, the chat shows its latest message, waiting card or result.

An unsent message stays with its chat while you switch, but it is lost if you reload or close the page. **Stop** on the activity bar stops only the chat you are looking at.

6.9 Move, back up or share a project

GeoForge signs the approval, the progress state and every run record with a key kept on this computer, outside the project folder (`C:\Users\you\.config\geoforge\flow-keys` on Windows, `~/.config/geoforge/flow-keys` on macOS). Each key belongs to one folder path.

So if you move, rename or copy a project folder, even on the same computer, or open it on another computer, GeoForge cannot verify that project's records there. The files themselves (outputs, plots, transcript) are unchanged and readable. Once a project has a plan, **Project status** in a moved copy shows **Project needs attention** with "Project evidence is unreadable; inspect status details."

You want to	Do this
Keep working on a project	Leave the folder where it is, with the same name.
Back it up	Copy the whole folder to a backup drive and leave the original in place. Treat the copy as a record to read. If you ever restore it on this computer, put it back at exactly the original path and name; its records then verify again.
Send results to a colleague	Send the files they need from <code>artifacts\</code> and <code>outputs\</code> , rather than the whole folder.
Free disk space	Archive the chat (6.10), then delete the archived folder yourself.

Caution: Do not delete or edit the `flow-keys` folder. Without it, GeoForge cannot verify the run records of any of your projects.

Before you send a whole project folder to anyone, know what it contains:

- the full chat history (`memory\transcript.jsonl`);
- Baidu Pan share links and extraction codes from the GeoForge Database, if the project used a manual download. They are saved, for example, in `.geoforge\manual-download-details.json` and in the `setup-request` files, so the download card survives a restart. They were issued to you; do not post them publicly;
- downloaded data, which may have its own licence terms.

Your AI keys and your GeoForge Database token are not stored in the project folder.

Look at a project someone sent you.

1. Copy the folder, with its name unchanged (it must end in `--` and the 12-character code), into your default projects folder, such as `C:\Users\you\kiss\projects` .
2. Look at the GeoForge sidebar. The chat appears within a few seconds, because the list refreshes by itself while the GeoForge window is visible.

Use it to read: its records cannot be verified on your computer, it keeps the sender's AI choice, and the model software must be installed on your computer before anything can run. Start a new chat for new runs.

6.10 Archive (delete) a chat

1. Hover over the chat in the sidebar. A **×** appears at the right end of its title line.
2. Click **×** (tooltip: "Archive chat; project files are kept").
3. Click **OK** in the dialog "Archive this chat? Its local project files will be kept."

The chat leaves the sidebar. Nothing is deleted: GeoForge moves the folder into an `_archived` folder next to it and adds `-deleted-<number>` to its name. For a project in the default location that is `C:\Users\you\kiss\projects\_archived\`.

The **×** is greyed out while the chat is working. Wait for the AI to finish its reply, or click **Stop** first.

To free the disk space, delete the folder in `_archived` yourself with File Explorer or Finder. Model outputs and downloaded data can be large.

Tip: GeoForge has no button to restore an archived chat. To bring one back, move its folder from `_archived` back to where it was and remove the `-deleted-<number>` ending. For a project outside the default location, also move `C:\Users\you\kiss\sessions\_archived\<id>-deleted-<number>.json` back to `C:\Users\you\kiss\sessions\<id>.json`. The chat reappears in the sidebar within a few seconds. Back at its original path, its records verify again.

If something goes wrong

What you see	What to do
Choose... opens nothing and the dialog says "Browser mode: type the folder path here..." (Windows)	This is expected. Type or paste the full folder path.
Create chat is greyed out and the preview reads "Enter a project name and location to preview its folder."	One of the two fields is empty. Fill in Project name and Create the project inside .
"Could not preview folder: project location must be an absolute folder path"	Type a full path that starts with a drive letter, such as <code>D:\GeoForge-Manual\projects</code> .
"Could not preview folder: project name must contain a letter or number"	Add at least one letter or digit to the name.
"could not create project location ..." in red under the preview	The drive does not exist or you cannot write there. Choose another folder. Your entries are kept so you can retry.
"Could not open the folder picker; enter the path directly." (macOS)	Type or paste the full folder path into Create the project inside .
After ☒ Folder , the line under the message box shows an error (for example "no such session") instead of "Opened ..."	The project folder is no longer at its path or under its name. Move it back to its original path and name.
"file larger than 300 MB"	Copy the file with File Explorer or Finder instead (6.6.3).
"This chat is working. Add the files to a new chat, or wait for this turn to finish."	Wait for the AI to finish its reply, then attach the files.
Approval stops with "These still need your decision before anything runs: ... Upload your file with the Upload button on the card (it goes to inputs/user/...)"	Click Not now on the dialog. Use Upload files on that row in Project status → Data in this plan , then click Respond and approve again (6.6.2).
"Your saved answers cannot be read (...). Fix or remove that file, then approve again."	<code>.geoforge\user-answers.json</code> was edited or damaged. Undo the change if you can. If you remove the file, approval works again, but GeoForge treats your earlier planning answers as never given; check the card carefully.
The chat says " GeoForge verification: not complete yet — ... these files are not vouched for by a passing receipted run (left by a failed attempt or written outside run-tool); regenerate or remove them: ..."	If a listed file is one you added, move it out of the project folder (keep a copy elsewhere). Then send a short message, such as "continue". If the agent made the file outside a recorded run, ask it to regenerate the file through the approved step. On Windows, a background job the agent started can keep running after Stop (a known issue) and write such files; end it in Task Manager first.
Project needs attention with "Project evidence is unreadable; inspect status details." after you moved, renamed or copied the folder	Move the folder back to its original path and name. A copy elsewhere can be read but not verified.
A chat disappeared from the sidebar after you moved its folder	Move the folder back to its original path and name. The sidebar shows it again within a few seconds.

7 More tools

In this chapter you will use the tools around the chat: the **KI Library**, the **KI Observatory**, **KI Studio**, skills, MCP servers, the in-app **Guide**, and model calibration.

Where to find each tool

The first three tools have buttons at the bottom of the sidebar and cards on the start page.

Tool	Open it from	Use it to
KI Library	Sidebar KI Library ; start page Browse & verify ; Guide → Open KI Library	see every KI, check whether its software runs here, import a KI, update the library
KI Observatory	Sidebar KI Observatory ; start page Observe KIs ; KI Library header	learn what KIs exist and how one works; ask the AI about a KI
KI Studio	Sidebar KI Studio ; start page Create a KI ; KI Library header Create KI	turn your own model or workflow into a checked KI
✦ Skills	Beside the message box	pin a reusable procedure, such as a plotting method, to a chat
MCP servers	Settings → Permissions	let a local agent in one chat use an outside service, such as GitHub
Guide	Sidebar Guide	read a three-step summary of GeoForge
Calibration	◇ Project status → Details → Calibration	tune model parameters against your observations

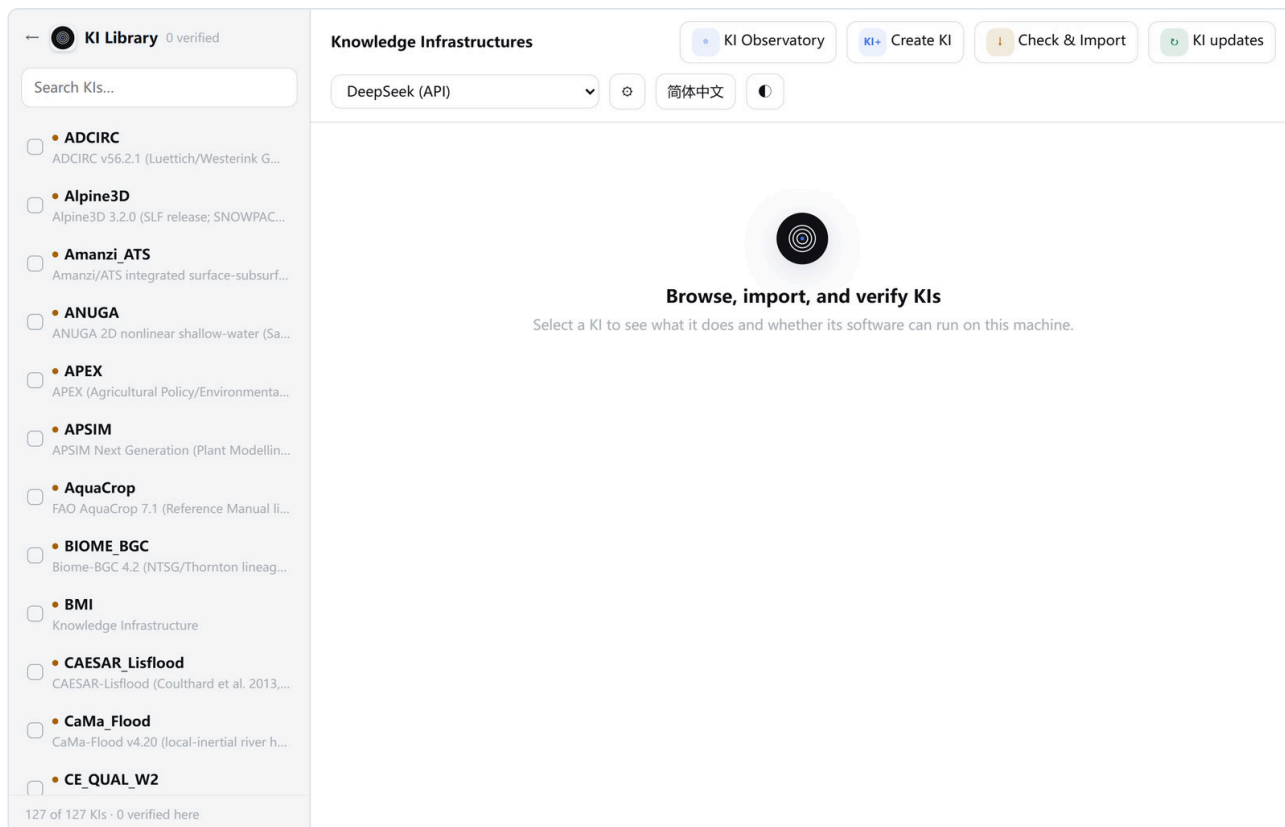
The ← or < at the top left of each page takes you back. From **KI Studio** it goes to the **KI Library**. From the **KI Library** and the **KI Observatory** it goes to the main window.

KI Library

A KI (Knowledge Infrastructure) is checked guidance for one scientific model: how to install, prepare, run and check it. The **KI Library** lists every KI on this computer and shows whether its model software has been checked here.

Read a KI's status

1. Click **KI Library** at the bottom of the sidebar.



Check the line at the bottom of the list, for example **127 of 127 KIs · 0 verified here**. On a new installation no KI is verified yet. That is normal.

2. Type part of a model name in **Search KIs...** to shorten the list.

3. Click a KI in the list. Its details open on the right.

Check the two badges under the KI name. The first is about the KI package. The second is about the model software on this computer.

The dot before each name gives the software state at a glance: green means verified here, red means the last check failed, and orange means the software still needs setting up.

Badge	Meaning
Bundled KI	The KI came with GeoForge or with an official KI library update.
Valid imported package	You imported this KI and it passed the package check.
Package not checked	GeoForge has not checked this package.
Verified on this machine	The model software passed the KI's own start-up check (its "preflight") on this computer. The date is under Last checked on this machine .
Setup needed or Ready to set up	The software is not verified here yet. You can still choose the KI: GeoForge has the agent set the software up after you approve a plan.
Manual setup	As above, but GeoForge cannot fetch the software by itself, so expect to provide some of it yourself.
Verification failed	The last check on this computer failed. Set it up again with the agent.

Keep three ideas apart. A KI being in the library means GeoForge can read its guidance. **Verified on this machine** means the model program runs here. Neither means that a project has the data and decisions it needs, and neither says anything about whether the model's results are scientifically right.

The four cards show the **Scientific software** and its version, **Last checked on this machine** (**Not yet** if never checked), **Language & licence**, and the **Source** (**Open project website**).

Start a chat or set up the software from the Library

- **Use in a new chat** creates a chat with this KI already chosen. It uses the AI shown in the selector at the top of the **KI Library**. It skips the **New project chat** dialog, so the project goes to the default location (Chapter 6) and the chat takes its name from your first message.
- **Set up with agent** opens **Agent setup** to install and verify the model software. Once the KI is verified, the button reads **Open setup & verification**. Chapter 4 explains this page.
- **Run data** shows what the KI needs: **Inputs** → KI → **Outputs**, a **Local data check**, the **Declared input groups** (**Forcing, Parameters, Initial conditions, Boundary & controls**) and any **Coupled KIs**. Click a group to see its items.

Run data is built from the KI's own files and from what GeoForge finds on this computer. It describes the KI, not one of your projects. A project's data is in ◇ **Project status** (Chapter 6, section 6.7).

To read **Run data** in plain words, click **Explain my data** under **Need this in simple words?**. The AI in the selector at the top of the page writes the explanation. It cannot change the requirements or the verification marks. If no AI is connected, the button reads **Connect AI first** and cannot be clicked.

Tip: "Local files are not machine-checkable yet." means this KI does not list file paths that GeoForge can check. The input groups still describe the model's interface.

Caution: The ⚙ button in the **KI Library** header opens an older, smaller Settings window. It has no **Permissions** page, and its API-key fields do not hide what you type. Use **Settings** in the main window's sidebar instead, and never share a screenshot of the ⚙ window while you type a key.

Import a KI package

You can add a KI that a colleague sent you as a `.zip` package. GeoForge checks the package before it adds anything. The KI takes its name from the file name, without `.zip`.

1. In the **KI Library** header, click **Check & Import**.
2. Choose the `.zip` file. The **KI package check** window opens and shows "Checking *name* before importing...".
3. Read the result. **BLOCK** lines stop the import. **WARN** lines are advice and do not stop it.

Check that the summary says the package is valid. Otherwise **Import valid KI** stays greyed out.
4. Click **Import valid KI**. The new KI is selected in the list with the badge **Valid imported package**.

Importing checks the package's structure only. The model software still has to be set up and verified on this computer, like any other KI. Click **Cancel** to leave without importing.

Update the KI library

GeoForge can download newer KI definitions from the official GeoForge repository. Before it uses a downloaded library, GeoForge checks every package in it. If the download or a check fails, the current library stays in use. An update never changes your chats, input data, installed model software, verification history, or KIs you imported yourself.

Platform	When GeoForge checks for KI updates
Windows	Only when you ask. The first check downloads and checks about 88 MB, so it is not started at launch.
macOS	Automatically at each launch. You can also check by hand.

To check by hand:

1. In the **KI Library** header, click **KI updates**. The **KI library updates** window opens.
2. Click **Check again**. While it runs, the header button reads **Checking KIs...**
3. Read the summary at the top of the window.

Check the summary. On Windows in 0.6.54 it normally says the current library was kept.

That result is expected on Windows, and it is safe. Every KI in the Windows build has Windows installation notes, and 23 also have Windows install recipes. The online repository does not have them yet. GeoForge refuses a library that would remove them and keeps the one you have. Updates resume once the repository has the Windows notes.

After a kept result, the header button reads **Library kept**. The main window may also show a notice, **Current KI library kept**, at the bottom right when it opens. Click **View report** to open this window, or **Dismiss** to close the notice.

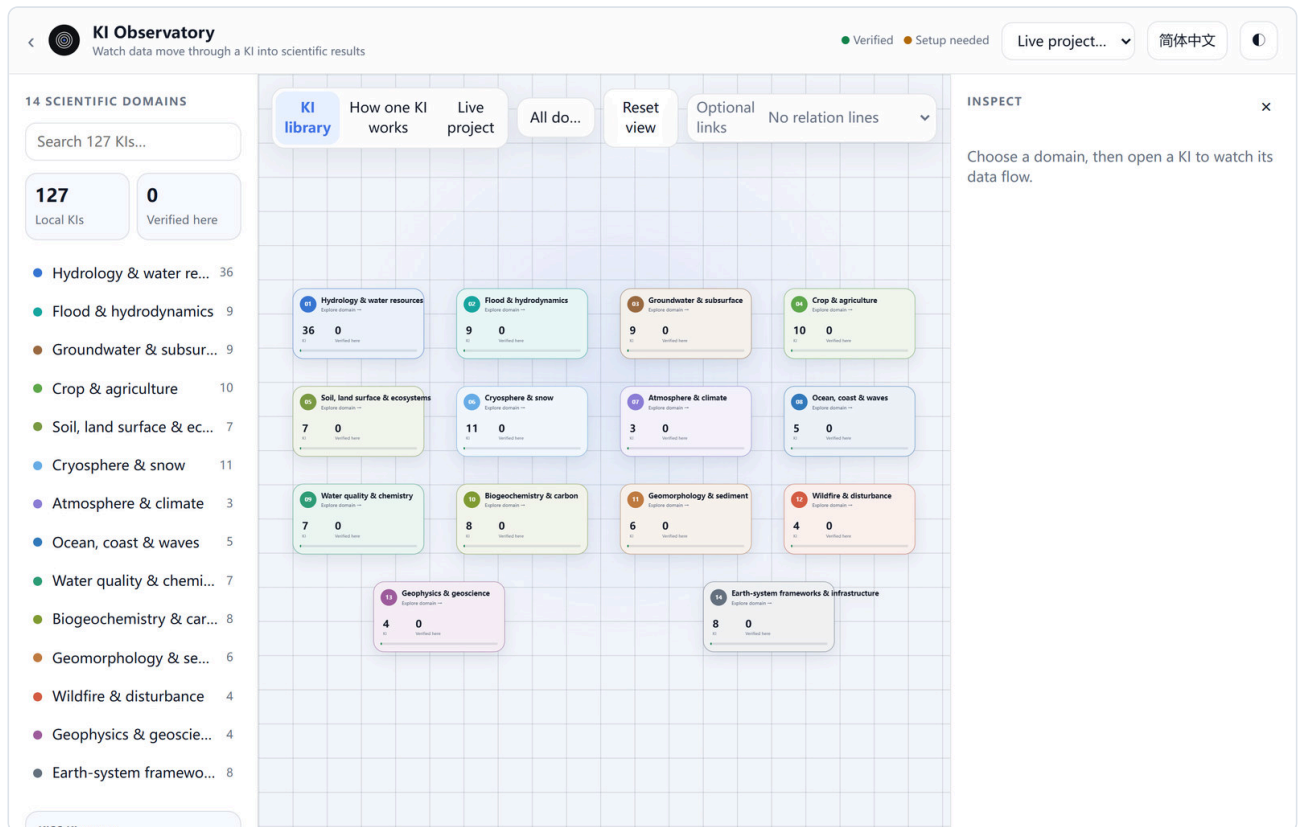
If GitHub can only be reached through a proxy, set it in the main window: **Settings** → **Network & proxy**, tick **GitHub & KI updates**, then click **Save** (Chapter 2). The **Network settings** button in this window opens the older ⚙ Settings window described above.

KI Observatory

The **KI Observatory** is a map for learning which KIs exist and how each one turns data into results. It only reads KI descriptions and never runs model code. The one thing it can create is a new chat, through **Ask Agent**.

Explore the domains and one KI

1. Click **KI Observatory** at the bottom of the sidebar.



Check the left column: **14 SCIENTIFIC DOMAINS** with a count for each, and two boxes, **Local KIs** and **Verified here**. The legend at the top right shows a green dot for **Verified** and an orange dot for **Setup needed**.

2. Click a domain card, or a domain name in the left column. The KIs of that domain appear as cards. Each card shows how many inputs, processes and outputs the KI declares, for example **4 IN · 6 PROCESS · 3 OUT**.
3. Click a KI card. The **How one KI works** tab opens for that KI.

Check the badge beside the domain name: **Verified** or **Setup needed** on this computer.

4. Click a stage card. The **INSPECT** panel on the right explains what happens in that stage.
5. Open **Technical evidence** in the **INSPECT** panel to see the exact names the KI uses.
6. To see the KI's complete workflow graph (its "DAG"), click **Full technical DAG**. It shows four lanes: **Inputs**, **Core engine**, **Verification gate** and **Results**.
7. Click **How it works** to go back to the stage view.

Below the graph, **Visualization capability** lists the kinds of displays the KI declares, such as plots or maps.

Tip: The moving current in **How it works** shows the order of the KI's scientific stages. It does not mean a model is running.

Other ways to move around:

- Type in **Search 127 KIs...** to find a KI in any domain.
- Drag the map to move it and use the mouse wheel to zoom.
- On the **KI library** tab, click **Reset view** to go back to all domains. In a KI's view, click ← **Back to domain** to go up one level.
- **Related KIs** shows the KIs linked to this one. **Optional links** chooses which links to draw: **No relation lines**, **Scientific coupling**, **Shared data** or **All evidence**.

Ask the AI about a KI

Ask Agent creates a **new** chat with this KI chosen and sends a question about it at once. It appears in the KI header, and in the **INSPECT** panel when a KI, a stage or a graph node is shown there.

1. Before you open the Observatory, check the **AI CONNECTION** (**Local** | **API**, provider and model) at the top of the main window. The new chat uses that choice, and a chat's AI cannot change after its first message.
2. In the Observatory, click **Ask Agent**. The button reads **Creating a new chat...**
3. Wait. GeoForge switches to the new chat and sends a question such as "Explain in plain language how data moves through the FSM2 KI into results."

Notes:

- The chat skips the **New project chat** dialog. Its project goes to the default location, named after the question.
- Each click creates one more chat and project folder. Archive the ones you do not need with ✕ in the chat list. Archiving keeps the project files (Chapter 6).
- The question asks for an explanation, so it usually gets an answer without starting a plan. To run the model, describe your task in this chat or in a new one.
- If no AI is ready, the question waits in the message box with the hint "The new chat is ready with the question filled in. Connect an AI, then press Send."

Watch a chat project

The **Live project...** list at the top right shows your chats. Choose one to open the **Live project** tab. It shows the chat's recorded stage in a row of eight steps, from "understanding" to "results", and what the agent is doing now. A **Waiting for you** badge means the chat needs your answer.

This tab is only for watching. To answer a question or approve a plan, go back to the chat. ◇ **Project status** in the chat remains the full record of a project (Chapter 6, section 6.7).

KI Studio

KI Studio turns your own model, or a repeatable task such as a data-cleaning or plotting procedure, into a KI that GeoForge can use. It is for advanced users. KI Studio uses a separate engine called KDT to read your source and check the result. An AI agent writes the KI, two independent checks test it, and you decide whether it joins your library.

Create a KI

1. Click **KI Studio** at the bottom of the sidebar.

The screenshot shows the 'KDT Workbench' interface. At the top, there's a header with 'KI Studio' and 'KDT-single' tabs, and a language selector set to '简体中文'. Below the header, a status bar indicates 'Checking KDT...'. The main area is titled 'KDT Workbench' and includes a description: 'Import source, documentation, papers and examples; dissect the KI, inspect its live structure and visualization contract, then decide whether a separately verified snapshot belongs in your KI Library.' The interface is divided into two main sections. The left section, 'New KI workspace', contains a 'Reviewed KDT engine' status box with an 'Install KDT engine' button. Below this, a form asks 'What are you turning into a KI?' with two options: 'Process-based model' (selected) and 'Task or workflow'. The form also includes fields for 'Name' (e.g., 'MyHydrologyModel'), 'Scientific domain' (a dropdown menu), 'Source' (with options for 'Public Git repository' and 'Local source folder'), and 'Workspace parent folder'. The right section, 'Your KI workspaces', shows 'No KI workspaces yet.' and a 'Four separate decisions' box detailing the workflow from draft to verified software.

Check the pill at the top right. On the first visit it reads **Checking KDT...**, then **KDT setup needed**. Once the engine is installed it reads **KDT ready**. Your earlier workspaces are listed under **Your KI workspaces**.

2. Under **Reviewed KDT engine**, click **Install KDT engine**. You do this once. GeoForge downloads the engine (KDT-single) from GitHub with Git. It is not shipped inside GeoForge, and its repository does not declare a licence.
3. Under **What are you turning into a KI?**, choose **Process-based model** or **Task or workflow**.
4. Type a **Name**.
5. Choose the **Scientific domain**. If yours is not listed, choose **Other — enter your own** and type it. The agent then works out the workflow from your source, examples and tests only.
6. Under **Source**, choose **Public Git repository** or **Local source folder**.
7. Paste the repository's main address (for example `https://github.com/organisation/model`), or the path of your source folder.
8. Check **Workspace parent folder**. GeoForge creates a new folder there and never edits your original source.
9. Choose the **AI agent**.
10. Choose the **Agent model**.
11. Click **Create workspace**. The **KI creation run** panel opens.

12. Optional: under **Source material for this KI**, paste a DOI, a web link or a local path to a paper, manual or example.
13. Click **Add**. Repeat steps 12 and 13 for each item.
14. Work through the five numbered steps of the panel in order:

Step	Button	What happens
1 Map the model source	Run KDT probe	KDT reads the source, build files, documentation and input/output paths.
2 Dissect and build with your agent	Build or repair KI	The agent writes the KI: workflow, tools, checks and plots.
3 Review the live Workbench preview	View above	Read the four lanes: Workflow DAG, Tools and data I/O, Diagnostics and preflight, Visualization functions . Repeat step 2 to repair.
4 Finish and run KI_verify	Finish and verify	KDT checks the KI, then GeoForge checks a fixed, read-only copy of it (KI_verify is this structural check).
5 You make the final decision	Accept and add to KI Library	Nothing is added until you click this.

Check that the result reads **Independent KI_verify passed** before you accept. If it reads **KI_verify needs revision**, read the listed problems, click **Continue modifying** and repeat step 2.

The panel **Four separate decisions** explains why these results are kept apart:

- **Workbench draft**: an editable candidate, nothing more.
- **KI_verify passed**: KDT and GeoForge both checked the KI's structure.
- **User accepted**: happens only when you click step 5.
- **Verified software**: comes later, when the model software passes its check on the computer that runs it.

None of the first three shows that the model program runs, and none of the four shows that the model is scientifically right.

Platform	Choosing folders in KI Studio
Windows	Choose folder only shows "Folder picking is available in the desktop app. Paste a path here in browser mode." Copy the path from File Explorer's address bar and paste it, for example <code>D:\GeoForge-Manual\kis</code> .
macOS	Choose folder opens the system folder picker.

Caution: Install KDT engine needs Git, which GeoForge does not include. If Git is missing, the message reads "Git is not installed. Install the Xcode command line tools, then retry." The Xcode tools are for macOS. On Windows, install Git for Windows and try again.

The Windows end-to-end test for 0.6.54 covered only the FSM2 example with the DeepSeek API. KI Studio was not part of that test.

Skills

A skill is a written procedure that an agent can follow, for example how to make a publication-quality plot. The agent chooses useful skills by itself. Pin a skill only when you want it used throughout one chat.

GeoForge does not ship skills. The list shows the skills already installed for your agents. Each skill is a folder containing a `SKILL.md` file in one of these places:

- `C:\Users\you\.agents\skills`

- C:\Users\you\.codex\skills
- C:\Users\you\.claude\skills

On macOS the same folders are in your home folder. Both local agents and API providers can use them.

Pin a skill with the Skills button

1. Open the chat.
2. Click **Skills** beside the message box. The window **Skills for this chat** opens.
3. Type in **Search skills...** if the list is long.
4. Click each skill you want. Click it again to remove it.
5. Click **Apply**.

Check the button. It now shows the number pinned, for example **1**. To remove all pinned skills, open the window again, click **Clear**, then click **Apply**.

Pin a skill with a slash command

1. Type **/** at the start of an empty message box. A menu opens with **/skills** (Browse all installed skills) and up to eight skills.
2. Keep typing to narrow the list, for example **/plot**.
3. Click a skill. The hint **"/name is active for this chat"** confirms it.

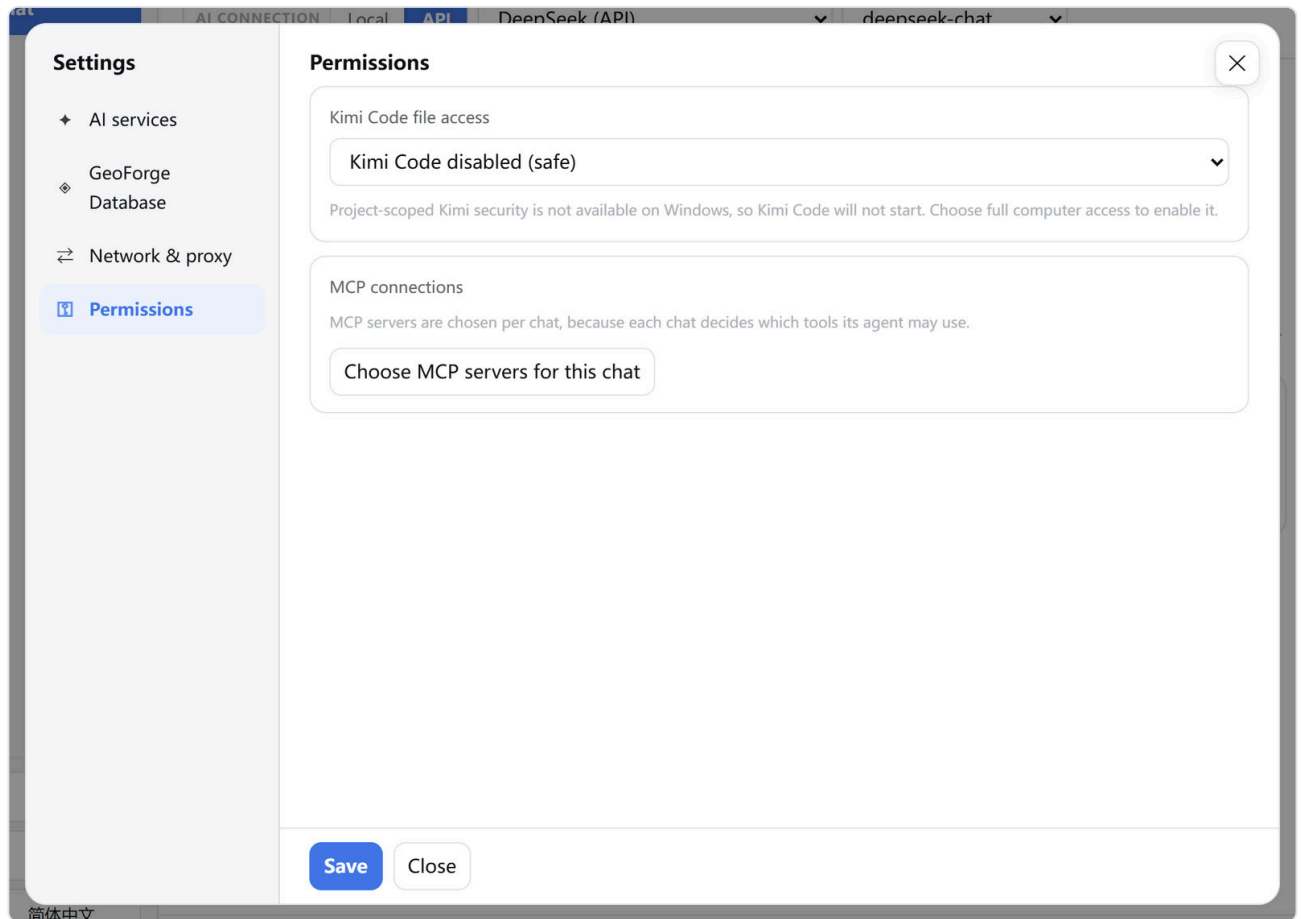
You can also type the skill name and your request in one message and press Enter. For example, if you have a skill named **hydrograph-plot**, type **/hydrograph-plot Plot daily discharge for 2015**. GeoForge pins the skill and sends the rest as your message. Sending **/skills** alone opens the **Skills for this chat** window.

MCP servers for a chat

MCP (Model Context Protocol) servers connect an agent to an outside service, such as GitHub. GeoForge does not run MCP servers itself. It lists the servers you have already set up for the local agents Claude Code and OpenAI Codex, and lets you choose which ones each chat may use.

Caution: Only the local agents Claude Code and OpenAI Codex (**Local** under **AI CONNECTION**) can use MCP servers. API providers, such as DeepSeek, use only GeoForge's own tools. Local agents were not tested end to end on Windows for 0.6.54.

1. Open the chat that should use the server. The button in step 4 does nothing when no chat is open.
2. Click **Settings** at the bottom of the sidebar.
3. Click **Permissions** on the left.



Check that **MCP connections** shows the button **Choose MCP servers for this chat**. **Kimi Code file access** is explained in Chapter 2.

Tip: If you changed anything else in Settings, click **Save** now. The next step closes the Settings window without saving.

4. Click **Choose MCP servers for this chat**. The Settings window closes and the **MCP connections** window opens.
5. Click each server this chat should use, so that it is ticked. Each line shows which agents it is set up for, such as **Configured for codex, claude · stdio**.
6. Click **Apply**.

Check that the servers you want are ticked before you click **Apply**. You can change the choice later in the same chat.

The **GitHub MCP** card offers **Configure for Codex** when the Codex CLI and either the official `github-mcp-server` program or Docker are installed. After it is configured, the message says that GitHub login opens the first time the server is used. Otherwise the card shows a link, **Official setup guide**.

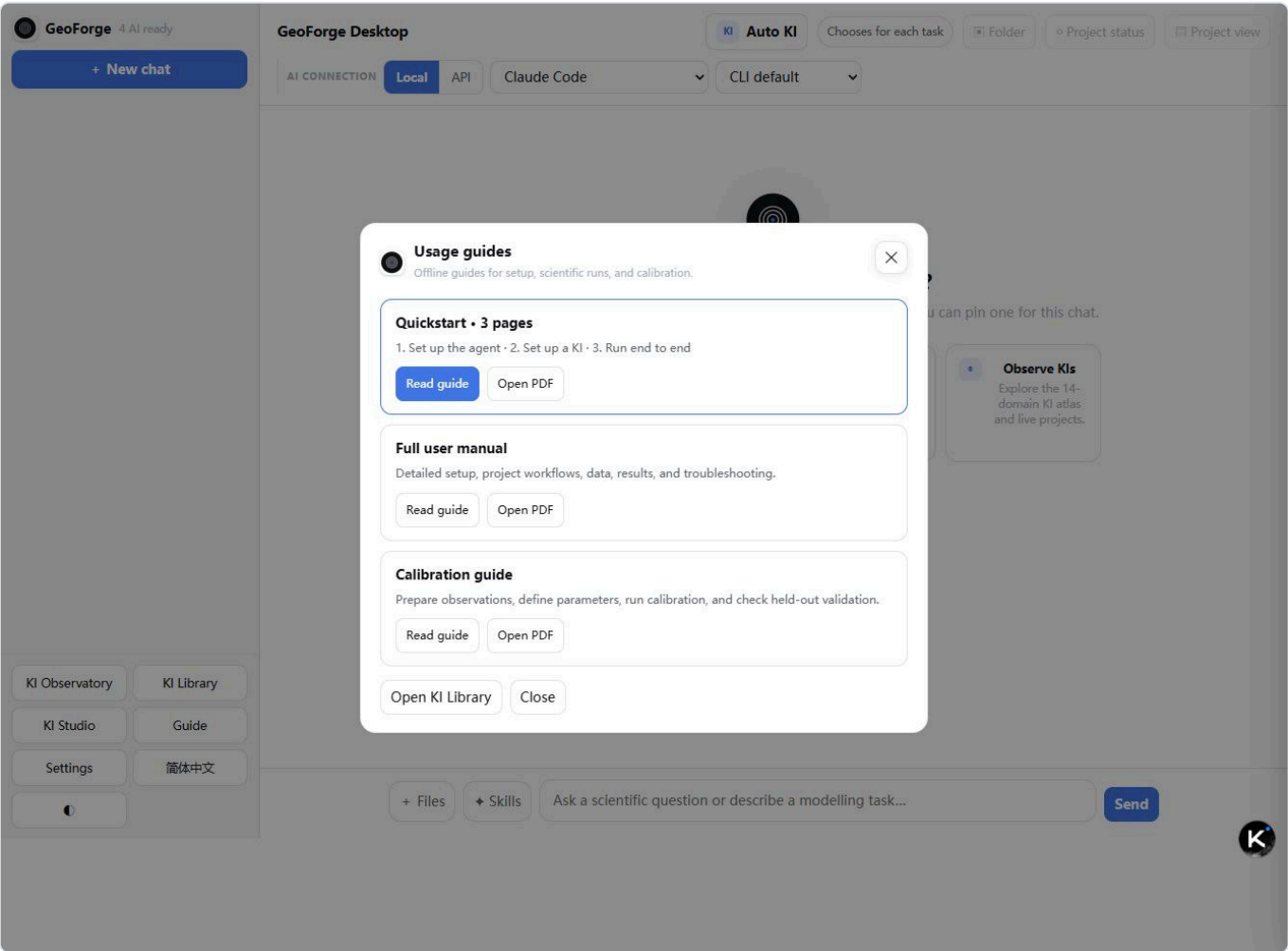
An MCP server gives the agent more reach. It does not replace the KI's checks or the project's run records.

Guide: three offline documents

Click **Guide** at the bottom of the sidebar. The menu opens three guides in the current interface language:

Guide	Use it for
Quickstart · 3 pages	One page each for agent setup, KI setup and an actual end-to-end SHAW run
Full user manual	Detailed workflows, real screenshots and troubleshooting
Calibration guide	Parameters, observations, training/holdout separation, real-model adapters and the tested reference-recovery benchmark

Each card has **Read guide** for local HTML and **Open PDF** for the printable document. The guides are bundled with the app and do not require an Internet connection. Provider login, software/data downloads and cloud AI calls still require their own connectivity.



Check that all three cards are visible. The quickstart PDF is exactly three pages; the manual and calibration guide are longer.

Calibration

Calibration must run the actual model through its KI adapter. Chapter 10 contains the full workflow, the meanings of the Calibration card, and a reproducible SHAW reference-recovery test. Open **Guide** → **Calibration guide** for the same material as a standalone document. A reference-model benchmark is not observational validation.

If something goes wrong

What you see	What to do
KI updates reports "GeoForge kept the current KI library...", or the notice Current KI library kept appears.	Nothing. This is expected on Windows in 0.6.54, and your library is unchanged. Click Dismiss .
"Could not update the KI library. GeoForge kept the last validated version." or the notice KI update not completed	The network or a package check failed, and your library is unchanged. If you need a proxy, tick GitHub & KI updates under Settings → Network & proxy , click Save , then click Check again .
" <i>name</i> cannot be imported yet. Fix the blocking findings and try again."	Read the BLOCK lines and send them to whoever made the package.
A BLOCK line says "a KI named '...' already exists; rename the zip and re-import".	Rename the <code>.zip</code> file, then click Check & Import again. The KI takes its name from the file name.
Explain my data reads Connect AI first .	Connect an AI in Settings → AI services (Chapter 2), then reopen the KI Library .
Ask Agent shows "Could not create chat" and "No existing chat was reused".	The question was not sent and no existing project changed. Close the message and try again.
After Ask Agent : "The new chat is ready with the question filled in. Connect an AI, then press Send."	Click API or Local under AI CONNECTION , pick a ready provider, then click Send .
How one KI works does nothing.	Open a KI first: click a domain, then a KI card.
KI Studio: "Git is not installed. Install the Xcode command line tools, then retry."	On Windows, install Git for Windows. On macOS, install the Xcode command line tools. Then click Install KDT engine again.
KI Studio: "This page cannot be cloned. Use repository root: ..." under the address field	Paste the address the message suggests: the repository's main page, not a file or folder inside it.
KI Studio: "Folder picking is available in the desktop app. Paste a path here in browser mode."	On Windows, paste the path from File Explorer's address bar.
Skills for this chat shows "No matching skills found."	No skills are installed in the skill folders, or none matches your search.
Choose MCP servers for this chat does nothing.	Open a chat first, then open Settings → Permissions again.
MCP connections shows "No MCP servers are configured for the local agents yet."	Set the server up for Claude Code or Codex first, following its own instructions.
You closed a black console window during a calibration.	That model run counts as failed. Leave the other windows open. When the turn ends, ask the agent to check the run in <code>calibration\runs\</code> .
Calibrate with agent in a completed chat only gets an explanation.	A completed chat is read-only. Start a new chat for the calibration.
The Calibration card shows Engine unavailable .	This build cannot load the engine, so no calibration can run. Report it (Chapter 9, section 9.18).

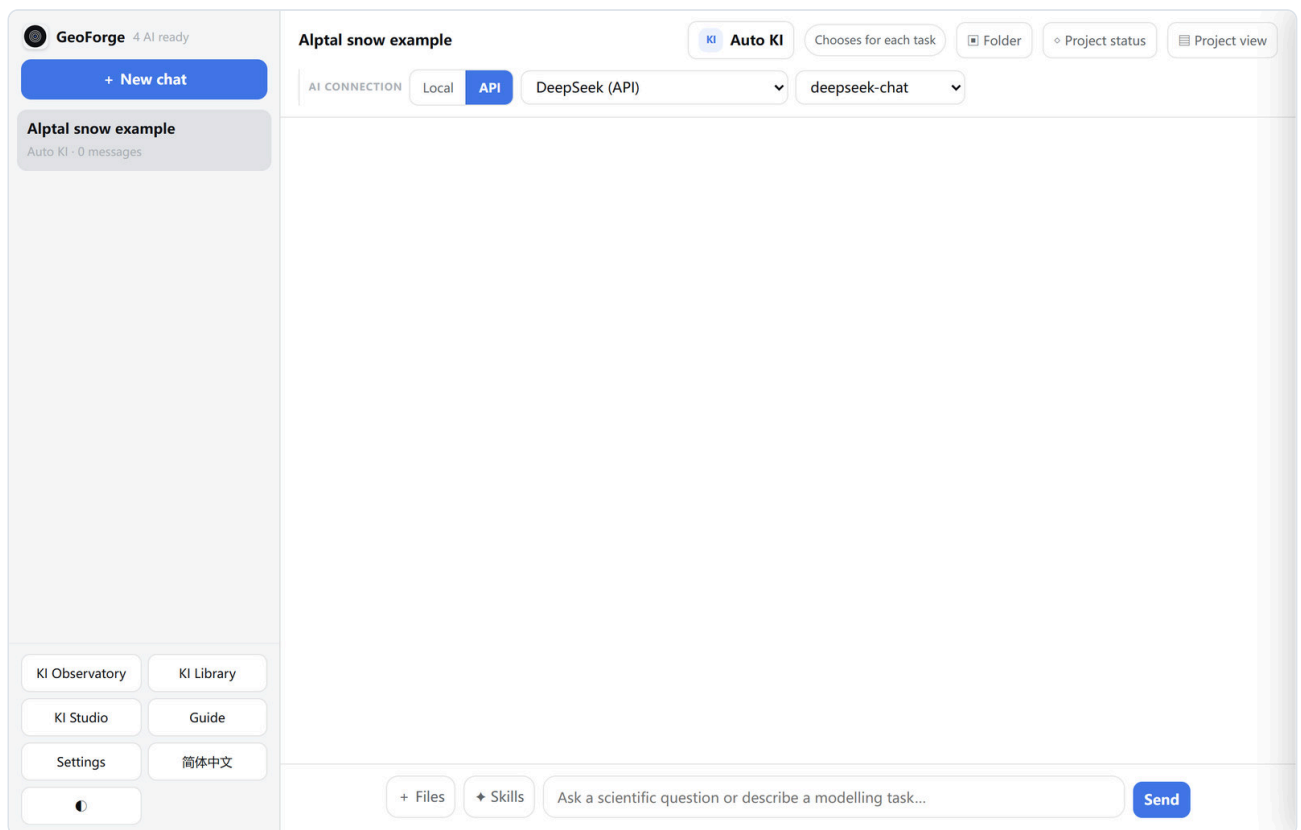
8 Language, appearance and keyboard

In this chapter you will switch GeoForge between English and Simplified Chinese, choose a light or dark look, and learn the few keys GeoForge responds to.

Switch the interface language

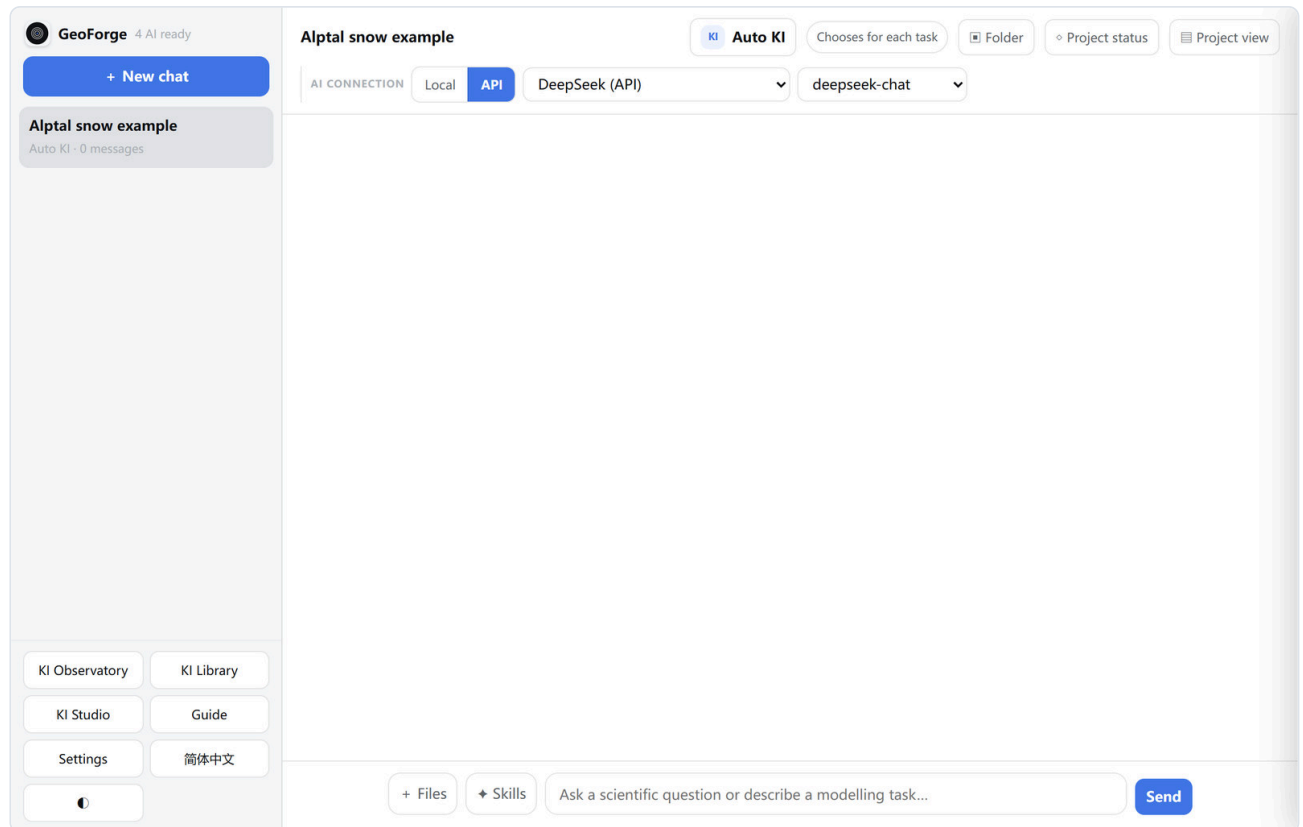
GeoForge has two interface languages: English and Simplified Chinese. The language button is at the bottom of the sidebar, to the right of **Settings**. It always shows the language you will switch to: **简体中文** while the interface is in English, **English** while it is in Chinese.

1. Send or copy anything you have typed in the message box but not sent. The next step reloads the page, and the reload clears unsent text and the list of files attached to that message. Files you already added stay in the project folder.
2. At the bottom of the sidebar, click **简体中文**.



Check that **简体中文** sits to the right of **Settings**, with the  button on the row below **Settings**.

3. Wait for the page to reload. Buttons, panel titles and hints are now in Chinese.



Check that the sidebar button now reads **English** and that **Send** has become 发送. The chat name, the provider *DeepSeek (API)* and the model *deepseek-chat* do not change.

4. To return to English, click **English** in the same place. The page reloads in English.

After either reload, GeoForge opens the chat at the top of the chat list. If you were working in another chat, click it in the list.

The language button is on every GeoForge page: the main window, **KI Library**, **KI Studio**, **KI Observatory** and **Agent setup**. One choice applies to all of them. A page that is already open in another tab picks up the change when you reload it.

The language at first start

When GeoForge has no saved choice, it uses the language that the browser reports. On Windows this is your default browser's language setting. Any Chinese setting, including Traditional Chinese, opens the Simplified Chinese interface. Every other language opens English.

On macOS the GeoForge window can open in English even on a Chinese system. If it does, click 简体中文.

Writing in Chinese switches the interface

When you send a message that contains any Chinese character, the interface switches to Chinese at once, without a reload. This happens even for a single Chinese place name in an English sentence, such as "Run FSM2 for 淮河".

The AI gets a matching instruction. GeoForge asks it to reply in the language of your latest message. When that message contains any Chinese character, GeoForge asks for a reply entirely in Simplified Chinese, including plot titles, captions and axis labels. It also asks the AI to keep commands, code, file paths, KI names and exact scientific identifiers unchanged.

To go back to English:

1. Click **English** at the bottom of the sidebar. The page reloads in English.

2. Write your next message without any Chinese characters. GeoForge then asks the AI to reply in English.

Tip: To stay in English, write place names in English or pinyin (for example *Huai River*, *Bengbu*). The GeoForge Database search boxes match words in the catalogue's English text, so English names and dataset IDs also find data that Chinese names miss. See Chapter 3, GeoForge Database.

What stays in English in Chinese mode

The Chinese interface translates most of GeoForge's own buttons, menus, panel titles, dialogs and hints. GeoForge never translates these:

- model names, KI names, file and folder paths, commands and units;
- chat and project names, which stay as you typed them;
- the AI's messages, which follow the language of your latest message;
- raw technical output, such as logs and reports in grey boxes.

Some screens show English text even in Chinese mode. Expect these:

Where	English text you will see
Plan approval card	The title Approve the plan? and the options Approve and start and Modify the plan
Data cards in the chat	Download N datasets from Baidu Pan, Some approved data could not be fetched, Retry the missing data
Project status summary and data labels	"Completed — every step has a verified receipt", "Stopped by you — send a message to continue", "Data needs attention"
Chat messages from GeoForge	"Software verified. Starting your approved plan in a new session..."
Banners above the chat	"Connect an AI to begin." and "... is not verified on this machine"
Chat list	The line under each chat name, for example "Auto KI · 0 messages"
Right-click menu in a text box	Cut , Paste and Select All (Copy is translated)
Settings → AI services	The report shown by Test AI & GitHub (the button itself is translated)
Settings → GeoForge Database	Error messages about the token, quota or network
Windows installer	Every page of the setup wizard
Windows system tray menu	Open GeoForge and Exit GeoForge

The approval card and data cards work the same way in both languages. Chapter 5, Your first project, explains each option.

Choose light or dark

1. At the bottom of the sidebar, click . The window switches between light and dark at once, with no reload.
2. To switch back, click  again.

Check that the sidebar, chat area and message box all turn dark. The theme changes only screen colours; plots and files in your project folder are not affected.

Until you click , each page follows the light or dark setting of your computer, or of your browser if you set one there. Your choice applies to the main window, **KI Library**, **KI Observatory** and **Agent setup**.

KI Studio has its own  button and its own setting. It starts in light mode, whatever the other pages use, until you click its .

Keyboard

Besides the usual browser keys (Tab moves between controls; Enter or Space presses the focused button), GeoForge responds only to the keys below. There is no shortcut for a new chat, for switching chats or for stopping a run. The Command-K shortcut from older manuals does not exist.

Key	Where	What it does
Enter	Message box	Sends the message, like Send
Shift+Enter	Message box	Starts a new line without sending
Enter	Project name in the New project chat dialog	Creates the chat, like Create chat
Enter	GeoForge Database search box in Project status → Details	Runs the search, like Search
Enter or Space	A domain or KI that you reached with Tab in KI Observatory	Opens it, like a click
Esc	A dialog in the main window (Settings , New project chat , Knowledge Infrastructures for this chat , Skills for this chat , MCP connections , How GeoForge works , a question card) or in KI Library	Closes it. Esc does not close the Project status or Project view panels.
Ctrl+C, Ctrl+X, Ctrl+V, Ctrl+A (macOS: Command)	Text boxes and selected text	Copy, cut, paste, select all

To copy a whole chat message, click **Copy** on the line with the sender's name. Right-clicking in a text box opens a small menu with **Cut**, **Copy**, **Paste** and **Select All**. Right-clicking on selected text elsewhere shows only **Copy** and **Select All**.

Caution: Pressing Esc, clicking **×** or **Not now**, or clicking outside a question card closes it without answering. The plan approval card is a question card too. The card does not come back by itself on that page. To answer it, open **Project status** and click **Respond**, or reload the page and open that chat again.

Tip: With a Chinese input method, the **Project name** field ignores the Enter that confirms a candidate, so the chat is not created half-named. The message box does not check for this, so with some input methods that Enter can send the message. Confirm candidates with Space or a number key there, and press Enter only to send.

Preferences after a restart

GeoForge keeps three preferences in the browser's storage for the page's address: the interface language, the theme, and the **Local | API** switch under **AI CONNECTION**. Each time GeoForge starts, it uses a new local address, such as `http://127.0.0.1:52814/` . The browser treats the new address as a new site, so these preferences go back to their defaults.

Preference	Default after a restart
Language	The language your browser reports
Theme	Your computer's or browser's light or dark setting (KI Studio : light)
Local API	The kind of AI used by the chat that opens; Local when that chat has no AI yet

Platform	What to expect
Windows	GeoForge opens in your default browser at a new address on every launch, so expect all three preferences to reset each time. Within one launch they are kept, including when you reload the page or reopen it with tray → Open GeoForge .
macOS	GeoForge opens in its own window, which also uses a new local address at each start. Expect the same reset.

Your chats, projects, API keys and other **Settings** are stored by GeoForge itself, not by the browser. They are kept across restarts.

Tip: If you use only an API key (for example DeepSeek), a new start may show "No local agent CLI is installed. Open AI Settings for setup details." ("AI Settings" is the old name of **Settings** → **AI services**.) Click **API** under **AI CONNECTION**, or open a chat that already uses your API provider. See Chapter 2, Connect an AI.

Because the address changes, a bookmark to GeoForge stops working after a restart. On Windows, while GeoForge is running, reopen the page with tray → **Open GeoForge**. After you have quit it, start it again from the Start menu. See Chapter 1, Install and first launch.

If something goes wrong

What you see	What to do
The interface switched to Chinese on its own.	You sent a message with a Chinese character, or your browser language is Chinese. Click English at the bottom of the sidebar.
The AI keeps replying in Chinese.	Your latest message contained Chinese characters. Write the next message without any, and GeoForge asks the AI for English.
A message you were typing disappeared after you switched language.	The language button reloads the page, which clears unsent text and the attachment list. Type it again. Files you had added are still in the project folder. Next time, send or copy the text before you switch.
After switching language, a different chat is shown.	The reload opens the chat at the top of the chat list. Click your chat in the list.
Language, theme or Local API went back to the default after a restart.	This is expected, because the address changes at each start. Set them again.
A question or approval card vanished after you pressed Esc.	Esc closes the card without answering. Open Project status and click Respond .
A Chinese-mode screen shows English text, such as Approve the plan?	This is expected. See the table in "What stays in English in Chinese mode".
KI Studio stays light while the other pages are dark.	KI Studio has its own setting. Click  in KI Studio .
Enter sent a message before you finished.	Use Shift+Enter for new lines. With a Chinese input method, confirm candidates with Space or a number key.

9 Troubleshooting, known issues and glossary

In this chapter you will find the fix for the problems people meet most often in GeoForge Desktop 0.6.54, read the known issues of this release in plain words, look up the words the app uses, and learn how to report a problem so that it can be fixed.

Tip: Before you retry or restart anything, read the last real message. Most projects that look stuck are waiting for you: an answer, an approval, a file, or one click.

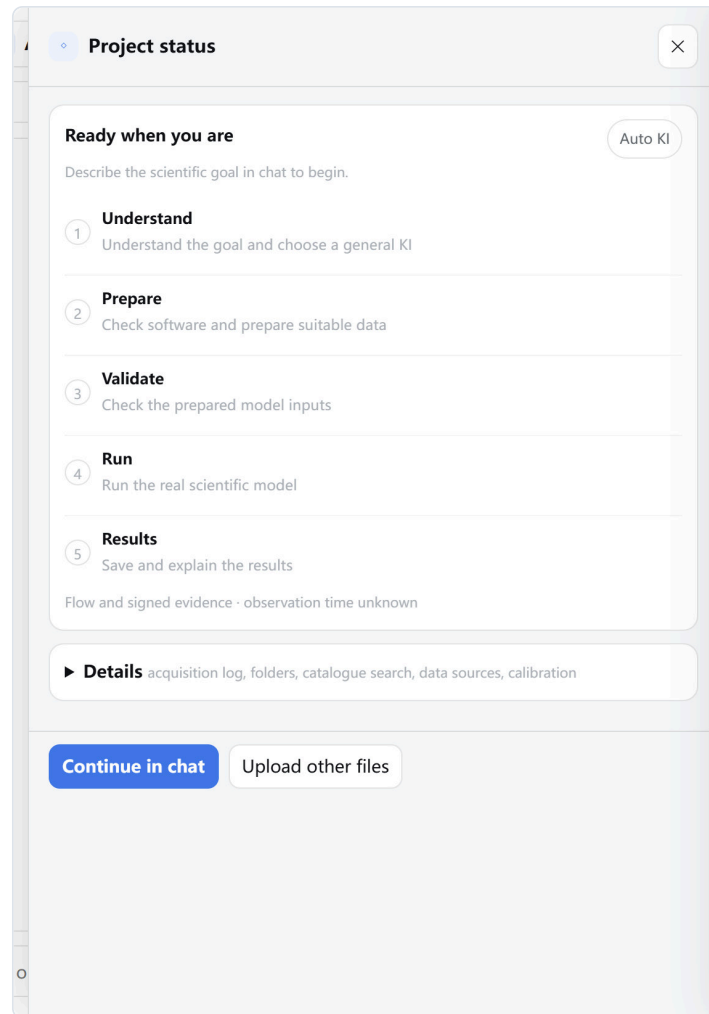
9.1 Where GeoForge tells you what is wrong

Look in these places, in this order:

Where	What it tells you
Top of the sidebar, next to GeoForge	How many AI connections work: N AI ready , or AI setup needed .
Banner above the chat	No usable AI (Connect an AI to begin.), no usable AI of the kind selected in AI CONNECTION , or a KI whose software is not set up yet (... is not verified on this machine.).
Activity bar below the message box, while the AI works	What the agent is doing, for how long, and the buttons Recheck , Stop and + New chat . The chat's entry in the sidebar shows a short state: Active , Running · quiet , Check status or Connection issue .
◇ Project status button in the chat header	A coloured dot. Hover over it to read the state and the summary.
◇ Project status panel	A headline, a one-line summary, the five stages, Data in this plan and Details . When something waits for you, a One thing needs you card sits on top.
Chat lines that start with GeoForge verification: , GeoForge needs you: or [GeoForge could not finish this turn: ...]	Messages from GeoForge itself, not from the AI. GeoForge verification: reports what the signed run records show.

To read the project's state:

1. In the chat header, click ◇ **Project status**.
2. Read the bold headline (for example **One thing needs you**, **GeoForge is working**, **Project needs attention**, **Project complete** or **Ready to continue**) and the grey summary line under it.
3. If the problem is about data, folders or logs, click **Details**.



Check: the headline (here **Ready when you are**) and the line under it say what GeoForge is waiting for. **Details** holds the acquisition log, folders, catalogue search, data sources and calibration.

Inside **Details**, the first card, **Status evidence**, lists where the status comes from. Its line **Agent/app report (not scientific proof)** ends with GeoForge's latest status sentence, for example "Running — 2 steps done, 0 to go". This chapter calls it the *status line*.

Tip: The five stages (**Understand**, **Prepare**, **Validate**, **Run**, **Results**) show roughly where the work is. During planning and data download the marker stays on **Understand**. Trust the summary line, not the marker.

9.2 Quick fixes: symptom, cause, fix

Find your symptom, then follow the fix. Longer fixes point to a section below.

Starting and installing

Symptom	Likely cause	Fix
Windows protected your PC when you run the installer.	GeoForge is not code-signed.	Check the SHA-256 first, then More info → Run anyway . See 9.12.
macOS refuses to open GeoForge Desktop.app .	The Mac app is not notarised by Apple.	Open Anyway in Privacy & Security, or the <code>xattr</code> command. See 9.13.
The portable Windows copy does not start.	The zip is not fully extracted, or <code>GeoForge Desktop.exe</code> was moved away from its <code>_internal</code> folder.	Extract the whole zip to a normal folder. Run <code>GeoForge Desktop.exe</code> inside the folder <code>GeoForge Desktop 0.6.55 Windows</code> .
You closed the browser tab and GeoForge is gone (Windows).	On Windows the tab is only a view. GeoForge keeps running in the tray.	Double-click the GeoForge tray icon, or right-click it → Open GeoForge . It may be under the <code>^</code> arrow next to the clock.
A bookmark to GeoForge stopped working (Windows).	GeoForge uses a new address <code>http://127.0.0.1:<number>/</code> at every launch.	Open GeoForge from the tray icon, not from a bookmark.
Two GeoForge tray icons, or two GeoForge tabs with different addresses (Windows).	GeoForge was started a second time.	Keep one copy and exit the other. See 9.14.

AI connection

Symptom	Likely cause	Fix
Sidebar shows AI setup needed ; banner Connect an AI to begin .	No AI provider is usable.	Add an API key or sign in to a CLI in Settings → AI services . See 9.3.
A CLI you just installed still shows not installed .	GeoForge reads the list of installed programs only when it starts.	Exit GeoForge from the tray, start it again, then Recheck local CLIs . See 9.4.
Kimi Code is signed in but never starts (Windows). The chat shows <code>[Kimi Code was not started: project-scoped security could not be applied ...]</code> .	On Windows, Kimi Code stays off until you allow full computer access.	See 9.3, "Kimi Code on Windows".
Gemini CLI or Qwen Code keeps an amber dot in Settings , and the AI list in the chat header shows ... — login checked on first use .	GeoForge cannot check their sign-in in advance.	Nothing to fix. A failed sign-in shows in the chat the first time you use them.
The chat ends with <code>[GeoForge could not finish this turn: ...]</code> , or the activity bar shows The live connection may be interrupted (sidebar entry: Connection issue).	Often the connection to the AI provider broke: network, proxy or provider outage. The text after the colon names the error.	Check Network & proxy , then send the message again. See 9.3.

Chat and cards

Symptom	Likely cause	Fix
Send is grey.	The AI is still working, a file is still uploading, the selected AI is not usable, or this chat's own AI stopped working.	See 9.5.
The AI CONNECTION buttons, the AI and model lists and the KI button are grey. Tooltip: Fixed for this chat — start a new chat to switch AI.	A chat keeps its AI, model and KI from the first message on.	Click + New chat and choose them before the first message. See 9.5.
The interface suddenly switched to Chinese.	You sent a message that contains Chinese characters.	Click English at the bottom of the sidebar. The page reloads in English. See Chapter 8.
A question card or the approval card disappeared.	Not now , ×, Esc or a click outside the card hides it for this page visit. The question is still waiting.	◇ Project status → One thing needs you → Respond . See 9.6.
You answered in the chat, but GeoForge still waits.	Only an answer given on the card is saved as your answer.	Reopen the card (9.6) and answer there.
Choose one option first. in red on the card.	You clicked Continue with this choice without selecting an option.	Click an option, then Continue with this choice .
Enter your answer or file location below.	You chose Use my own answer / file and left the note empty.	Type your value, date range or file location in Optional note for the agent .
Return to the chat that asked this question.	The card belongs to another chat.	Open that chat in the sidebar, then answer.

Approval

Symptom	Likely cause	Fix
You clicked Approve and start and nothing happened.	That click only selects the option.	Click Continue with this choice .
After you approved, a new approval card appeared. The chat says The updated card is in the chat; ...	GeoForge re-issued the card: your data pick was re-pinned, an uploaded file was linked to its input, or a server clip size estimate changed.	Read the new card. Approve it again if it still fits.
These still need your decision before anything runs: ...	An input marked please provide has no file, or a decision is open.	See 9.7.
Not ready to execute yet on the approval card.	A step has no tool, or an input is still missing.	Choose Modify the plan . See 9.8.
The card's Data list names a different source than Data sources, your pick or a Waiting on you line.	Known issue 1.	Do not approve. Choose Modify the plan . See 9.16.
GeoForge Database access is off or not activated, but this plan still needs it for ... Nothing was approved or downloaded.	The plan needs Database data, but the Database is off or not activated.	Activate it (Chapter 3) and approve again, or choose Modify the plan to use public sources or your own files.
Your saved answers cannot be read (...). Fix or remove that file, then approve again.	The saved-answers file <code>.geoforge\user-answers.json</code> was edited or damaged.	Undo any edit you made to it. If you cannot, report the problem (9.18) before you remove the file: removing it deletes your saved answers.

Data

Symptom	Likely cause	Fix
Card Some approved data could not be fetched , or a row in Data in this plan says Fetch failed .	Network, server, daily quota, or files already in the target folder.	See 9.9.
You clicked Files are in place, continue , but the Baidu Pan card stays open.	The files are not exactly in the Place at folder, or the copy has not finished. Partial files (for example <code>.part</code> , <code>.crdownload</code> , or Baidu's <code>.bc!</code>) are ignored.	Click Copy path , paste it into the File Explorer address bar, move the complete files there, then click the button again. See Chapter 3, 3.7.
Banner Approved data acquired · run not started (or software setup needed).	The automatic download finished in the background. GeoForge waits for you before it runs the model.	Click Start the approved run in the banner.

Running and stopping

Symptom	Likely cause	Fix
Activity bar: No new work evidence; the turn may be stuck (sidebar entry: Check status).	Nothing new for about 90 seconds. Compiles and model runs can be quiet for a long time.	Click Recheck . If nothing changes for much longer than the step should take, click Stop , then send a message that says what you saw.
After Stop or Exit GeoForge , the computer stays busy, or new files keep appearing.	Known issue 2: a job the agent started through Git Bash escaped Stop.	End it in Task Manager. See 9.10.
After Stop , the chat shows [...] and the headline reads Ready to continue .	You pressed Stop . The stopped attempt is recorded as stopped, not as failed. The status line (9.1) reads "Stopped by you — send a message to continue".	Send a message when you want to continue.
After Approve and start , GeoForge spends a long time setting up software.	The KI's software is not yet verified on this computer. A first setup can download and compile it.	Wait and watch the activity bar. Known issue 3: setup may also run the model's example.
Many black console windows open during calibration (Windows).	Known issue 4.	Leave them open. Closing one counts that model evaluation as failed.

Results

Symptom	Likely cause	Fix
A GeoForge verification : line in the chat, and the project is not Completed .	Some approved steps have no passing run record, or files exist that no passing run produced.	See 9.11.
Headline Project needs attention , summary An execution attempt failed; agent diagnosis is needed .	A run's output failed GeoForge's automatic checks.	Send "Please diagnose the failed step and retry." See 9.11.
Completed . followed by "N earlier failed attempts were superseded by a passing retry".	An earlier attempt failed and a later one passed.	Nothing to fix. The failed attempts stay listed in <code>runs\evidence.json</code> .
A Completed project still shows Produced; scientific checks pending , or Project View panels end in "— experimental / unvalidated".	Completed means every approved step ran and passed GeoForge's automatic checks. It does not mean the results are validated.	Expected. See Completed in the glossary (9.17).
After Completed , the agent will not run anything more.	Completed is final. The chat can still answer questions and draw plots from the existing files.	Start a new chat for another period, scenario or calibration.

GeoForge Database

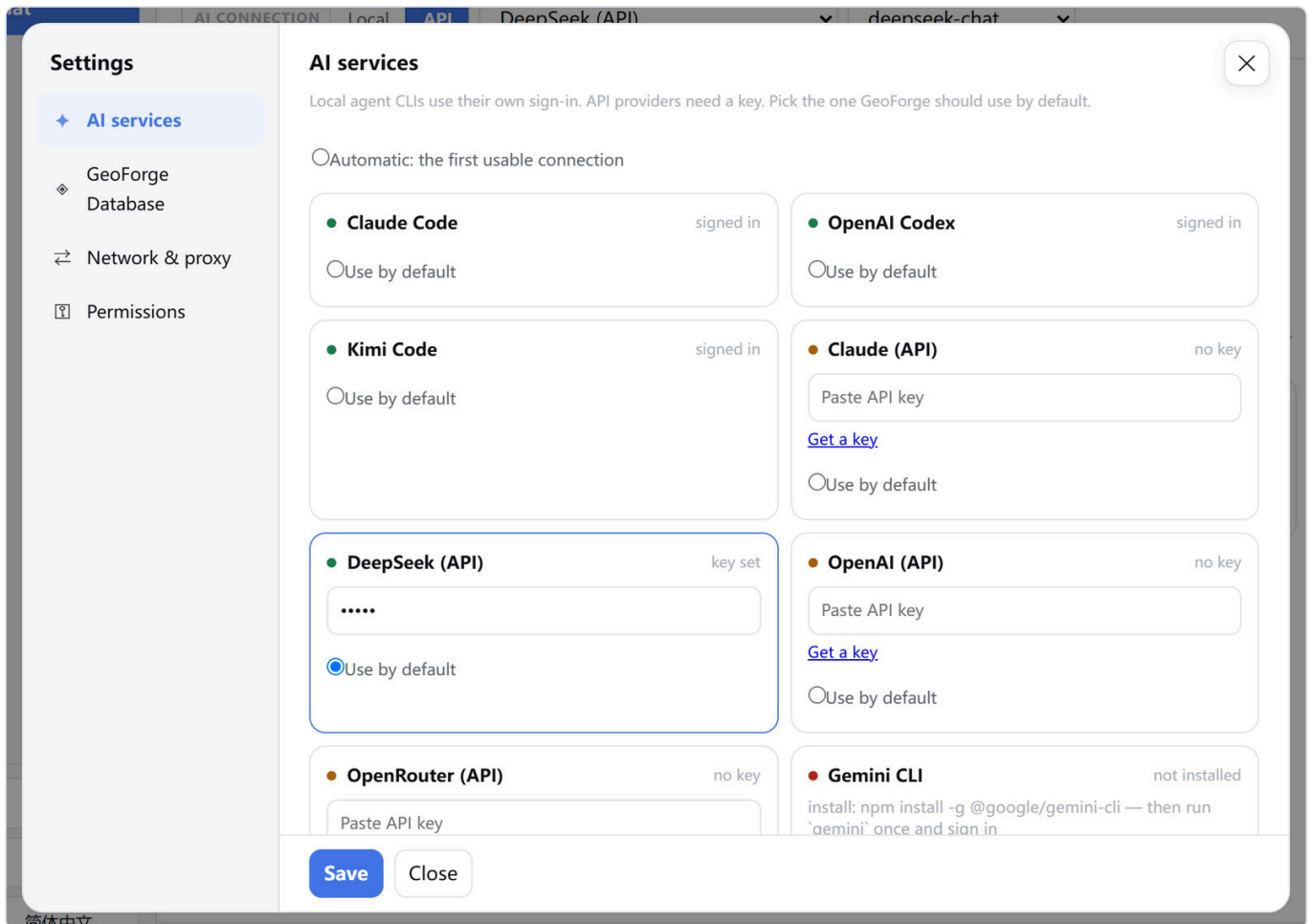
Symptom	Likely cause	Fix
Settings → GeoForge Database shows Not configured , Off , Unavailable or Not synced , or a token message.	Token missing, wrong, expired or revoked; access set to off; or no connection.	See 9.15 and Chapter 3.

9.3 No AI is ready

The sidebar shows **AI setup needed**, and a banner says **Connect an AI to begin**. "Open AI Settings" in the banner means the **Settings** button in the sidebar.

A second banner can describe the kind selected under **AI CONNECTION: No API connection yet**. when **API** is selected and no key works, or **No local agent CLI is installed**. (or **Local CLIs are installed but not ready**, with a **Recheck** button) when **Local** is selected.

1. In the sidebar, click **Settings**.
2. Click **AI services**.
3. Read the dot and the status word on each card.



Check: a green dot with **signed in** or **key set** means usable. **no key** or **not installed** means not usable yet.

4. For an API provider, paste your key into the **Paste API key** box. Use **Get a key** if you have none.
5. For a local CLI, install it with the command shown on its card and sign in once, then see 9.4. Chapter 2 has the exact commands.
6. Click **Save**. Closing the window without **Save** discards your changes.

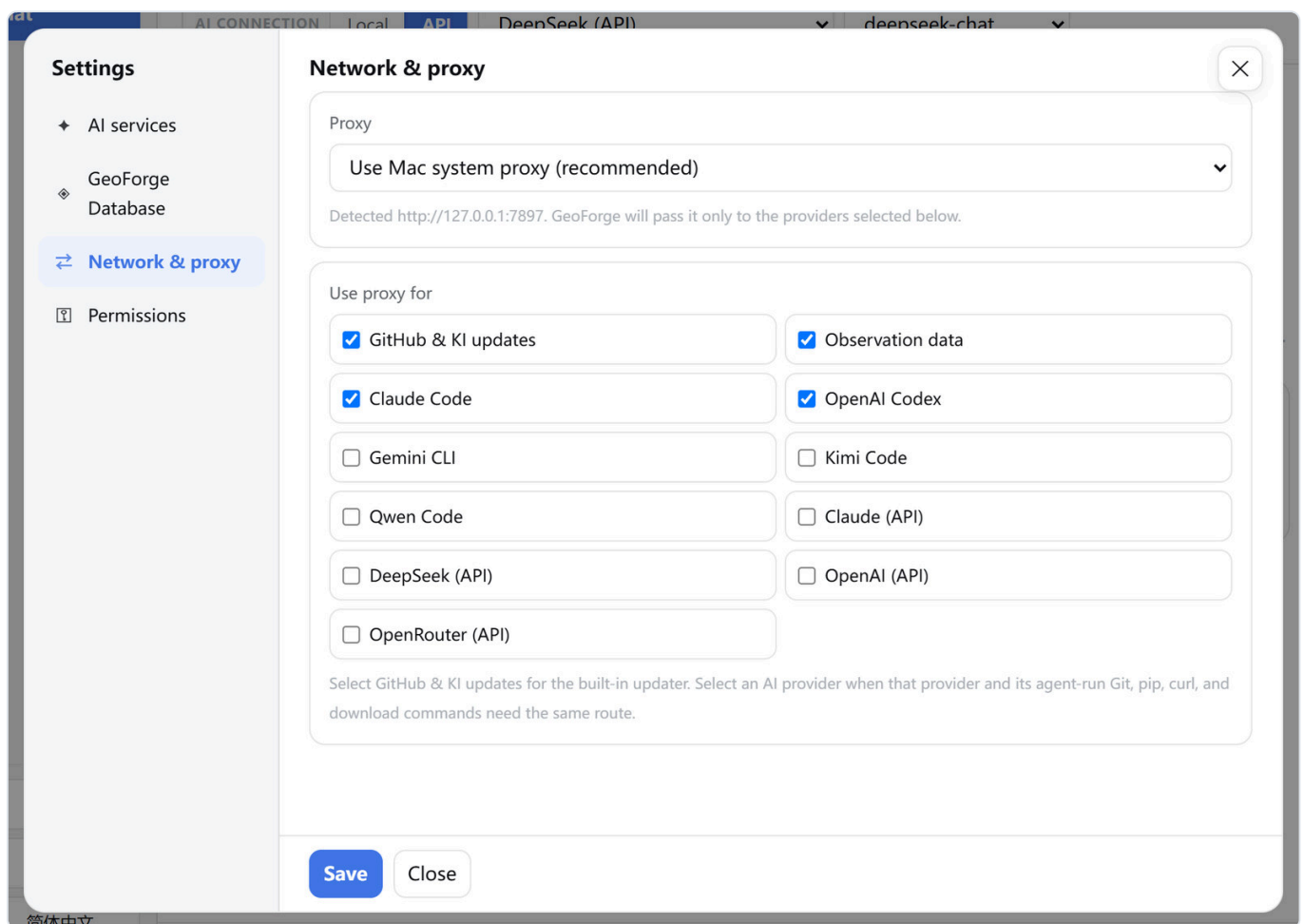
Chapter 2 explains each provider and has a longer error table.

The connection breaks during a turn

A *turn* is one round of work that starts when you send a message. The chat ends with [GeoForge could not finish this turn: ...] (for example a `ConnectionAbortedError`), or the activity bar shows **The live connection may be interrupted**.

1. In **Settings**, click **Network & proxy**.
2. Under **Proxy**, check the detected address. On Windows, **Use Mac system proxy (recommended)** means your Windows system proxy.

- Under **Use proxy for**, tick the provider you use if your network reaches it only through the proxy. API providers such as **DeepSeek (API)** are not ticked by default.
- Click **Save**.



Check: the address under **Proxy**, and a tick next to the provider your chat uses.

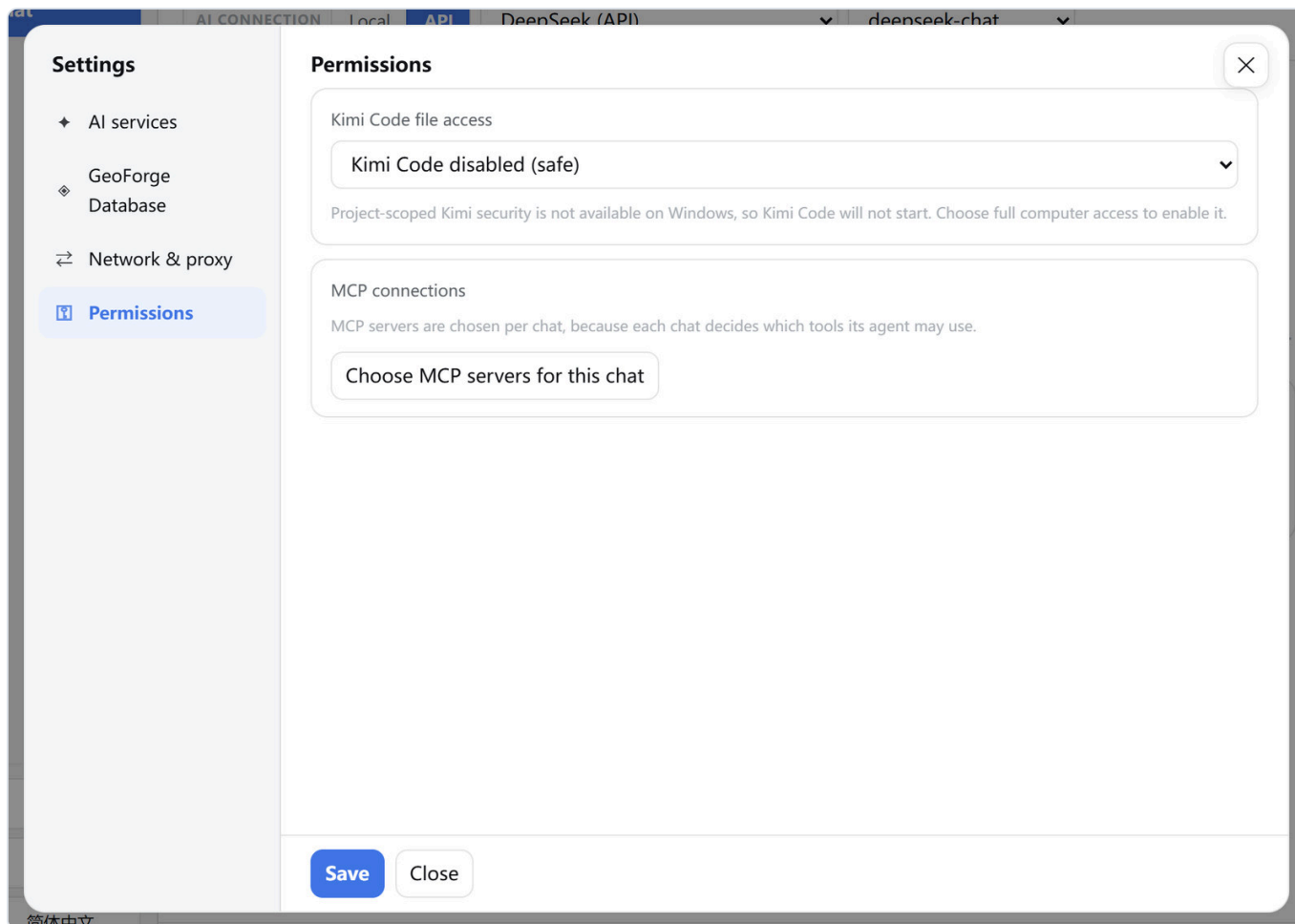
- Send your message again. The chat keeps its history.

Tip: Test AI & GitHub on the **AI services** page checks the connection step by step. Its last step sends one tiny real message to the provider, which the provider may bill. If step [2/6] (Python) fails on Windows, it prints macOS advice (`brew install python`). That advice does not apply to Windows; report the failure instead (9.18).

Kimi Code on Windows

Project-scoped Kimi security is not available on Windows, so Kimi Code does not start by default.

- In **Settings**, click **Permissions**.
- Under **Kimi Code file access**, choose **Enable Kimi with full computer access** only if you accept that Kimi Code can read and change every file your Windows account can access.
- Confirm the warning with **OK**.
- Click **Save**.



Check: the note under the list. With **Kimi Code disabled (safe)** it says Kimi Code will not start.

9.4 A CLI is not detected after you install it

GeoForge reads the list of installed programs (the Windows program path) only when it starts. A CLI installed while GeoForge runs can stay **not installed**, even after **Recheck local CLIs**.

1. Finish installing the CLI.
2. Sign in once in PowerShell, as the CLI's instructions say.
3. Wait until no chat is working.
4. Right-click the GeoForge tray icon.
5. Click **Exit GeoForge**.
6. Start GeoForge again from the Start menu.
7. Click **Settings** → **AI services** → **Recheck local CLIs**.
8. Check that the card now shows a green dot and **signed in**. **Gemini CLI** and **Qwen Code** keep an amber dot even when they work.

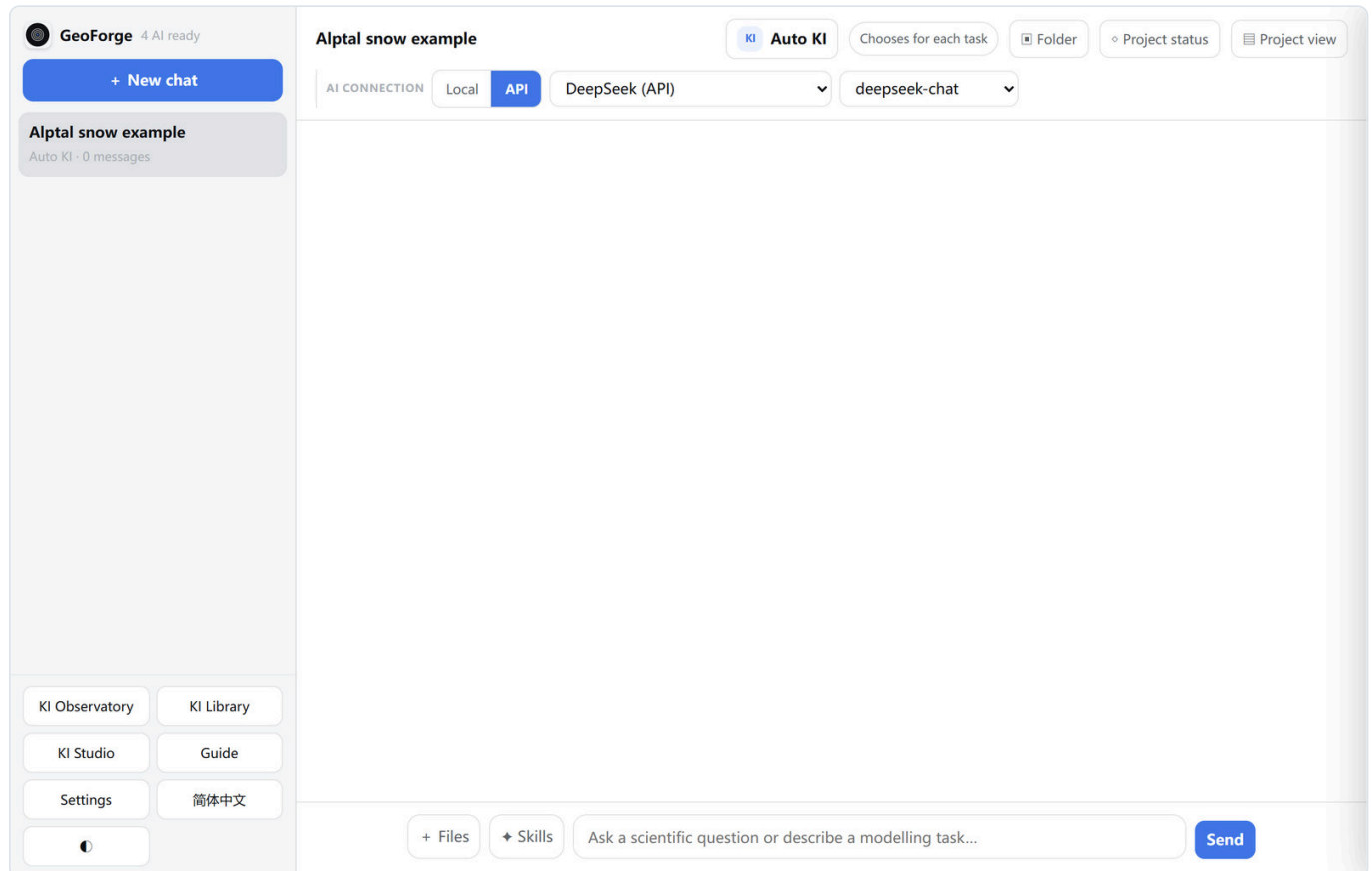
On macOS, quit GeoForge and open it again, then recheck.

Tip: CLIs installed with `npm` need Node.js. Claude Code on Windows also needs Git for Windows. OpenAI Codex older than 0.144 shows "update required". Chapter 2 lists the commands.

9.5 Your message is not sent, or the chat's AI is locked

Check these causes in order.

1. **The AI is still working.** The **Send** button reads **Working...**. Wait, or click **Stop** in the activity bar. You can work in another chat meanwhile: click **+ New chat**.
2. **The wrong kind of AI is selected in a new chat.** Look at the **AI CONNECTION** row in the header. **Local** uses a CLI on this computer; **API** uses a key. If you have only an API key and **Local** is selected, **Send** stays grey. Click **API**, then choose the provider and model.



Check: **AI CONNECTION** shows the kind, provider and model you want (here **API**, **DeepSeek (API)**, **deepseek-chat**), and the KI button shows the KI you want (here **Auto KI**). All of them are fixed when you send the first message.

Caution: On Windows the address changes at every launch, so the browser does not remember your last **Local** / **API** choice. A new chat takes the kind of the chat that is open when you click **+ New chat**. Right after a launch with no chat open, it starts on **Local**. If a CLI is ready, **Send** works, but the chat would then run on that CLI. Check the row before your first message.

3. **The chat is locked to its AI.** After the first message, the **AI CONNECTION** buttons, the AI and model lists and the KI button turn grey. The tooltip says **Fixed for this chat — start a new chat to switch AI**. The conversation lives with the AI that started it, so it cannot move to another one. Click **+ New chat** and choose again.
4. **This chat's own AI stopped working.** For example, its API key was removed or the CLI signed out. **Send** is grey and you cannot switch. Repair that same AI in **Settings** → **AI services** (paste the key again, or sign in and click **Recheck local CLIs**), or start a new chat.

9.6 A question card or the approval card disappeared

Not now, **X**, **Esc** and a click outside the card all hide it. GeoForge does not show the same card again by itself while the page stays open. The question or approval is still waiting.

1. Make sure the chat that asked the question is open.
2. In the chat header, click ◇ **Project status**.
3. At the top, find **One thing needs you**. Its bold line is the question, or **Approve the plan?** for an approval card.
4. Click **Respond**. The card opens again.
5. Select an option.
6. Click **Continue with this choice**.

Reloading the page (F5 in the browser tab) and opening the chat again also brings a waiting card back.

Caution: Answer on the card, not by typing in the chat. Only a card answer is saved as your answer and can be cited on the approval card ("From your answer: ..."). A typed reply leaves the question open.

9.7 Approval is refused because items still need you

After you approve, GeoForge answers **These still need your decision before anything runs:** and lists inputs. Nothing has started, and GeoForge shows the approval card again. If an input expects your own file, the message also says "Upload your file with the Upload button on the card". That text is out of date: the **Upload files** button is in ◇ **Project status**, not on the card.

1. Close the card with **Not now**.
2. Click ◇ **Project status** and find **Data in this plan**.
3. Under **You**, find the row named in the message. It shows **Waiting for you**.
4. Click that row's **Upload files**.
5. Choose your file. GeoForge saves it in `inputs\user\<input name>\` and says **Saved and queued for your next message (not sent, approved, or verified)**. Uploading does not approve or check anything.
6. In **One thing needs you**, click **Respond** to reopen the approval card.
7. Select **Approve and start**.
8. Click **Continue with this choice**. GeoForge links the file to the input and issues an updated card. The chat says **Your file is now named as the input: ...** and **The updated card is in the chat; approve it to start**.
9. Check that the updated card names your file, then approve it the same way.

If the open item is a data source, pick it under **Data sources, your pick** on the card. For anything else, choose **Modify the plan** and say in the note what to use. Chapter 6 (6.6.2) explains uploads for plan inputs.

Caution: Files added with **+ Files** or **Upload other files** go to `inputs\uploads\` and are never linked to a plan input. Use the row's own **Upload files** button.

9.8 "Not ready to execute yet" on the approval card

The approval card can end with a section **Not ready to execute yet**. It lists lines such as "step ... has no tool — not ready to execute" or "step ... input ... is still missing". In **Steps**, such a step shows **no tool — cannot execute** in red.

GeoForge does not block **Approve and start** here. If you approve anyway, a step without a tool cannot produce a run record, so the project cannot reach **Completed** without a plan change.

1. On the approval card, select **Modify the plan**.
2. In **Optional note for the agent**, say exactly what should change. For example: "Step stage_inputs has no tool. Use the KI's run tool to stage the forcing inside the project, and keep all outputs in the project folder."
3. Click **Continue with this choice**. GeoForge sends the plan back to the agent with your note.
4. Answer the question card if the agent asks one.

5. On the new approval card, check that **Not ready to execute yet** is gone.
6. Check that every step in **Steps** names a tool, **run by GeoForge (preflight)** (GeoForge's own check that the model software works), or **written by the agent**. Then approve.

When in doubt, choose **Modify the plan** rather than approving.

9.9 Approved data could not be downloaded

Signs: a card **Some approved data could not be fetched** lists each failed input with its error; the row in **Data in this plan** says **Fetch failed**; the pill of **Data in this plan** reads **Data needs attention**. The card offers **Retry the missing data**, which fetches only the missing items, and **Modify the plan**, which sends the plan back to the agent with your note.

1. Read the error after each input name on the card.
2. Remove the cause:

Error on the card	What to do first
"GeoForge could not reach GeoForge Database."	Check your internet connection, and Observation data under Use proxy for in Network & proxy .
"GeoForge Database server or gateway is unavailable."	Wait and try again later. The problem is on the server side.
"Daily download limit reached; resets at midnight server time."	Wait until after midnight server time.
"The downloaded file failed integrity verification twice."	Try again later. If it happens again, report it to the database owner.
"... already holds files with no receipt ... Existing files were not overwritten."	Move the files out of the named folder (do not delete them), then retry. Or choose Modify the plan to use them as your own input.
A token message ("That token isn't right...", "Your token expired...", "This token was revoked...")	Fix the token first. See 9.15.

3. On the card, select **Retry the missing data**. To use other data instead, select **Modify the plan** and name the source in the note.
4. Click **Continue with this choice**.

If you closed the card, reopen it with ◇ **Project status** → **Respond** (9.6). Chapter 3, section 3.10, lists every database message.

Caution: GeoForge Database downloads were not tested end to end on Windows for 0.6.54. If a retry keeps failing with the same error, report it (9.18).

9.10 Stop did not end a background job (Windows)

Stop ends the current turn of the chat: the agent, any KI tool or setup command, and the programs they started. Tray **Exit GeoForge** does the same before it quits. Model runs that GeoForge records are stopped normally.

The exception (known issue 2): a local CLI agent such as Claude Code runs commands through Git Bash (the command window that comes with Git for Windows). If it starts a job with `nohup`, `timeout`, `env`, `sh -c` or a bash script, Windows loses the link to that job, and **Stop** cannot find it. Signs: the computer stays busy after **Stop**, or new files keep appearing in the project folder. Files such a job writes later block **Completed**.

1. Press Ctrl+Shift+Esc to open Task Manager.
2. Click **Details**.

3. Right-click any column heading.
4. Click **Select columns**.
5. Tick **Command line**, then click **OK**.
6. Look for processes such as `bash.exe`, `sh.exe`, `nohup.exe`, `timeout.exe`, `python.exe`, or your model's program (for example `FSM2.exe`) whose command line contains your project folder or model folder.
7. Right-click that process.
8. Click **End task**. For a shell (`bash.exe`, `sh.exe`) that started other programs, click **End process tree** instead.
9. Back in GeoForge, send a message when you want to continue.

Caution: Do not end `GeoForge Desktop.exe` or `geoforge-agent-bridge.exe` (that quits GeoForge without stopping its work cleanly), and do not end processes whose command line does not point to your project or model folder: they may belong to other programs.

9.11 The project does not reach Completed

A project is **Completed** only when all of these hold:

- the plan you approved is still valid;
- every approved step has a passing run record, and its outputs passed GeoForge's automatic checks;
- no result file exists that no passing run produced;
- no step used an input file that was rewritten afterwards.

Read the last **GeoForge verification:** line in the chat, and the status line under ◇ **Project status** → **Details** → **Status evidence** (9.1). In these messages, a *receipt* is a run record; "unreceipted" or "not vouched for" means no passing run record covers the file.

You see	What it means	What to do
Status line Running — N steps done, M to go	M approved steps have no passing run record yet.	Ask the agent to run the remaining approved steps.
Status line ending "..., K unreceipted files", and GeoForge verification: "not complete yet — these files are not vouched for by a passing receipted run ... regenerate or remove them: ..."	Files exist that a failed attempt left or that were written outside GeoForge's recorder: for example by an earlier failed attempt or outside an approved executable step.	Reply: "Please regenerate these files through the approved steps, or remove them if they are not needed." If the agent will not remove them, and they are not your own inputs, move them yourself to a folder of their own inside the project, for example <code>notes\</code> , and send a message so GeoForge checks again. GeoForge checks only <code>outputs\</code> , <code>artifacts\</code> , <code>inputs\</code> and <code>calibration\</code> .
Ready to continue , rather than Completed , after a run	One or more required checks or steps still need work.	Read the host verification message and Project status; repair the named step or evidence instead of relying on the AI summary.
GeoForge verification: "not complete yet — these steps passed on input files that were rewritten afterwards; rerun them: ..."	An input changed after the step passed.	Ask the agent to rerun the named steps.
GeoForge verification: "this turn ran no receipted step and downloaded nothing."	The agent described work it did not run through GeoForge.	Do not use numbers from that turn. Ask the agent to run the approved step.
Headline Project needs attention ("An execution attempt failed; agent diagnosis is needed."); status line A run failed validation — see runs/evidence.json	A step's latest attempt failed GeoForge's automatic checks. This also happens when the agent reruns a step that had already passed and that attempt fails.	Send "Please diagnose the failed step and rerun it exactly as approved." GeoForge reruns under the same approval and reuses data it already has. The latest attempt counts, and the agent's own words do not: check the status line again after its reply.
Status line The agent needs a plan change , then The agent is revising the plan	The agent found it needs another tool, data source, period or model.	Wait for the new approval card, read it, and approve it if it fits. Your earlier approval is withdrawn.
Headline Ready to continue ; status line Stopped by you — send a message to continue	You pressed Stop .	Send a message to continue.

Project status

5

Results

Save and explain the results

Flow and signed evidence · observed 12m 58s ago

Data in this plan · 6

2/6 · Waiting for your input

From the submitted data inventory; status comes from receipts and files on disk, not from the agent's words.

You (1)

fsm2_alptal_forcing

FSM2

The KI expects forcing (8 declared inputs) from a dataset (DRIV1D=1 space-delimited ASCII met file, no header, one row per timestep (col 5)); none is pinned yet. Pick one in the plan.

Waiting for you

The run prepares itself (5)

fsm2_source_repo

FSM2

Prepared during the run.

Pending

fsm2_alptal_namelist

FSM2

Prepared during the run.

Pending

fsm2_alptal_outputs

FSM2

Produced; scientific checks pending

fsm2_alptal_snow_series

FSM2

Produced; scientific checks pending

fsm2_alptal_mass_balance

FSM2

Pending

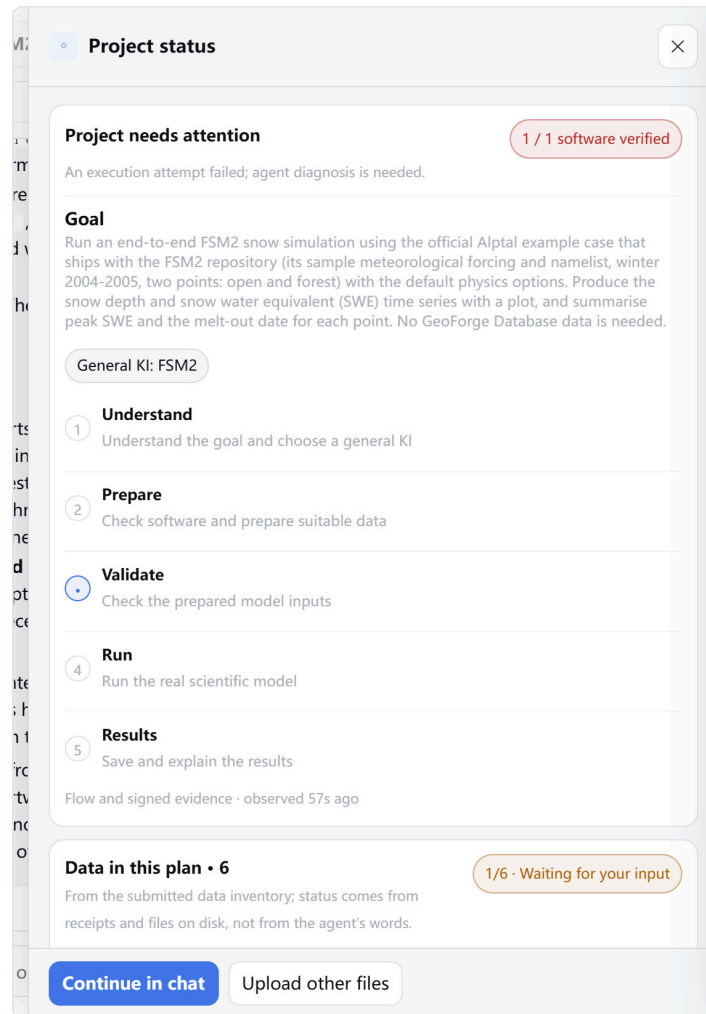
► Details

acquisition log, folders, catalogue search, data sources, calibration

Continue in chat

Upload other files

Check the data list: here the outputs read **Produced; scientific checks pending** and one input still reads **Waiting for you**, although the model ran. **Project status** shows what the run records prove, not what the agent says.



Check the headline **Project needs attention** and its line "An execution attempt failed; agent diagnosis is needed." Chapter 5 shows how this project reached this state.

Caution: Do not edit or delete anything in `.geoforge\`, or the files `runs\plan.json`, `runs\data-inventory.json` and `runs\approval.json`. A plan changed after approval forces a new plan and a new approval.

9.12 Windows SmartScreen blocks the installer

GeoForge is not code-signed, so Windows SmartScreen may stop the installer the first time.

1. Download the installer only from the `windows-v0.6.55` release page: <https://github.com/lzwei196/KISS-Knowledge-Infrastructure-for-Scientific-Simulation/releases/tag/windows-v0.6.55>.
2. Check its SHA-256 (a fingerprint of the file) against the value on the release page, as Chapter 1 describes. For example, in PowerShell: `Get-FileHash "C:\Users\you\Downloads\GeoForge-Desktop-Setup-v0.6.55-Windows-x64.exe" -Algorithm SHA256` If the value differs, delete the file and do not run it.
3. Double-click the installer.
4. If **Windows protected your PC** appears, click **More info**.
5. Check that the app name is `GeoForge-Desktop-Setup-v0.6.55-Windows-x64.exe`.
6. Click **Run anyway**.

The installer needs no administrator rights. It installs for your user only, into `%LOCALAPPDATA%\Programs\GeoForge Desktop`.

Caution: Use only the Windows files from `windows-v0.6.55`. The tested Windows build is the one on that page.

9.13 macOS blocks the app

The Mac app (release `v0.6.54`) is not notarised by Apple. macOS blocks it the first time, and again after every update. Do this only for an app you downloaded from the official release page and checked (Chapter 1).

With System Settings:

1. Try to open **GeoForge Desktop** once.
2. Close the warning.
3. Open the Apple menu → **System Settings** → **Privacy & Security**.
4. Scroll to **Security**. Next to the message about GeoForge Desktop, click **Open Anyway**.
5. In the dialog that follows, confirm that you want to open the app.
6. Enter your password or use Touch ID if macOS asks.

With Terminal: run this once, using the place where the app is:

```
xattr -dr com.apple.quarantine "/Applications/GeoForge Desktop.app"
```

9.14 Opening, quitting, and a second copy of GeoForge

	Windows	macOS
Where GeoForge shows	In your default web browser, plus a tray icon	In its own window
Closing the tab or window	GeoForge keeps running, and so does its work	GeoForge quits, and running work stops
Reopen	Double-click the tray icon, or right-click → Open GeoForge	Open the app again
Quit	Right-click the tray icon → Exit GeoForge	Close the window, or quit the app

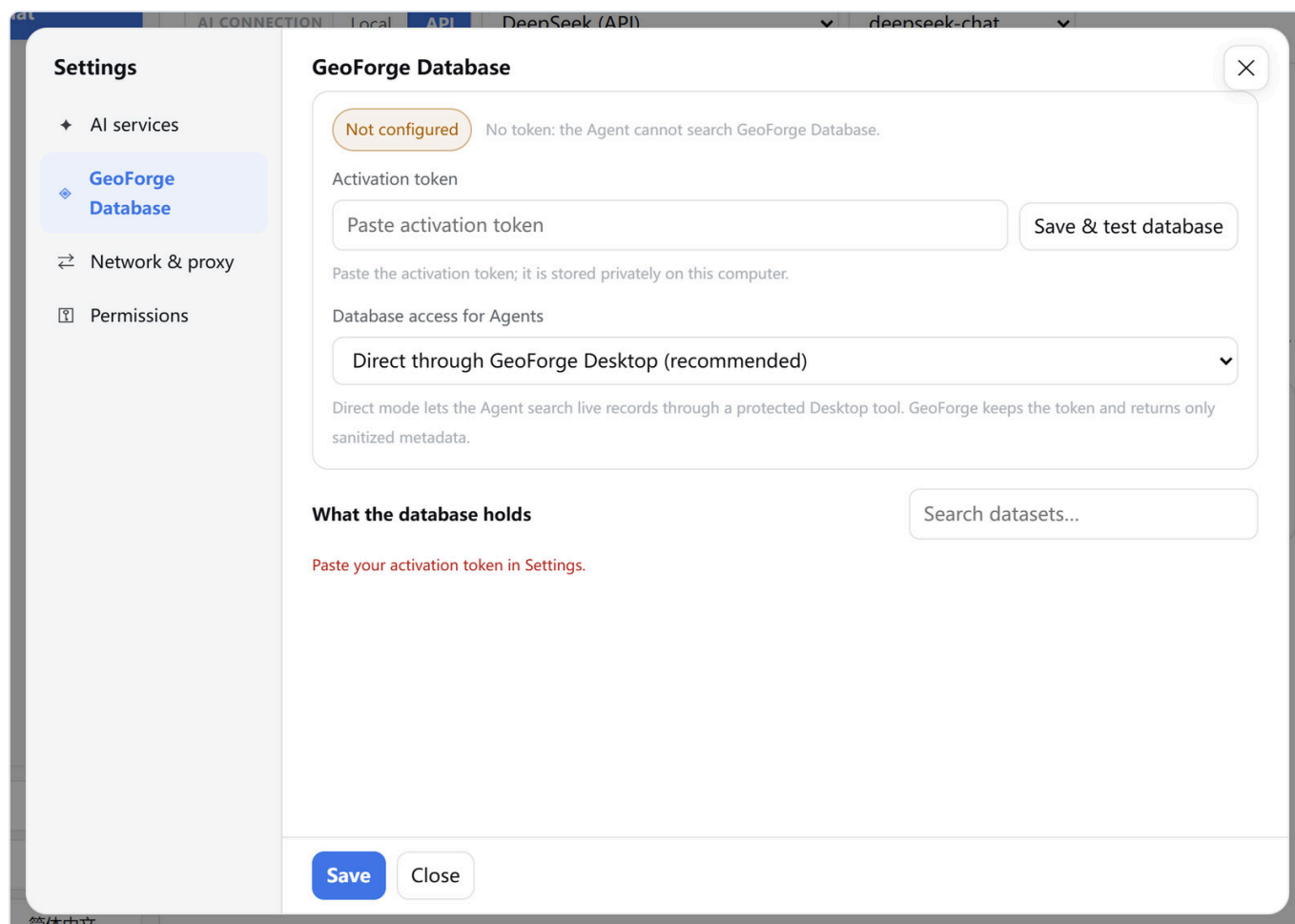
On Windows there is no guard against starting GeoForge twice. If you start it from the Start menu or desktop icon while it runs, a second copy starts with its own tray icon and its own address. The two copies do not know about each other: a chat that is working in one copy does not look busy in the other, and **Stop** or **Exit GeoForge** in one copy does not end work started by the other.

1. Find the browser tab where your work runs, and note its address, for example `http://127.0.0.1:52814/`.
2. Double-click each GeoForge tray icon in turn. Each one opens its own address in the browser.
3. Right-click the tray icon whose address you do not need.
4. Click **Exit GeoForge**.
5. Next time, reopen GeoForge from its tray icon instead of the Start menu.

Caution: **Exit GeoForge** stops all running work of that copy first. A model run in progress is interrupted.

9.15 GeoForge Database token problems

Open **Settings** → **GeoForge Database**. Read the pill at the top and the sentence next to it. After you click **Save & test database**, the result appears under the token box.



Check: the pill. **Not configured** means no token is saved, so the agent cannot search the database.

Pill	Meaning	What to do
Not configured	No token is saved.	Paste your token under Activation token and click Save & test database .
Off	Database access for Agents is set to Do not provide database access .	Choose Direct through GeoForge Desktop (recommended) , then Save .
Unavailable	The token cannot be used: the server rejected it, or GeoForge could not read the stored token.	Read the sentence next to the pill, fix the token, and click Save & test database .
Not synced	The token is saved, but the local copy of the catalogue has not been downloaded yet.	Click Save & test database , which also downloads the catalogue. Reopen Settings after a minute.

Common token messages are "That token isn't right — check for a copy/paste miss.", "Your token expired — ask the group owner for a new one." and "This token was revoked — ask the group owner." Chapter 3 explains how to get a token (3.2), how to enter it (3.3), and every other message (3.10).

Without the database, plans that use public sources or your own files still work.

9.16 Current Windows behaviour and remaining limits

Fixes included in 0.6.55

- Approval checks that the offered data-source choice agrees with the actual inventory. External sources remain visible. Choosing another source requires a revised plan and fresh approval.

- Steps that declare results must have an executable tool. Help/version probes cannot replace scientific run evidence. Real failed reruns still invalidate the affected step.
- API setup is installation-only with bounded startup probes. The host runs preflight before scientific execution. Local CLI setup does not yet have equivalent direct command enforcement.
- Closing/reloading a browser tab no longer terminates the running chat turn. Reopen the app and chat for saved progress. Use **Stop** to stop work.
- The host's verification verdict is shown after execution that immediately follows setup. An AI's success claim does not override failed or stale evidence.
- Windows archive failures now explain when a file is open elsewhere. Close the other application and retry.
- SHAW/CRHM/VIC installation, parsing, timezone and completion repairs are documented in Chapter 11. Missing matplotlib is now checked in the affected model environments.

What remains to check on your project

Windows process-tree snapshots may miss children detached through shell wrappers; if **Stop** leaves one, inspect its command line and project path before ending it (9.10). Some calibration model evaluations may open console windows; use **Stop**, not a manual close. The engine benchmark in Chapter 10 is not acceptance of every model's console behaviour.

DeepSeek API completed genuine FSM2, CRHM, VIC and SHAW official examples. That does not establish live acceptance for every local CLI, every KI, or authenticated GeoForge Database downloads. The calibrated SHAW reference-recovery benchmark is separate from observational validation.

Check GeoTIFF spatial coverage independently if the packaged reader cannot verify it. Some NetCDF compression plugins may be unavailable. A catalogue's whole-product size can differ from the amount your project needs; review the approved subset. Windows browser mode may require pasting folder paths rather than choosing them in a native picker. Older macOS builds may differ from this Windows edition.

The new **Guide** menu provides local HTML and PDF manuals. No token or account is required to read them; this does not grant access to paid AI services or the optional GeoForge Database.

9.17 Glossary

Term	Meaning
Agent	The AI working in your chat. It is either a local CLI on your computer (for example Claude Code, OpenAI Codex, Kimi Code) or an API model (for example DeepSeek (API)). GeoForge decides what it may do at each stage.
Local / API	The two kinds of AI under AI CONNECTION . Local uses a CLI (command-line program) installed and signed in on this computer. API uses a cloud AI service through a key you paste in Settings .
Turn	One round of work that starts when you send a message and ends when the AI's reply is saved. Stop ends the current turn.
KI (Knowledge Infrastructure)	GeoForge's package for one scientific model or workflow: how to install and check the software, which inputs it needs, which tools run it, and which checks its outputs must pass. All KIs are listed in KI Library .
Auto KI	The default: the header shows Auto KI and Chooses for each task . GeoForge reads your first request, picks the KI, and announces it in the chat, for example "Task understood · KI: FSM2. Building the plan...". If it picked the wrong model, start a new chat with the right KI pinned.
Pinned KI	A KI you choose yourself in Knowledge Infrastructures for this chat (the KI button in the header) before the first message. It is fixed for the chat. Loose word matches and negated mentions ("do not use CaMa-Flood") do not add other KIs, but a catalogue model you name plainly in your first message is added.
Verified on this machine	The KI's scientific software passed its check on this computer. Otherwise the KI shows Setup needed . It says the software runs here. It does not say your data are ready or your results are valid.
Flow	GeoForge's fixed path for a scientific chat: understand the task, plan (one question at a time), approval, data, software setup, the approved steps with run records, checks, Completed . In a chat, nothing is downloaded, installed or run before you approve. ◇ Project status names its source as "Flow and signed evidence".
Approval card	The card Approve the plan? showing the whole plan: What I understood, Data, Data sources, your pick (when offered), Steps, Scientific decisions, Waiting on you and, if any, Not ready to execute yet . Its options are Approve and start and Modify the plan . Your approval is signed into <code>runs\approval.json</code> .
Run record (receipt)	The signed record GeoForge writes for every approved step and every acquired input: the command, a fingerprint of the program, the exit code, fingerprints of the input and output files, and the approval it belongs to. Messages call it a "receipt". Only run records count as proof, never the agent's words. They are kept in <code>.geoforge\receipts\</code> .
Completed	Panel headline Project complete ; status line "Completed — every step has a verified receipt". Every approved step has a passing run record, and its outputs passed GeoForge's automatic checks (for example: the files exist and are not empty, contain no missing or infinite values, cover the expected time steps, and are not one constant value). It does not mean the model was validated against observations. It is final: the chat can still answer questions and draw plots from the existing files, but a new experiment needs a new chat.
Acquisition	GeoForge bringing in the approved data after you approve: by direct download, server clip, or Baidu Pan manual delivery. Acquired data are not checked science: rows read Acquired; scientific checks pending .
Server clip	For a large GeoForge Database dataset, the server cuts out your study area (and period or variables), and GeoForge downloads only that part, up to 2 GB. Direct downloads are for files up to 100 MB.
Baidu manual delivery	For data the server cannot send, ◇ Project status shows a Baidu Pan card with the link, the Extraction code , the file to take from the share and a Place at path. You download, place the files, and click Files are in place, continue . The link and code are shown only to you, never to the agent. The run record proves which files you placed; there is no server checksum for them.
Project folder	The folder one chat owns, named <code><date>-<project name>--<chat id></code> , by default in <code>C:\Users\you\kiss\projects</code> . It holds <code>inputs</code> , <code>outputs</code> , <code>artifacts</code> , <code>runs</code> , <code>memory</code> and <code>.geoforge</code> , among others. Open it with  Folder . Installed model software is kept outside it, in a shared folder such as <code>C:\Users\you\kiss\fsm2</code> .

9.18 How to report a problem

A good report lets someone else see what you saw. Collect the items below, read every file before you send it, and remove private data.

What to include

Item	Where to find it
GeoForge version and release, for example "0.6.54, windows-v0.6.55 installer"	Windows: Windows Settings → Apps → Installed apps → GeoForge Desktop . The portable copy's folder is named <code>GeoForge Desktop 0.6.55 Windows</code> . macOS: the release you downloaded (<code>v0.6.54</code>).
Operating system	Windows: press Win+R, type <code>winver</code> , press Enter; note the edition and version, and whether Windows runs in Chinese. macOS: Apple menu → About This Mac ; note the macOS version and the chip.
AI and KI	The AI CONNECTION row (Local or API , provider, model) and the KI shown in the header.
What you did, what you expected, what happened	Your own words, with the approximate time.
A screenshot of Project status	The whole panel. If the problem is about data, open Details too. Add a screenshot of the card or message you are reporting.
The chat text	Point at a message; a Copy button appears next to the sender's name. Paste the text into your report. Exact text helps more than a picture. The whole conversation is in <code>memory\transcript.jsonl</code> ; send it only if asked, and read it first.
Files from <code>runs\</code>	Click  Folder in the chat header to open the project folder. See the table below.

File	What it shows
<code>runs\project-events.jsonl</code>	The project's timeline
<code>runs\evidence.json</code>	What GeoForge verified, step by step
<code>runs\flow-state.json</code>	The current Flow state
The newest files in <code>runs\logs\</code>	The output of each recorded tool run
<code>runs\plan.json</code> , <code>runs\data-inventory.json</code>	The plan and its data list, if the problem is about the plan or data

For a software-setup problem started from the **Agent setup** page, also include `setup-agent.log` from the model's installation folder (for example `C:\Users\you\kiss\fsm2`), if it exists. For a connection problem, include the text of the **Test AI & GitHub** report.

Never include

Caution: Never put any of these in a report, a screenshot or a shared folder:

- API keys, and the GeoForge Database activation token;
- Baidu Pan links and extraction codes, and the file `.geoforge\manual-download-details.json`, which stores them;
- `%APPDATA%\KISS\settings.json` (macOS: `~/Library/Application Support/KISS/settings.json`), which stores API keys in plain text;
- the whole project folder or the whole `.geoforge` folder;
- screenshots of the small settings box inside **KI Library**, which shows the end of each API key.

Replace your user name in paths with `you`, for example `C:\Users\you\kiss\projects\...`.

Where to send it

- Open an issue at <https://github.com/lzwei196/KISS-Knowledge-Infrastructure-for-Scientific-Simulation/issues>. Issues are public, so check again that nothing private is included.
- For token, quota or Baidu Pan delivery problems, contact the GeoForge Database owner who gave you the token (Chapter 3, 3.2).

10 Calibration

Calibrate a real model

Calibration searches for parameter values that improve a model's fit to a chosen target. GeoForge supplies a common search engine, while a KI adapter must write the parameters into the real model, execute it and score its outputs. The engine must never replace the scientific model.

Separate three claims. A working engine, a successful calibration run and a model validated against independent observations are different results. The SHAW benchmark in this guide tests recovery of the publisher's reference output; it is not field validation.

Prepare a project

1. First run the uncalibrated model on the intended input data. Check dates, units, water/energy balances and physical plausibility before optimizing anything.
2. Start a new chat if the earlier project is already **Completed**. Choose the intended KI and AI before sending the first message.
3. Provide authentic observations with timestamps, units, locations and missing-value rules. Specify which simulated output corresponds to each measured variable.
4. State parameter names, meanings, units, defensible bounds, fixed parameters, fitting period and a separate held-out period. Keep the held-out data out of parameter search.
5. Set the search method, run budget and random seed. Ask for both baseline and fitted scores on both periods.

Example request for your own observations

Calibrate my selected model against the observation file I will upload. First check the baseline and the time/space/unit alignment. Propose a small set of physically defensible parameters and bounds; keep forcing and measurement corrections fixed unless independently justified. Fit only the agreed training period and reserve the later period as holdout. Use DDS with a budget of 100 evaluations and a recorded seed. Report baseline and fitted metrics on both periods, rejected runs and the exact model/input versions. Show the plan before running.

The dates, bounds and metrics are scientific decisions for your case. They are not supplied by this example request. Answer GeoForge's planning cards and use the input's **Upload files** button to bind required observations.

Prepare the adapter without executing it

If the selected KI needs a project adapter, planning can write only that KI's calibration contract and runner into its project calibration folder. This prepares the two files for review; it does not authorize optimization, native model execution or arbitrary project edits.

Review and run

1. In **Project status** → **Details** → **Calibration**, check whether an adapter and project case exist. **Adapter needed** means preparation is still required; it is not an engine failure.
2. **Calibrate with agent** (or **Continue calibration**) sends a request to the chat. **Add observations** attaches files; bind any required plan input through its own upload row.
3. On **Approve the plan?**, review inputs, parameter bounds and defaults, objective, train/holdout split, algorithm, requested budget, seed and executable calibration step. The plan binds the exact adapter and calibration contract that will run. Change the plan if they differ from your study.

4. Select **Approve and start**, then **Continue with this choice**. GeoForge records approved execution; a verbal claim from the AI does not replace that record.
5. Keep the app running. **Stop** ends the current attempt and preserves its recorded outcome. A stopped or failed attempt must not be presented as a successful fit.

Changing the adapter, contract, data, parameter bounds or invocation after approval requires the relevant plan to be reviewed again. Both API agents and local CLI agents must use the approved calibration step and its recorded run; switching agent type does not bypass approval. Do not change files underneath a running optimization.

Choose a defensible protocol

Engine, adapter and case

Part	Responsibility	Project location
Shared engine	Runs the selected optimizer and checks the holdout	Bundled with GeoForge
KI adapter	Applies parameters, launches the native model, reads real output and returns a score	<code>calibration/kis/<KI>/</code>
Project case	Defines parameters, targets, periods, objectives and budget	<code>calibration/cases/</code>
Run evidence	Records requests, logs, scores, parameters and acceptance	<code>calibration/runs/<run id>/</code>

A shipped adapter is a starting point, not proof that it fits your particular basin, variable or observation format. For a KI without one, the agent prepares a project-owned adapter; the shared optimizer remains unchanged. The tested SHAW benchmark uses this route.

Search methods

Target	Available methods	Practical choice
One scalar loss	DDS, SCE-UA, DREAM	Choose a loss whose direction and units you understand. Record the seed and budget.
Several competing objectives	NSGA-II, NSGA-III, MOEA/D	Review trade-offs among candidate solutions rather than choosing by a single hidden average.

The method names describe search algorithms, not a guarantee of convergence. A large parameter space needs more evidence and computation than a small, identifiable one. The app accepts a requested optimizer budget from 1 to 10,000. This is not a strict ceiling on native model calls: optimizer initialization or population batches, baseline checks and holdout checks can add work. Review the actual evaluation count and allow for the cost of each native run. Begin with a small smoke run before committing a large budget.

Before approving

- **Target and alignment:** match the same variable, units, time support and spatial support. A point observation does not automatically validate a basin mean.
- **Bounds:** justify them from model documentation, measurements or literature. Keep physically linked parameters consistent; do not optimize nuisance parameters merely because they move the answer.
- **Forcing:** do not tune precipitation, elevation or observation transformations simply to hide a model-data disagreement.

- **Split:** use a held-out period that the search cannot see. Warm-up may precede a scoring period but must be documented.
- **Metrics:** state which metric determines acceptance and which are diagnostic. Include baseline and fitted values, sample counts and missing-data handling.
- **Reproducibility:** retain the native executable version/hash, original inputs, applied parameters, seed, budget and scoring code.

For Celsius temperature, percentage bias is inappropriate because Celsius has an arbitrary zero; the SHAW reference test reports NSE and RMSE instead. For other variables, choose metrics using that model's validation convention rather than copying this temperature protocol.

An improved training score with worse holdout performance is evidence of overfitting or a deficient protocol. It is not a reason to reuse the holdout as more training data and report the same holdout score as independent.

Read the result and keep evidence

Interpret the Calibration card

Status	Meaning and next action
Engine unavailable	The bundled engine cannot load; repair the installation before running.
Adapter needed	The selected KI needs a project adapter.
Project case needed	Define the target, parameter bounds, observations and held-out period.
Workflow prepared	A case exists; the agent must still check it against this study.
Holdout passed	The result satisfied the configured held-out check. Read the report and scientific limitations.
Holdout not passed	Do not promote the fitted parameters as a validated result. Diagnose the protocol or model/data mismatch.
Run needs review	Inspect the failed or interrupted attempt and its logs.

Validated on the latest calibration means it passed the configured check. It does not certify all model physics, identify a unique parameter set, or establish validity at other sites. **Not promoted** means the fit is not accepted for use by that workflow.

Open the files

Use the project's files/Results view or its folder on disk. Under `calibration/runs/<run id>/`, keep:

- `report.json` : the host's result and acceptance summary.
- `engine-report.json` : optimization and holdout details.
- `engine-request.json` : the actual engine request, including the declared case and protocol.
- `engine.log` : execution and diagnostic messages.

Each run retains its own immutable copy of the calibration runtime, so a later run cannot overwrite the earlier run's recorded runtime evidence. Keep the project adapter, case, original observations and native model inputs with these reports. Individual evaluation files should make it possible to verify which values were applied to the model. A successful process exit without readable scientific output is not a valid evaluation.

Decide whether the fit is usable

1. Confirm the real model ran and every reported parameter was applied. Check the command, executable identity and generated parameter/input files.

2. Compare baseline versus fitted metrics on **both** training and holdout periods; verify the sample counts and dates.
3. Inspect time series and residuals, not only a headline score. Check physical constraints and balances required by the KI.
4. Report uncertainty, identifiability limitations and any boundary-hitting parameter. A single best point does not quantify uncertainty.
5. Check GeoForge's run records and final project status. A missing or rejected required holdout check is a warning, not scientific completion. The AI's narrative cannot override failed validation, stale inputs or missing evidence.

Use GeoForge's **Stop** control to interrupt an optimization. Windows launches non-interactive calibration subprocesses without requesting a console window. Any model-specific graphical interaction still needs separate verification.

If a run fails, preserve its reports and error text, fix the actual cause and obtain a fresh passing run. Never rename a failed report as a success or substitute reference output for model execution.

Tested example: SHAW reference recovery

This Windows benchmark exercises the real SHAW 3.03 native executable repeatedly through its KI wrapper and the bundled calibration engine. It asks whether optimization can recover a known parameter setting from the publisher's own output.

Scope: the target is an official model reference, not measured field data. The result demonstrates parameter application, native execution, scoring and holdout handling. It must not be quoted as observational model skill.

Reproducible setup

Use the five unchanged SHAW Trial inputs listed in the three-page quickstart and the publisher's temperature reference table from the [official SHAW 3.03 archive](#). The reference file is `Shaw303/Output/Trial/temp.out` inside that archive. Preserve the original files. The project adapter creates a separate candidate input set for each evaluation.

Setting	Tested value
Parameter	<code>bulk_density_scale</code> , applied to all 11 soil nodes
Bounds	0.8 to 1.5
Deliberately perturbed baseline	1.3
Known reference setting	1.0
Method / seed / search budget	DDS / 20261002 / 60 requested optimizer evaluations
Training rows	155 hourly rows × 11 nodes; 1986-12-04 13:00 to 1986-12-10 23:00
Held-out rows	145 hourly rows × 11 nodes; 1986-12-11 00:00 to 1986-12-17 00:00
Metrics	NSE and temperature RMSE in °C; no Celsius PBIAS

These bounds and the artificial baseline are a **software reference-recovery protocol**, not recommendations for estimating field soil density. Other Trial parameters and forcing remain fixed.

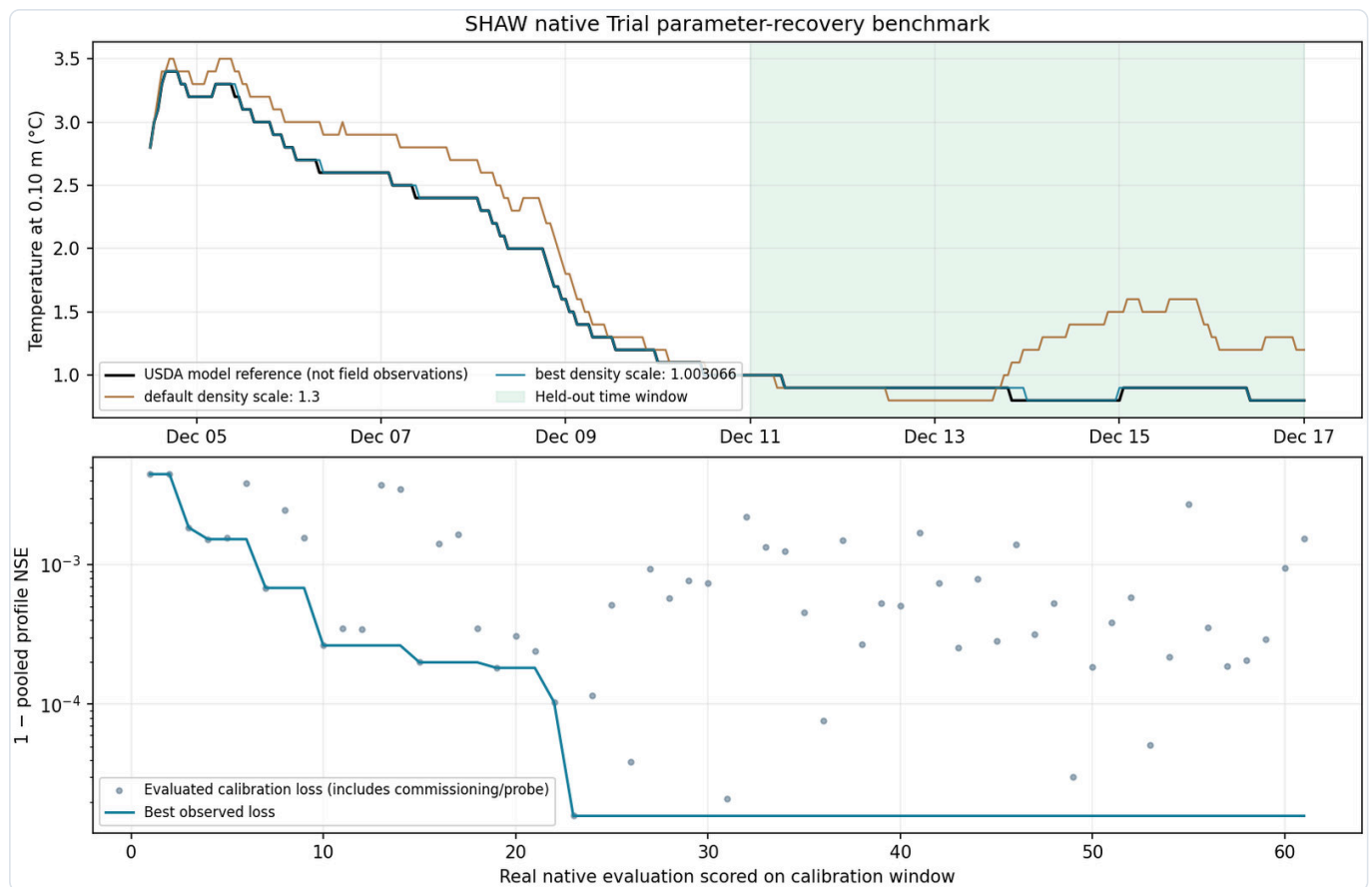
Actual result

The fitted scale was **1.003065603**. Training NSE was **0.99998413** and RMSE **0.011098 °C**. Held-out NSE improved from **0.98713640** at the perturbed baseline to **0.99998794**. The configured held-out check passed, and the engine marked the result promotable within this benchmark.

The shared engine was pinned at revision `579162102f71f2fa4619b874a21bd219caaed25e`. The adapter was project-owned (`calibration/kis/SHAW/calibration.yaml` and `tools/calib_run.py`), not a newly claimed universal SHAW adapter. Full execution evidence was retained in the test project and `D:/GeoForge-Calibration-20261002/native-calibration-result.json` on the validation machine.

The source and frozen Windows engine produced the same fitted result. The test made 64 actual native calls, including the probe, baseline and holdout checks. A separate run at the known reference setting 1.0 reproduced the publisher's temperature output byte for byte. All six optimizer backends also passed a synthetic interface test; those synthetic tests do not demonstrate scientific performance.

Inspect the actual model output



The upper panel compares the publisher reference, the deliberately perturbed baseline and the fitted native run at 0.10 m soil depth. Shading marks the held-out period. The lower panel shows actual scored evaluations and the best training loss. These are model reference outputs, not field observations.

Ask the AI for this benchmark

Prepare a project-owned SHAW calibration adapter that runs the real native Trial for every evaluation. Use the official temperature output only as a reference-recovery target, not observations. Recover `bulk_density_scale` on all 11 nodes in `[0.8, 1.5]`, with baseline 1.3, DDS seed 20261002 and budget 60. Train through 1986-12-10 23:00 and hold out 1986-12-11 00:00 through 1986-12-17 00:00. Report applied parameters, actual run records, NSE and RMSE for both splits and the uncalibrated baseline. Preserve original inputs and present a plan before execution.

For a scientific study, replace this reference-recovery target with suitable independent observations and justify a new protocol. Do not reuse these near-perfect reference scores as evidence for a real catchment or field site.

11 Verified Windows examples and current behaviour

The earlier FSM2 walkthrough shows real 0.6.54 screens and remains useful for learning the detailed workflow. This chapter records the additional 0.6.55 checks and directs new users to a short, reproducible SHAW case.

Reproduce SHAW without a private data source

Open **Guide** → **Quickstart** • 3 pages. It covers AI setup, software setup and the full approved run. The official input download is [USDA-ARS Shaw303.zip](#). The five original files are directly inside its Shaw303/ folder:

Trial.303.inp , Trial.30.sit , Trial.30.wea , Trial.moi , Trial.tem .

Upload these five inputs to a fresh SHAW chat. Do not upload precomputed files from Shaw303/Output/Trial/ as if they were new model results. That folder contains the publisher's references for comparison. Keep inputs and reference output clearly separate.

Request genuine native execution through the KI, CSV parsing and plotting, with all original control choices preserved. Review the plan and approve the three executable steps. After running, inspect **Project status**, the raw tables, five CSVs and profile figure. A model's software-verification badge alone does not establish a completed scientific run.

What was checked

Model / official case	Real output checked	Acceptance
SHAW 3.03 / Trial	Six tables: temperature 301 rows, moisture 301, liquid 14, energy 300, water 13, frost 48; 11 profile nodes; all end 1986-12-17 00:00	Printed values matched the official reference; five CSVs and the plot checked; all three Desktop steps completed with passing signed records
VIC 5.1.0 / Stehekin	16 cells × ten days, 1949-01-01 to 1949-01-10; 48 flux/snow/snowband files	Original forcing/parameters preserved, finite output and precipitation aggregation checked; Desktop Completed
CRHM / Bad Lake	8,760 hourly records, 1973-01-01 01:00 to 1974-01-01 00:00; three HRU SWE series	Original parameters/forcing preserved; raw result matched independent native run; Desktop Completed

CRHM peak SWE was 36.38742, 65.86527 and 64.42825 mm for the three HRUs. The native executable reported 4.7_16 ; an older manifest's 1.3.5 label is not evidence of the executable version. Executable identity was recorded independently.

These are official forward examples, not calibration or validation against field observations. Installs needed assistance: the first CRHM setup was blocked by acquisition/build work, VIC's first Windows build crashed on actual input, and SHAW needed runtime/dependency/path repairs. The failed attempts were preserved. Later successful results do not rewrite that history or promise unattended installation on every PC.

Repairs that matter when using the app

- **Startup:** Windows loader/crash exits cannot pass merely because a file or banner exists. Preflight checks the configured interpreter and genuine executable.
- **SHAW:** managed binaries, all four compiler DLLs and matplotlib are checked. The wrapper stages original uploaded inputs, retains PEST/control settings and handles an authorized overwrite using the native model's own prompt. Every enabled output must still be fresh and complete.
- **VIC:** the Windows configuration cursor/newline and error-reporting repairs were reproduced on final 5.1.0; LF and CRLF inputs produce identical results. The phrase "classic driver" no longer selects the unrelated CLASSIC KI.

- **CRHM:** `TZ=UTC0` is applied only to the model process so host timezone does not change model civil-time calculations and SWE. It does not convert a station's geographical timezone. Merely shifting old output labels cannot fix a timezone-affected simulation.
- **Review:** materialized tools are checked against the selected project; generated output paths are not described as existing inputs. Tool or plan changes can invalidate an old approval.
- **Evidence:** named stdout/stderr diagnostic logs remain hashed evidence without being parsed as numeric output. Real scientific files still undergo numeric/completeness checks. Failed, stale or unrecorded results cannot be made successful by the AI's summary.

The fresh SHAW project reached **Completed** with three passing steps and no missing, stale, rejected or unrecorded outputs. Its older unsuccessful conversation was retained. If your project has a long history of incompatible plan changes, preserve it and use a fresh project with the same authentic inputs and a newly reviewed plan.

For parameter fitting and the separately tested native SHAW reference-recovery benchmark, read Chapter 10 or **Guide** → **Calibration guide**.